18.03 Class 1, Feb 3, 2010

Introduction and first methods

[1] Introduction
[2] What and Why
[3] Separable equations
[4] Geometric methods

[1] Welcome to 18.03.

I hope you've picked up an information sheet and syllabus, and a problem set when you came in. All this is also available on the web.

On the PSet you will see reference to:

EP = Edwards and Penney, (6th or 5th ed). The very similar to the each other and to the 4th. I'll try to be sure to give the numbering from both.

Notes = Notes and Exercises. Available from CopyTech in the basement of Building 11, though it is also available (in small pieces) through the course website.

SN = Supplementary Notes. Available in sections through the course website.

The first PSet is due Friday, Feb 12, at 12:45, in the cubbies at 2-106, next to the UMO.

I also hope you went to recitation on Tuesday, where these yellow booklets were handed out. More are available here. Usually I ask students to manufacture the booklet themselves, but the UMO was kind enough to do that for you already. Yea! We'll use them for a primitive but effective form of communication between us. It's private; really I'm the only one who can see the numbers you put up. Today, only one question. More later.

If you need to change recitation, go to the website and follow the link to the "grade management system," which is on Stellar. The sizes are limited.

There are two lecture times for this course, 1:00 and 2:00. You can attend either lecture, BUT you should register in the hour at which you plan to take the hour exams.

The information sheet and the website contain lots of other information. For example, my OFFICE HOURS are Wednesdays 3:15 - 5:15 : e.g., this afternoon.

Any questions?

You should also have picked up the listing of the 10 essential skills at your recitation. Teachers of these courses know the list of skills. They expect you will know how to do these things.

Here's a list of some of the larger courses listing 18.03 as a pre-requisite or co-requisite.

2.001 Mechanics and Materials I

2.003 Dynamics and Vibrations

2.005 Thermal-Fluids Engineering

2.016 Hydrodynamics

3.23 Electrical, Optical, and Magnetic Properties of Materials

6.002 Circuits and Electronics

6.021 Quantitative Physiology: Cells and Tissues

8.04 Quantum Physics I

10.301 Fluid Mechanics

12.005 Applications of Continuum Mechanics to Earth, Atmospheric, and Planetary Sciences

16.01 Unified Engineering

18.100 Analysis I

18.330 Introduction to Numerical Analysis

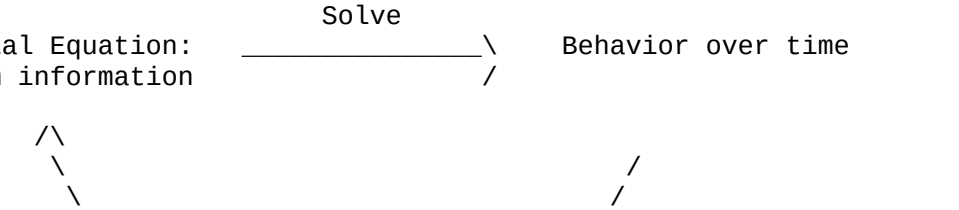
20.309 Biological Engineering II

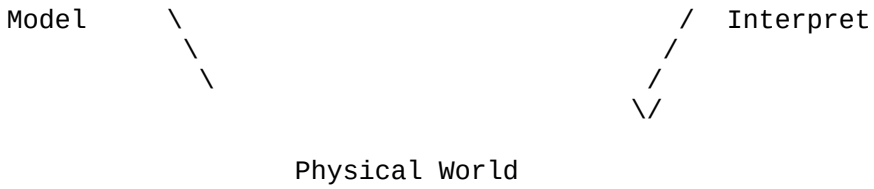
22.05 Neutron Science and Reactor Physics

Altogether 140 courses at MIT list 18.03 as a prerequisite or a co-requisite.

[2] A DIFFERENTIAL EQUATION is a relation between a function and its derivatives.

Differential equations form the language in which the basic laws of science are expressed. The science tells us how the system at hand changes "from one instant to the next." The challenge addressed by the theory of differential equations is to take this short-term information and obtain information about long-term overall behavior. So the art and practice of differential equations involves the following sequence of steps: one "models" a system (physical, chemical, biological, economic, or even mathematical) by means of a differential equation; one then attempts to gain information about solutions of this equation; and one then translates this mathematical information back into the scientific context.





A basic example is given by Newton's law, F = ma. a = acceleration, the second derivative of x = position. Forces don't effect x directly, but only through its derivatives. This is a second order ODE, and we will study second order ODEs extensively later in the course.

[3] In this first Unit we will study ODEs involving only the first derivative: first order: y' = F(x,y) .

Example 1: y' = 2x Solution by integrating: y = x^2 + c.

Notice that there are many solutions.
An expression like this, involving a constant (c here) is called the GENERAL SOLUTION. The constant is a "constant of integration."

Example 2: y' = ky. General Solution: y = Ce^{kx} . MEMORIZE THIS

- it's the central example in this course.

(In fact, a good definition of the exponential function e^x is that it is the solution of the differential equation y' = y such that y(0) = 1 .)

Q1: What is the general solution to the ODE dy/dx = 2y+1 ?

1. y = Ce^{2x} - 1

2. y = Ce^{x/2} - 2

3. x = y^2 + y + c

4. y = e^{x/2} + C

5. y = Ce^{2x} - 1

6. y = Ce^{2x} - 1/2

7. y = e^{2x} + c

8. None of the above

Blank: Don't know

The method: "separation of variables." You studied this in recitation yesterday. Recall the method: Put all the x 's on one side and y 's on the other (if possible):

dy/(2y+1) = dx

Integrate both sides:

(1/2) ln|2y+1| + c1 = x + c2

Amalgamate the constants and (if possible) solve for y in terms of x :

ln|2y+1| = 2x + c , |2y+1| = e^c e^{2x} , 2y+1 = C e^{2x}

y = C e^{2x} - 1/2

So the answer is: 6

We can check this!: y' = 2 C e^{2x} = 2(y + 1/2) = 2y+1

It's a nice feature of differential equations in general: it's easy to check your answer!

Q1.2. Is y' + xy = x separable?

1. Yes

2. No

Blank: don't know.

Well, y' = x - xy = x(1-y) so dy/(1-y) = x dx : YES.

We could go on to solve this, but you can do that on your own.

We will see many other methods of solving various types of equations. Unfortunately, most real life equations aren't explicitly solvable, and often you don't actually care as much about the explicit solution as about the general properties. That's one place where today's topic is helpful.

[4] Graphical approach

The ODE y' = F(x,y) specifies a derivative - that is, a slope - at every point in the plane. This is a DIRECTION FIELD or SLOPE FIELD.

Eg y' = 2x : I drew some of the direction field. Notice that the slope F(x,y) does not depend on y here: It is invariant under vertical translation.

A SOLUTION of the differential equation is a function whose graph has the given slope at every point it goes through. I drew some. The graphs of solutions are INTEGRAL CURVES. They are vertically nested parabolas. The translation invariance of the direction field is reflected in the fact that a vertical translate of a solution is another solution.

To specify a particular solution of an ODE you have to give an INITIAL CONDITION: when x takes on a certain value, y takes on a specified value.

Eg $y' = y$: I drew some of the direction field. Notice that the slope $F(x,y)$ does not depend on x here: It is invariant under horizontal translation.

Graphs of solutions are now horizontal translates of each other.

Example 3: $y' = y^2 - x$.

This equation does not admit solutions in elementary functions. Nevertheless we can say interesting things about its solutions.

To draw the direction field, find where $F(x,y)$ is constant, say m. This is an ISOCLINE. Eg

$m = 0 : x = y^2$. I drew in the direction field.

$m = 1 : x = y^2 - 1$

$m = -1 : x = y^2 + 1$.

I invoked the Mathlet Isoclines and showed the example.

I drew some solution curves. We have seen in action the

EXISTENCE AND UNIQUENESS THEOREM FOR ODEs:
$y' = F(x,y)$ has exactly one solution such that $y(a) = b$,
for any (a,b) in the region where F is defined.
(The solution $y(x)$ may only exist for x near to a .)

(You actually have to put some technical conditions on F -- see EP.)

The E and U theorem says that there is just one integral curve through each point:

EVERY POINT LIES ON JUST ONE INTEGRAL CURVE: NO CROSSING ALLOWED.

The applet makes it look like many solutions coalesce, but this is just a pixel problem. In reality they are separate, but very close to each other.

Many seem to bunch up along the bottom
branch of the parabola. Can we explain this?

I cleared the solutions and drew in just the isoclines $m = -1$ and $m = 0$. Once a solution gets between these two parabolas, it can never escape. The poor thing can't cross the $m = -1$ parabola because it would have to have slope greater than -1 when it does; and it can't cross the $m = 0$ parabola because it would have to have slope less than zero when it does. This is a FUNNEL.

So solutions in that region stay in that region: they are trapped

between those two parabolas, which are asymptotic as x ---> infinity.
All these solutions become very close to the function - sqrt(x)
for large x . This is an ideal situation! - we know, approximately,
but with increasing accuracy, about the long term behavior of these
solutions, and the answer doesn't depend on initial conditions (as long
as you are in this range). This is "stability." (Of course if the solution
doesn't get trapped, it's a different story.)

Direction fields let you visualize the qualitative behavior of solutions
to differential equations, and this is often what you want to know.
But we also want to be able to solve ODEs "analytically," that is,
using formulas.

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18.03 Differential Equations

Spring 2010

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18.03 Class 2, Feb 3, 2010

Numerical Methods

[1] How do you know the value of e?
[2] Euler's method
[3] Sources of error
[4] Higher order methods

[1] The study of differential equations has three parts:

. Analytic, exact, symbolic methods

. Quantitative methods (direction fields, isoclines ....)

. Numerical methods

Even if we can solve symbolically, the question of computing values remains. The number e is the value y(1) of the solution to y' = y with y(0) = 1. But how do you find that in fact e = 2.718282828459045.... ? The answer is: numerical methods. (Euler already computed e to at least 18 decimal places. It is now known to some 200 billion places, but I won't write them all out here.)

As an example, take the first order ODE

y' = y^2 - x = F(x,y) with initial condition y(0) = -1

-- an Initial Value Problem, IVP.

For example: what is y(1) ?

I invoked the Euler's Method Mathlet, and selected F(x,y) = y^2 - x . The slope field appears. I selected the initial condition (0,-1) , and then invoked "actual."

This solution is one of those trapped in the funnel, so for large x , the graph of y(x) is close to the graph of - \sqrt(x) : y(100) is very close to -10 .

But what about y(1) ?

The listing says y(1) = -.83 . How did the computer know this?

[2] The tangent line approximation gives one approach.

Recall: We have a function y(x), and a point (a,y0) on its graph (so y(a) = y0). The tangent line to the graph at this point has slope y'(a), which I will denote m_0 . The tangent line approximation is

y(a+h) ~ y0 + m0 h

The right hand side is the y-coordinate of a point on the tangent line directly above or below (a+h , y(a+h)).

In our case, we do know the derivative of y at x = 0 :

y'(0) = F(0,-1) = (-1)^2 + 0 = 1

So the tangent line at (0,-1) has slope 1 , and we find the tangent line approximation y(1) ~ -1 + 1 = 0. I showed this on the applet.

Of course this is a really bad appromation, because we took such a big hop. The tangent line approximation works much better if h is small.

The idea then is to take a *small* hop, and then use the direction field at the end point of that line segment to decide where to go next. Keep it up till you get where you want to go.

Let's set this up more generally. You have an initial value problem

y' = F(x,y) , y(a) = y0 .

You want to estimate y(b) where b > a . Divide the interval from a to b into small hops each of width h . Give names to the hash marks:

x0 = a , x1 = a+h , ... , xk = a+kh , ...   b

Then we build a polygon starting at (x0,y0). The first segment ends at (x1,y1) where y1 = y0 + m0 h , m_0 = F(x0,y0) . The second ends at (x2,y2) where y2 = y1 + m1 h , m1 = F(x1,y1) . And so on.

I call the line segments "Euler struts" and the whole thing the Euler polygon. It's supposed to be an approximation to the graph of y.

Notes: (1) x_1, ... and y1, ... depend on h .
(2) y0 = y(x0) , but y1 is not y(x1); it is only an approxiation to it.

Let's do this for our example, with h = 1/2 : so two steps.

Keep this calculation organized in a table.

h = .5

k	xk	yk	mk = yk^2 - xk	mk h
0	0	-1	1	.5
1	.5	-.5	-.25	-.125
2	1.0	-.625		

I checked this on the applet and then showed some more Euler polygons for this equation.

h = 1 estimate of y(1) is 0.00
h = .5 -.62

h = .25	-.75
h = .125	-.80
"actual"	-.83

[3] Euler's method is rarely exact. Let's identify some sources for error. Much of numerical analysis is understanding and bounding potential error.

(1) Round off accumulation. Each computation involves a small error, from round off, and these errors accumulate. So you want to use the fewest number of steps possible. On the other hand making the stepsize small keeps the polygonal approximation close to the actual solution. There is a tension here.

Usually more important than this is:

(2) The Euler polygon is not an exact solution, and the direction field at its vertices differs more and more from the direction field under the actual solution. At places where the direction field is changing rapidly, this produces very bad approximations quickly.

The variation of the direction field causes the integral curves to bend away from the Euler strut.

Can we predict whether the Euler approximation is too big or too small? In this example they are too big. This is because the solution if curving down and leaving the polygons above it.

How would we know without a picture whether the solution is

convex = bending up or
concave = bending down ?

Well there's $y''(0) > 0$ convex at (0,-1)
$y''(0) < 0$ concave at (0,-1)

How to find y'' ? well, $y' = y^2 - x$
$y'' = 2yy' - 1$

so $y''(0) = 2(-1)(1) - 1 = -3$: concave : as we see in the picture.

If $y''(x_0) < 0$, Euler's estimate is too high
if $y''(x_0) > 0$, Euler's estimate is too low.

[4] Theory shows that for small h , the error in Euler's method is at most $C1\ h$ for some constant $C1$. It is a "first order method" (no relation with "first order equation.") Of course if you knew in advance exactly what the error was, you'd be rich and famous.

Making h smaller is one way to decrease the error caused by the variability of the direction field. But there are more clever ways which turn out to be even better.

Here is one: RK2 = improved Euler = Heun:
Poll the slope field at the end of the Euler strut and use the average.

Q2.1 What is the first slope for the RK2 polygon for y' = y^2 - x
with h = 1/2 and initial condition y(0) = -1 ?

1. 1
2. 1/2
3. -1/4
4. 3/8
5. 5/8
6. Don't know

The needed data is present in the h = 1/2 Euler table:
F(0,-1) = 1. The end of this Euler strut is (1/2,-1/2).
F(1/2,-1/2) = -1/4 . The average is 3/8 : [4] is correct.

[Almost everyone offering an answer got this right.]

This brings us down closer to the actual solution curve.

RK2 is *second order*: for small h the error is at most C2 h^2 for
some constant C2 .

Each evaluation of the direction field costs money - it takes time.
Euler polls once per step, RK2 polls twice per step.

So if we want to compare efficiencies, we should compare Euler with h
and RK2 with 2h . The error with Euler is around C1 h .
The error with RK2 is around C2 (2h)^2 = 4C2 h^2 . If h is small enough,
the the RK2 error will definitely be smaller than the Euler error,
even if C2 is larger than C1. In fact, C2 tends to be smaller than C1,
so RK2 wins for two reasons.

There is an RK4 method too, which polls the direction field 4 times before
deciding which way to head out. It's a "fourth order method," and its error
is approximately C4 h^4. To make a comparison with Euler(h) you should use
RK4(4h), but again C4 (4h)^4 = 64 C4 h^4 will be less than 4C2 h^2
or C1 h for h small enough; and in fact C4 itself is usually smaller
than C2 and C1.

Here's some data using the ODE y' = y, y(0) = 1; the solution is
y = e^x, and we study y(1) = e. Number of evaluations of the
direction field = 1000.

Method	Stepsize	Error
RK1 = Euler	.001	1.3 x 10^{-3}
RK2 = Heun	.002	1.8 x 10^{-6}
RK4	.004	8.0 x 10^{-15}

From this you can conclude that the constants (for this particular IVP)
are about

C1 ~ 1.3 C2 ~ 0.45 C4 ~ .00012

The moral is that for good accuracy Euler is essentially useless,
and RK4 will always win. There are still higher order methods,
but they involve more overhead as well and experience has shown
that RK4 is a good compromise.

Further note: The initial value problem $y' = f(x)$, $y(a) = y0$ has solution

$$y(x) = y0 + \int_a^x f(t)\, dt$$

When you apply these methods to approximate $y(x)$,

Euler's method is the left end-point Riemann sum
RK2 is the trapezoidal rule
RK4 is Simpson's rule.

Here's an example of how bad the errors can get: Catastrophic overshoot.

I selected Euler 1.00 again and followed the Euler polygon. The very fact
that the slope field is funnelling causes catastrophic overshoot.

ODE solvers are tricky and avoid things like this. One trick:
when the direction field is steep, use smaller stepsizes.

Please complete and hand in the Muddy Cards!

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18.03 Differential Equations

Spring 2010

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18.03 Class 3, Feb 8, 2010

First order linear equations; systems and signals perspective

[1] First order linear ODEs
[2] Bank Accounts; rate and cumulative total
[3] Systems and signals language
[4] RC circuits

[1] If I had to name the most important general class of differential equations it would be "linear equations." They will occupy most of this course. Today I'll show you how to model two real world systems by first order linear equations. Both of them involve systems evolving in time. The independent variable is time, t. If we write x or $x(t)$ for the dependent variable, we'll write x-dot for its time-derivative. In these ascii notes I'll continue to write x' the deriviative, though.

Definition: A "linear first order ODE" is one that can be put in the "standard form"

$$r(t)\, x'(t) + p(t)\, x(t) = q(t)$$

There is a general analytic method for solving this equation. I'll talk on Wednesday about that. For now note that if p, q, and r are *constant* then the equation is separable --

$$r\, dx / (q - px) = dt$$

Example 0: From recitation: Oryxes with constant growth rate, hunting allowed

$$x' = kx - h$$ h = harvest rate

as in Recitation 1. Notice that in the physical system both k and h can vary with time, and that the modeling process is fine with that.

The slide shows an Oryx.

[2] Bank account: I have a bank account. It has x dollars in it. x is a function of time. I can deposit money in the account and make withdrawals from it. The bank pays me rent for the money I deposit! This is called interest.

In the old days a bank would pay interest at the end of the month on the balance at the beginning of the month. We can model this mathematically:

With $\Delta t = 1/12$, the statement at the end of the month will read:

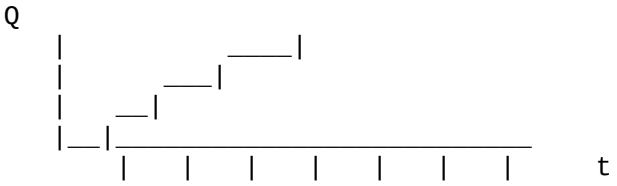
$$x(t + \Delta t) = x(t) + I\, x(t)\, \Delta t + [\text{deposits - withdrawals between } t \text{ and } t+\Delta t]$$

I has units (year)^{-1} . These days I is typically very small, say
1% = 0.01 . You don't get 1% each month! you get 1/12 of that.

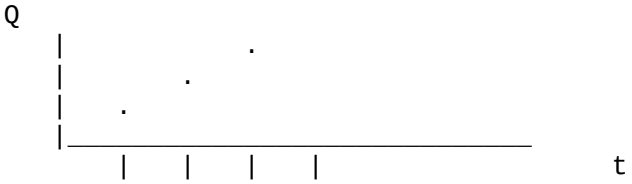
You can think of a withdrawal as a negative deposit, so I will call everything
a deposit.

Nowadays interest is usually computed daily. This is a step on the path
to the enlightenment afforded by calculus, in which Delta t ---> 0 .

In order to reach enlightenment, I want to record deposits minus withdrawals
as a *rate*, in dollars per year. Suppose I contribute $100 sometime every
month, and make no withdrawals. So my total deposits up to time t - my
"cumulative total" deposit - has a graph like this



In keeping with letting Delta t ---> 0 , we should imagine that I am making
this contribution continually, at the constant rate of $1200/year: graph like
this

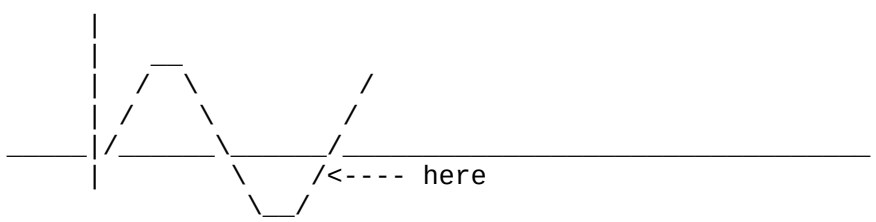


-- a straight line with slope 1/1200 : Q'(t) = q(t) is constant.

In general, say I deposit at the rate of q(t) dollars per year.
q(t) might vary over time, and might be negative too, from time to time,
because withdrawals are merely negative deposits.

Q3.1. At the indicated point of the graph of Q(t) am I making

(1) deposits
(2) withdrawals



Answer: when the slope is positive, not necessarily when Q is positive: (1).

So (assuming $q(t)$ is continuous)

$$x (t + \Delta t) \sim x(t) + I x(t) \Delta t + q(t) \Delta t$$

Now subtract $x(t)$ and divide by Δt :

$$\frac{x (t + \Delta t) - x(t)}{\Delta t} \sim I x + q$$

Now is the moment to let the interest period Δt tend to zero:

$$x' = I x + q$$

Note: $q(t)$ can certainly vary in time. The interest rate can too.
In fact the interest rate might depend upon x as well: a larger account will probably earn a better interest rate. Neither feature affects the derivation of this equation, but if I does depend upon x as well as t , then the equation we are looking at is no longer linear.
So let's say $I = I(t)$, $q = q(t)$.

We can put the ODE into standard form:

$$x' - I x = q$$

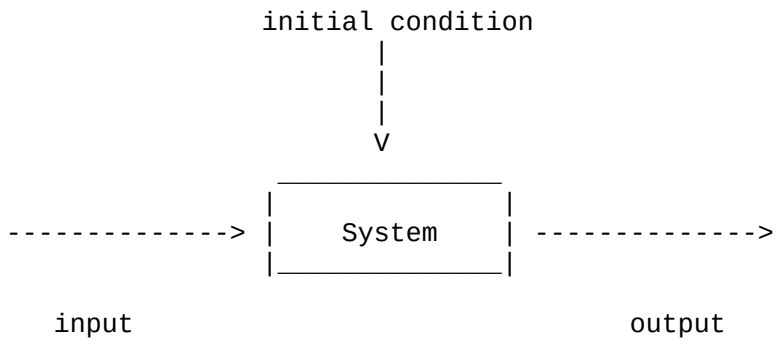
Each symbol represents a function of t .

[3] Systems and signals language

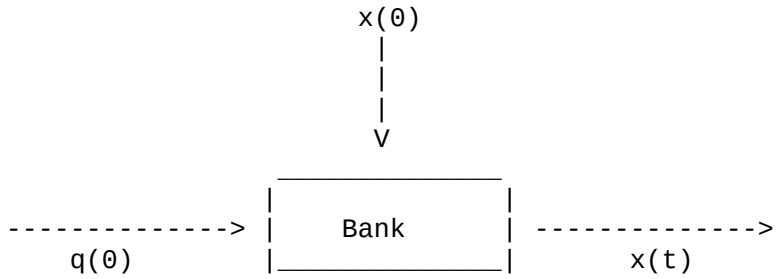
In the standard form, the left hand side represents the SYSTEM: the bank.

The right hand side represents an outside influence on the system:
it's a SIGNAL, the "input signal." A "signal" is just a function of time.
Here it is my rate of savings.

The system responds to the input signal and yields the function $x(t)$,
the "output signal" or SYSTEM RESPONSE. Here's a picture:



For example,



We will develop a theory of linear equations, complete with an algorithm for solving them. It's important to recognize them when you see them.

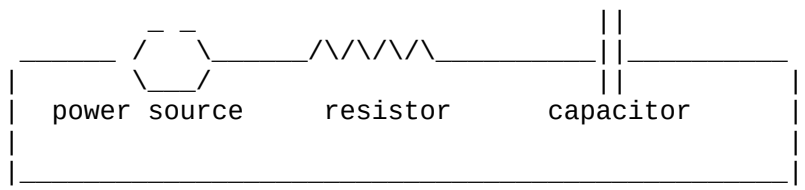
Question 3.2. Which of the following are linear ODE's?
(a) $\dot x+x^2=t$
(b) $\dot x=(t^2+1)(x-1)$
(c) $\dot x+x=t^2$

1. None
2. (a) only
3. (b) only
4. (c) only
5. All
6. All but (a)
7. All but (b)
8. All but (c)

Blank. Don't know.

Answer: (b) and (c) are linear, (a) is not: 6

[4] RC circuits. Suppose we have an electrical circuit like this:



It has a resistor, a capacitor, and a voltage source: it's an RC circuit. This is not a course in electromagnetism or in circuits, but I will use words from that subject. Let's pretend we understand what they mean!

Current flows around the circuit. (Confusingly, if the current flows to the right, the actual electrons flow to the left, because they are negatively charged. It's confusing. Sorry, I didn't invent this.) The current is measured in "amperes" and is denoted by I. (I don't know what language has a word for current starts with an I!) In this "series" circuit, the current is the

same everywhere but it may vary with time.

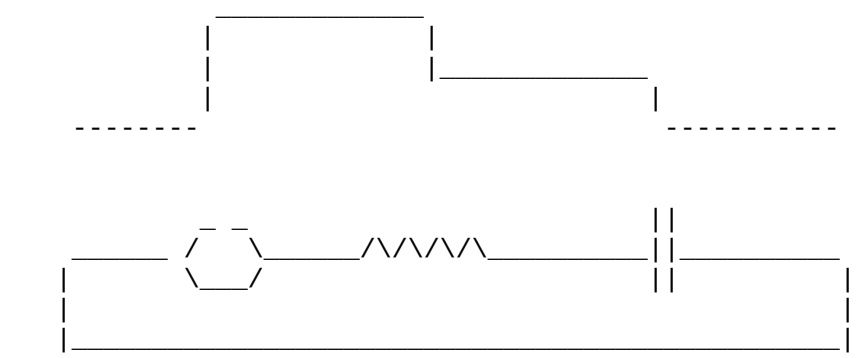
Let's say the positive direction in the circuit is clockwise (ie to the right over the top, for digital clock users). So if current is flowing counterclockwise along the wire, an ammeter would give a negative reading.

The system is powered by a variable power source, which creates a "voltage increase" across it. This what makes current move. Write V(t) for the voltage INCREASE from the bottom to the top of the source. Write V_R and V_C for the voltage DROPs across resistor and capacitor.

"Kirchhoff's voltage law" states that

$$V(t) = V_R(t) + V_C(t)$$

I drew a little graph like this:



to illustrate the KVL.

There is a relationship between the voltage drop across each circuit element and the current flowing through it. The relationship is different for resistors and capacitors:

Resistor: $V_R(t) = R\, I(t)$ for a constant R, the "resistance"

Capacitor: $V'_C(t) = (1/C)\, I(t)$ for a constant C, the "capacitance"

So:

- The voltage drop across the resistor is proportional to the current flowing through it. High resistance means big voltage drop.

- The voltage drop across the capacitor is proportional to the *integral* of the current; it results from a buildup of charge on the two plates of the capacitor. High capacitance means lots of space for the charge. A very large capacitor is like no capacitor at all.

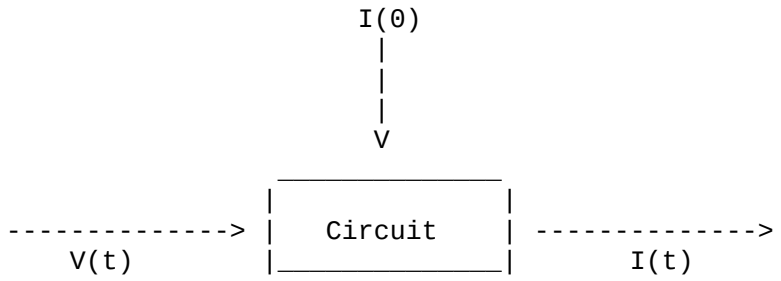
To relate these, differentiate KVL:

$$V'(t) = V'_R(t) + V'_C(t) = R\, I'(t) + (1/C)\, I(t)$$

This is a first order linear differential equation for $I(t)$:
In standard form:

$$R\, I'(t) + (1/C)\, I(t) = V'(t)$$

The circuit is the system and it is represented by the left hand side.
The input signal is V , the voltage increase across the power source.



The *derivative* of the input signal is what occurs on the right of the equation.

The system response is the current.

The system is the circuit.

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18.03 Differential Equations

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18.03 Class 4, Feb 10, 2010

First order linear equations: integrating factors

[1] First order homogeneous linear equations
[2] Newtonian cooling
[3] Integrating factor (IF)
[4] Particular solution, transient, initial condition
[5] General formula for IF

Definition: A "linear ODE" is one that can be put in the "standard form"

$r(t)x' + p(t)x = q(t)$	$x = x(t)$

r(t), p(t) are the "coefficients" [I may have called q(t) also a coefficient also on Monday; this is not correct, fix it if I did.]

The left hand side represents the "system," and the right hand side arises from an "input signal." A solution x(t) is a "system response" or "output signal."

We can always divide through by r(t), to get an equation of the Reduced standard form:

$x' + p(t)x = q(t)$	$x = x(t)$ (*)

The equation is "homogEneous" if q is the "null signal," q(t) = 0 .
This corresponds to letting the system evolve in isolation:
In the bank example, no deposits and no withdrawals.
In the RC example, the power source is not providing any voltage increase.

The homogeneous linear equation

$$x' + p(t)\, x = 0 \qquad (*)_h$$

is separable. Here's the solution,
in general on the left, with an example (with p(t) = 2t) on the right:

$x' + p(t)x = 0$	$x' + 2tx = 0$
Separate: $dx/x = - p(t)\, dt$	$dx/x = - 2t\, dt$
Integrate: $\ln\|x\| = - \int p(t)\, dt + c$	$\ln\|x\| = - t^2 + c$
Exponentiate: $\|x\| = e^c\, e^{ - \int p(t)\, dt }$	$\|x\| = e^c\, e^{-t^2}$

Eliminate the absolute value and reintroduce the lost solution:

$$x = C\, e^{-\int p(t)\,dt} \qquad\qquad x = C\, e^{-t^2}$$

In the example, we chose a particular anti-derivative of k, namely kt. That is what I really have in mind to do in general. The constant of integration is taken care of by the constant C.

So the general solution to $(*)_h$ has the form $C\,x_h$, where x_h is *any* nonzero solution:

$$x_h = e^{-\int p(t)\,dt} \quad , \quad x = C\,x_h$$

We will see that the general case can be solved by an algebraic trick that produces a sequence of two integrations.

[2] Example: Diffusion, e.g. of heat.

About this time of year I start to think about summer. I put my rootbeer in a cooler but it still gets warm. Let's model its temperature by an ODE.

$x(t)$ = root beer temperature at time t.

The greater the temperature difference between inside and outside, the faster $x(t)$ changes.

Simplest ("linear") model of this:

$$x'(t) = k\,(\,T_{ext}(t) - x(t)\,)$$

where $T_{ext}(t)$ is the "external" temperature. Sanity check: When $T_{ext}(t) > T(t)$, $x'(t) > 0$ (assuming $k > 0$). We get a linear equation:

$$x' - k\,x = k\,T_{ext}$$

This is "Newton's law of cooling." k could depend upon t and we would still have a linear equation, but let's suppose that we are not watching the process for so long that the insulation of the cooler starts to break down!

Systems and signals analysis:
The system is the cooler.
The output signal = system response is $x(t)$, the temperature in the cooler.
The input signal is the external temperature $T_{ext}(t)$.

Note that the right-hand side is k times the input signal, not the input signal itself.

What constitutes the input and output signals is a matter of the interpretation of the equation, not of the equation itself.

Question 4.1: k large means
1. good insulation
2. bad insulation
Blank. don't know.

k is small when the insulation is good, large when it is bad.
It's zero when the insulation is perfect. k is a COUPLING CONSTANT
When it is zero, the temperature inside the cooler is decoupled from
the temperature outside. In the construction industry, a number like
k is pasted on windows; it's called the U-value of the window.

Let's take k = 1/3 , for example.

Suppose the temperature outside is rising at a constant rate: say

$$T_{ext} = 60 + 6t \quad \text{(in hours after 10:00)}$$

and we need an initial condition: let's say x(0) = 32 .

So the IVP is $x' + (1/3)\,x = 20 + 2t$, $x(0) = 32$. (cooler)

This isn't separable: it's something new. We'll describe a method which
works for ANY first order linear ODE.

[3] Method: Integrating factors (Euler)

This method is based on the product rule for differentiation:

$$(d/dt)(ux) = ux' + u'x$$

For example, suppose we have the equation

$$t\,x' + 2\,x = t$$

(This is not separable; it is linear and in standard form, but not reduced
standard form.) Here's a *trick*. Multiply both sides by t :

$$t^2\,x' + 2t\,x = t^2$$

The left hand side is now the derivative of a product:

$$(d/dt)(t^2\,x) = t^2$$

We can solve this by integrating:

$$t^2\,x = t^3/3 + c$$

so $x = t/3 + c\,t^{-2}$

[In the first lecture, I posed this (with a different righthand side)
as a flashcard problem, but I did it just after describing the calculation
of an integrating factor for a *reduced* equation. The reduced equation is
x' + 2x/t = 1 , and this has integrating factor t^2 . So it was a poorly
placed question.]

That was great! The factor t we multiplied by is an "integrating factor."
I guessed it here. Often you can. The factor to use in the cooler equation
and other equations may not be so obvious. Here's a calculation, for a linear
equation in reduced form,

$$x' + p(t)x = q(t)$$

Multiply both sides by u

$$u\,x' + p\,u\,x = u\,q$$

In order for the right hand side to be $(d/dt)(ux) = ux' + u'x$,
the function u must satisfy the differential equation

$$u' = p\,u$$

This is separable, and we'll carry out the separation in general in a minute.
In the cooler equation, the coefficient $p(t)$ is constant.
In that case we have the natural growth equation!

$$u = e^{pt}$$

(I am choosing a value for the constant of integration, because I need just one integrating factor, any one.)

In the case of the cooler problem, $p = 1/3$, so we have:

$$(d/dt)\,(e^{t/3}\,x) = (20 + 2t)\,e^{t/3}$$

Integrate:

$$e^{t/3}\,x = 60\,e^{t/3} + \int 2t\,e^{t/3}\,dt$$

Um. Parts: $\int u\,dv = uv - \int v\,du$

$$u = 2t\ ,\quad dv = e^{t/3}\,dt$$
$$du = 2dt\ ,\quad v = 3\,e^{t/3}$$

$$e^{t/3}\,x = 60\,e^{t/3} + 6\,t\,e^{t/3} - 18\,e^{t/3} + c$$

$$= (\,42 + 6\,t\,)\,e^{t/3} + c$$

Solve for x:

$$x = (\,42 + 6t\,) + c\,e^{-t/3}$$

That's the general solution. Remember, you can check it easily.

u is an "integrating factor."

[4] We still should finish the IVP process:

$$32 = x(0) = 42 + c \quad \text{so} \quad c = -10 :$$

$$x = 42 + 6t - 10\,e^{-t/3}$$

We just want one u , not the general u : so the exponent could be any antiderivative of p . In the example, $p = 1/3$ was constant and we took $u = e^{t/3}$.

Note the structure of the genearal solution:

$$x = x_p + c\, u^{-1}$$ where

. x_p is a solution, *any solution*. It's called a PARTICULAR SOLUTION but this is a very poor name, because there is nothing particular about it. I this case we chose one with a pretty simple formula -- $x_p = 42 + 6t$.)

. u is an integrating factor.

Very often x_h approaches zero with time, as this one does. It is then called a TRANSIENT. All solutions come to look more and more alike as time goes on. This is a funnel!

I graphed the solutions $42 + 6t$ and x , and some others along with T_{ext} . If the temperature in the cooler is more than 60 degrees at the start, then it declines at first, crosses the nullcline $x = 60 + 6t$ where it is momentarily in equilibrium with the outside, and then rises to become asymptotic to $42 + 6t$ like every other solution.

[5] Let's compute an integrating factor for the general first order linear equation (*) : we are to solve $u' = up$.

This is a separable equation: $du/u = p\, dt$

$$\ln|u| = \int p\, dt$$

The constant of integration is in the indefinite integral.

$$|u| = e^{\int p\, dt}$$

Now there is a choice of sign. Pick one and go with it; say

$$u = e^{\int p\, dt}$$

That gives you an integrating factor. Any nonzero multiple serves as well.

Note that this is the reciprocal of a solution to the homogeneous equation:

$$u = x_h^{-1}$$

This gets fed into the solution for x :

$$x = u^{-1} \int u\, q\, dt$$

and the constant of integration in the integral lets us write

$$x = x_p + c\, x_h$$

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18.03 Differential Equations

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18.03 Class 5, Feb 15, 2008

Complex Numbers, roots of unity

[1] Complex algebra
[2] Complex conjugation
[3] Polar multiplication

Complex numbers provide a tool for expressing aspects of the real world. Anything with an amplitude and a phase is secretly a complex number, and it is worthwhile bringing that secret identity out in the open.

[1] Complex Algebra

We think of the real numbers as filling out a line. The complex numbers fill out a plane. The point up one unit from 0 is written i (but written j by many engineers). Addition and multiplication by real numbers is as vectors. The new thing is $i^2 = -1$. The usual rules of algebra apply. For example FOIL:

$$(1 + i)(1 + 2i) = 1 + 2i + i - 2 = -1 + 3i.$$

Every complex number can be written as $a + bi$ with a and b real.

$a = Re(a+bi)$ the real part
$b = Im(a+bi)$ the imaginary part: NB this is a real number.

Maybe complex numbers seem obscure because you are used to imagining numbers by giving them units: 5 cars, or -3 degrees. Complex numbers do not accept units. Also, there is no ordering on complex numbers, no "<."

Question 1. Multiplication by i has the following effect on a complex number.

1. It rotates the number around the origin by 90 degrees counterclockwise.

2. It rotates the number around the origin by 90 degrees clockwise.

3. It takes a number to the number pointing in the opposite direction with the same distance from the origin.

4. It reflects the number across the imaginary axis.

5. It reflects the number across the real axis.

6. None of the above.

Blank: Don't know.

Compute: $i(a + bi) = -b + ai$,
which is rotated by 90 degrees counterclockwise: Ans: 1.

[2] Complex conjugation

Notice the formula

$$(a+bi)(a-bi) = a^2 - (bi)^2 = a^2 + b^2 \qquad (*)$$

(which is the square of the distance to the origin).

This lets us divide by a complex number: For example

$$i/(1+2i) = (i/(1+2i))\ ((1-2i)/(1-2i))$$

$$= (i+2)/(1+4) = (2+i)/5 .$$

(*) suggests that we give a special name to the quantity $a-bi$:
it is the "complex conjugate" of $a+bi$ and is denoted by an overline:

$$\overline{a+bi} = a-bi :$$ reflect across the real axis.

Conjugation satisfies $\overline{w+z} = \overline{w} + \overline{z}$, $\overline{wz} = \overline{w}.\overline{z}$, $\overline{\overline{z}} = z$

Proofs: $w = a+bi$, $z = c+di$.

$$\overline{w} + \overline{z} = (a-bi) + (c-di) = (a+c) - (b+d)i = \overline{w+z}$$

$$\overline{w}.\overline{z} = (a-bi)(c-di) = (ac-bd) - (ad+bc)i$$

$$w.z = (ac-bd) + (ad+bc)i$$

which conjugates to the line above.

Question 2: If $\overline{z} = -z$, then

1. z is purely imaginary

2. z is real

3. z lies on the unit circle

4. z = 0

5. None of the above

Blank. Don't know

Ans: 1.

[3] Polar multiplication

Being points in the plane, complex numbers have polar descriptions.
The distance of z from zero is

$|z|$ = "absolute value" = "modulus" = "magnitude" of z.

$\overline{}$

The angle up from the positive real axis is

Arg(z) = "argument" = "angle" of z. As usual, it's only well defined up to adding multiples of 2π.

For example $Arg(1) = 0$, $Arg(i) = \pi/2$, $Arg(-1) = \pi$.

$|z|^2 = z\bar{z}$ and $\tan(Arg(z)) = b/a = Im(z)/Re(z)$.

- Magnitudes Multiply : $|wz| = |w||z|$.

proof: It's not pleasant to compute absolute values - they involve square roots - but it's easy to compute squares:

$$|wz|^2 = (\overline{wz})(wz) = \bar{w}\bar{z}wz = \bar{w}w\bar{z}z = |w|^2|z|^2 = (|w||z|)^2$$

It follows that $|wz| = |w||z|$ since both sides are positive.

For example, the product of complex numbers on the unit circle lies again on the unit circle. I drew a picture.

- Angles Add: $Arg(wz) = Arg(w) + Arg(z)$

I'll check this in case w and z are both on the unit circle. Then:

$(\cos a + i \sin a)(\cos b + i \sin b) =$

$((\cos a)(\cos b) - (\sin a)(\sin b)) + i((\cos a)(\sin b) + (\sin a)(\cos b))$

$= \cos(a+b) + i \sin(a+b)$

using the angle addition formulas for cos and sin .

In fact multiplication of complex numbers contains in it the angle addition formulas for sin and cos , and if you understand complex numbers you'll never have to memorize those formulas again.

This checks with our question about multiplication by i above.
Here's another question about multiplication:

Question 3. $(1+i)^4 =$

1. - 1

2. 4

3. - 4

4. - sqrt(2)

5. 4 i

6. None of the above

Blank: Don't know

	Modulus	Argument	a+bi
(1+i)^0	1	0	1
(1+i)^1	sqrt(2)	pi/4	1+i
(1+i)^2	2	pi/2	2i
(1+i)^3	2 sqrt(2)	3pi/4	-2+2i
(1+i)^4	4	pio	-4

These powers of (1+i) spiral away from the origin.

The powers all lie on a spiral emanating from the origin. Ans: (3) .
Thus 1 + i is a fourth root of -4.

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18.03 Class 6, Feb 16, 2010

Complex exponential

[1] Complex roots
[2] Complex exponential, Euler's formula
[3] Euler's formula continued

[1] More practice with complex numbers:

Magnitudes Multiply: $|zw| = |z||w|$
Arguments Add: $\text{Arg}(zw) = \text{Arg}(z) + \text{Arg}(w)$

so: $|z^n| = |z|^n$ and $\text{Arg}(z^n) = n\,\text{Arg}(z)$

For example:

Let's take some powers of $z = 1+i$.

$\|z\| = \sqrt{2}$	$\text{Arg}(z) = \pi/4$
$\|z^2\| = 2$	$\text{Arg}(z^2) = \pi/2$
$\|z^3\| = 2\sqrt{2}$	$\text{Arg}(z^3) = 3\pi/4$
$\|z^4\| = 4$	$\text{Arg}(z^4) = \pi$

Notice that these numbers march out along a spiral. This continues for all powers of $1+i$, even negative ones.

How about fractional powers? i.e. roots: of unity, first.

What are the cube roots of 1? Well 1 is one. If z is one, then $|z|^3 = |z^3| = 1$, so it lies on the unit circle.

The argument has to be so that 3 times it is zero - or 2π, or 4π, or ....
Thus the argument is
0 , giving 1 , or
$2\pi/3$, giving $(-1+\sqrt{3}i)/2$ or
$4\pi/3$, giving $(-1-\sqrt{3}i)/2$.

The nth roots of unity divide the unit circle into n equal pieces.

How about cube roots of -8 . If $z^3 = -8$, then $|z|^3 = |z^3| = 8$.
$|z|$ is a positive real number so $|z| = 2$: all cube roots of -8 lie on the circle with center 0 and radius 2.

The argument of -8 is π , giving a cube root with argument $\pi/3$
or 3π giving a cube root with argument π
or 5π giving a cube root with argument $5\pi/3$

These again divide the circle evenly. The second is -2, and the others are $1+\sqrt{3}i$ and $1-\sqrt{3}i$.

I gave a little demo with the Complex Roots applet.

We have been studying roots of equations like $z^n - a = 0$.

A "polynomial" is a function of the form

f(z) = an z^n + a{n-1} z^{n-1} + ... + a1 z + a0 , an not 0

The "degree" is the top nonzero coefficient ak .

Fundamental theorem of algebra (Gauss,...): Any polynomial of degree > 0 has a complex root.

In fact any polynomial factors as

f(z) = an (z - r1) (z - r2) ... (z - rn)

It has n complex roots (some of which may coincide).

[2] Complex exponential

The complex valued function z(t) = a(t) + i b(t) parametrizes a directed curve in the plane. For example

z = t + i parametrizes a line running horizontally through i , directed towards the right.

The derivative is computed for each component, and gives you the velocity vector. Here this is 1 : horizontal.

Here's an ODE we can try to solve: z' = iz , z(0) = 1. (**)

In lecture 1 we saw that e^{kt} is the solution to x' = kx, x(0) = 1. So we will write the solution to (**) as z = e^{it} .

On the other hand we found out that multiplication by i is rotation by 90 degrees; so the solution is a curve such that the velocity vector is always perpendicular to the radius vector. This is a circle, and if we add the initial condition it is the unit circle, traversed counterclockwise with speed 1:

z = cos t + i sin t

To check, compute

z' = - sin t + i cos t

iz = i cos t - sin t

and they agree. Thus:

e^{it} = cos t + i sin t. "Euler's formula."

This function parametrizes the unit circle, directed counterclockwise. This is the *definition* of e^{it} .

In fact, for any complex number a+bi you can compute that the solution to z' = (a+bi)z , z(0) = 1 , is

e^{(a+bi)t} = e^{at} (cos(bt) + i sin(bt))

That is, the magnitude of $e^{(a+bi)t}$ is e^{at}
and the argument of $e^{(a+bi)t}$ is bt

When $a > 0$ and $b > 0$, we have a spiral moving counterclockwise away from the origin.

Q6.1 The curve in the complex plane traced out by $e^{(1+2\pi i)t}$ most closely resembles which of the following?

(1) A straight ray along the positive real axis
(2) A circle of radius e and center at the origin.
(3) A circle of radius 1 and center at the origin.
(4) A spiral moving inwards and counterclockwise
(5) A spiral moving outwards and counterclockwise
(6) A spiral moving inwards and clockwise
(7) A spiral moving outwards and clockwise

(Blank) Don't know.

Ans: (5) .

[3] This definition (*) satisfies the expected exponential rule:

$$e^{(z+w)t} = e^{wt}\, e^{zt}$$

You can see this using the usual rule for real exponentials together with the angle addition formulas, or by using the uniqueness theorem for solutions to ODEs. See the Supplementary Notes.

General fact about complex numbers:

$$\frac{z + \bar{z}}{2} = \mathrm{Re}(z) \qquad \frac{z - \bar{z}}{2i} = \mathrm{Im}(z)$$

Proof by diagram.

Apply this to $z = e^{it}$. I will need to know what $\overline{e^{it}}$ is. Since complex conjugation preserves modulus and reverses angle,

$$\overline{e^{it}} = e^{-it} .$$

From Euler's formula (**) and the "general fact" we find

$$\cos(t) = \frac{e^{it} + e^{-it}}{2} \qquad \sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

Sometimes these are also called Euler's Formulas.

Anything you want to know about sines and cosines can be obtained from properties of the (complex) exponential function.

```
Neat examples:  e^{pi i}    = -1
                e^{2k pi i} = 1    for any whole number  k
               e^{pi i / 2} = i
           e^{(k/n)2 pi i}  = nth roots of unity.
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18.03 Class 7, Feb 17, 2010

Exponential and Sinusoidal input and output

[1] Sinusoidal functions
[2] Trig sum formula
[3] Integration of complex valued functions
[4] Linear equations with sinusoidal input signal
[5] Complex replacement

Euler: Re $e^{(a+bi)t} = e^{at} \cos(bt)$
Im $e^{(a+bi)t} = e^{at} \sin(bt)$

[1] Sinusoids

A "sinusoidal function" $f(t)$ is one whose graph is a (co)sine wave.

I drew a large general sinusoidal function, $f(t)$.

I drew the graph of $\cos(\theta)$; this is our model example of a sinusoid.

A sinusoidal function is entirely determined by just three measurements, or parameters, which determine it in terms of $\cos(\theta)$.

A = Amplitude = height of its maxima = depth of its minima

P = Period = elapsed time till it repeats
(or, in spatial terms, λ = wavelength = the distance between repeats)

t_0 = Time lag = time of first maximum

$f(t)$ can be written in terms of cosine. Clearly, $f(t) = A \cos(\theta)$.
To work out how, express θ as a function of t . I started drew a t-axis horizontally and a θ-axis vertically.
When $t = t_0$, $\theta = 0$. When $t = t_0 + P$, $\theta = 2\pi$.
I marked these data points on all three graphs.

The graph of θ as a function of t is a straight line; otherwise the cosine would get distorted, bunched up. So:

$$\theta = (2\pi/P) (t - t_0)$$

and so $f(t) = A \cos((2\pi/P) (t - t_0))$

The frequency is $\nu = 1/P$, measured in "cycles/unit time." More useful is: the circular frequency $\omega = 2\pi/P = \omega$, measured in radians/unit time.

So $\theta = \omega (t - t_0)$

The "phase lag" is

$$\phi = \omega t_0 = (2 \pi/P) t_0 .$$

It measures the radian measure corresponding to $t = 0$.
In terms of ω and ϕ ,

f(t) = A cos(omega t - phi)

For example,

sin(omega t) = cos(omega t - pi/2) .

-- the sine lags one quarter cycle behind the cosine.

Question 7.1. A graph of a sinusoidal function is displayed.
The problem is to express it in the "standard form" above.

1. 2 cos(4pi t + pi/4)
2. 2 cos((pi/4)t + pi/4)
3. 2 cos(4pi t - pi/4)
4. 2 cos((pi/4)t - pi/4)
5. 2 cos(4t+1)
6. 2 cos(4t-1)

P = 8, t_0 = -1 , A = 2 :

f(t) = 2 cos((2pi/8) (t + 1)) = 2 cos((pi/4) t + pi/4)

omega = pi/4 , phi = - pi/4 .

Ans: 2.

[2] Trig sum: a cos(omega t) + b sin(omega t)

I showed the Trig Id applet. This sum seems always to be another sinusoidal function! How can we find its "standard form" A cos(omega t - phi) ?

Recall the cosine difference identity:

cos(theta - phi) = cos(phi) cos(theta) + sin(phi) sin(theta)

cos(omega t - phi) = cos(phi) cos(omega t) + sin(phi) sin(omega t)

Now construct a right triangle with hypotenuse the segment in the plane joining (0,0) to (a,b) . If A is the hypotenuse and phi the angle at the origin, then

A = sqrt(a^2 + b^2)
a = A cos(phi)
b = A sin(phi)

and

A cos(omega t - phi) = a cos(omega t) + b sin(omega t)

Question 7.2. What are the amplitude, circular frequency, and phase lag:
A , omega , and phi in A cos(omega t - phi) , for the sinusoid

cos(omega t) + sqrt(3) sin(omega t)

1. 2 cos(\omega t-\frac{\pi}{4})
2. sqrt(3) cos(omega (t - pi/3))
3. 2 cos(omega(t- pi/3))
4. 2 cos(omega t - pi/3)
5. sqrt(3) cos(omega t - pi/3)
6. sqrt(3) cos(omega t - pi/4)

Blank. Don't know.

Ans: A = 2, phi = pi/3 : 4.

[3] Integration

Remember how to integrate $e^{2t} \cos(t)$?

Use parts twice. Or:

Differentiating a complex valued function is done separately on the real and imaginary parts. Same for integrating.

$$e^{2t} \cos(t) = \text{Re } e^{(2+i)t} \qquad \text{so}$$

$$\int e^{2t} \cos(t)\, dt = \text{Re} \int e^{(2+i)t}\, dt$$

and we can integrate exponentials because we know how to differentiate them! -

$$\int e^{(2+i)t}\, dt = (1/(2+i))\, e^{(2+i)t} + c$$

We need the real part.

Expand everything out: $1/(2+i) = (2-i)/5$

$$e^{(2+i)t} = e^{2t} (\cos(t) + i \sin(t))$$

so the real part of the product is

$$(1/5)\, e^{2t} (2 \cos(t) + \sin(t)) + c$$

More direct than the high school method!

[4] Linear constant coefficient ODEs with exponential input signal

Let's try $x' + 2x = 4 e^{3t}$

We could use our integrating factor, but instead
let's use the method of "optimism," or the inspired guess. The inspiration here is based on the fact that differentiation reproduces exponentials:

$$\frac{d}{dt} e^{rt} = r e^{rt}$$

Since the right hand side is an exponential, maybe the output signal x will be too: TRY $x = A e^{3t}$. This is not going to be the general

solution, so I'll write x_p for it. I don't know what A is yet, but:

$$2\,x_p = 2\,A\,e^{3t}$$

$$x_p' = A\,3\,e^{3t}$$

$$4\,e^{3t} = A\,(3+2)\,e^{3t}$$

which is OK as long as $A = 4/5$: $x_p = (4/5)\,e^{3t}$ is one solution. The general solution is this plus a transient:

$$x = (4/5)\,e^{3t} + c\,e^{-2t}\ .$$

[6] Replacing sinusoidal signals with exponential ones

Let'e go back to the original ODE

$$x' + 2x = 2\cos(t)$$

This equation is the real part of a complex valued ODE:

$$z' + 2z = 2\,e^{it}$$

This is a different ODE, and I use a different variable name, $z(t)$.

We just saw how to get an exponential solution: $z_p = A\,e^{it}$

$$2\,zp = 2\,A\,e^{it}$$

$$zp' = i\,A\,e^{it}$$

$$2\,e^{it} = A\,(2+i)\,e^{it}$$

so $z_p = 2/(i+2)\,e^{it}$

To get a solution to the original equation we should take the real part of this! Expand each factor in real and imaginary parts:

$$z_p = (2(2-i)/5)\,(\cos(t) + i\sin(t))$$

$$x_p = \mathrm{Re}(z_p) = (4/5)\cos(t) + (2/5)\sin(t)$$

This is the only sinusoidal solution. To get the general solution we add a transient:

$$x = x_p + c\,x_h$$

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18.03 Differential Equations

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18.03 Class 8, Feb 19, 2010

Autonomous equations

[1] Logistic equation
[2] Phase line
[3] Extrema, points of inflection

Announcements:

Final Tuesday, May 18, 9:00 - 12:00, Johnson Track

Hour exam next Wednesday: Rooms to be announced Monday.
Please go to the hour you are registered for. 50 minutes.

[1] Back to qualitative study of differential equations.

I'll use (t,y) today. The general first order equation is

$$y'(t) = F(t,y)$$

Autonomous ODE: $y'(t) = g(y)$.

Autonomous means conditions are constant in time, though they may depend on the current value of y .

Eg [Natural growth/decay] Constant growth rate: so $y' = k0\ y$.

Eg Bank account with interest rate NOT depending on time but possibly depending upon current balance, and constant savings rate: $y' = I(y)\ y + q$.

Extended example: Population model with variable growth rate $k(y)$ depending on the current population but NOT ON TIME; so $y' = k(y)\ y$.

Suppose that when y is small the growth rate is approximately $k0$, but that there is a maximal sustainable population p , and as y gets near to p the growth rate decreases to zero. When $y > p$, the growth rate becomes negative; the population declines back to the maximal sustainable population.

In the simplest version of this, when you graph of $k(y)$ against y you get a straight line with vertical intercept $k0$ and horizontal intercept p :

$$k(y) = k0\ (1 - (y/p)).$$

so $k(0) = k0$, and $k(p) = 0$. When $y > p$, $k(y) < 0$.

The Logistic Equation is $y' = k0\ (1 - (y/p))\ y = g(y)$.

This is more realistic than Natural Growth when you want to account for limits to growth. It is nonlinear but it IS autonomoous.

Autonomous equations are always separable, but we aim for a qualitative grasp of solutions. I sketched the nullcline: $y = 0$ and $y = p$.
Clearly all the isoclines will be collections of horizontal straight lines - the differential equation is constant in time.

To get a clear idea of the other isoclines, I will draw a graph of $g(y)$ as a function of y . It's a parabola opening downward, meeting the horizontal axis at $y = 0$ and $y = p$.

This says that for $y < 0$ the slopes are negative
for $0 < y < p$ the slopes are positive
for $y > p$ the slopes are positive.

I drew more isoclines and some solution curves. The bounded solutions are "logistic curves" or "S-curves." The represent the population drifting from just above one equilibrium (no oryx) towards the stable population (p oryx).

If the population exceeds the maximal stable population, it falls back towards it. The max stable population is a "stable equilibrium;" the zero population is "unstable."

Values of y for which $g(y) = 0$ are called "critical points" of the differential equation $y' = g(y)$. The are also called "equilibria."

[2] Phase line

Since the direction field is constant horizontally, its essential content can be conveyed more efficiently. Draw a vertical line. Mark on it the equilibria, where $g(y) = 0$. In between them, draw an upward pointing arrow if $g(y) > 0$ and a downward pointing arrow if $g(y) < 0$. This simple diagram tells you roughly how the system behaves. It's called the "phase line."

Question 8.1. In the autonomous equation $y' = g(y)$, where $g(y)$ has a graph which I sketch, looking like $g(y) = y^3 - y$, is the rightmost critical point

1. Stable
2. Unstable
3. Can't tell, could be either

Blank: don't know

This can be made clear by sketching the phase line. Ans: Unstable.

In terms of the graph of $g(y)$, the stable equilibria occur when $g'(p) < 0$, unstable when $g'(p) > 0$.

Now, the Kenyan government wants to establish a game preserve on which it will allow the hunting of oryx. It wants to sell licenses to kill 250 oryx each year and wants to know how big a preserve to establish to guarantee that this is sustainable. It's known that the population (measured in kilo-oryx, without hunting) follows the logistic equation

$$y' = (a-y)y$$

Here a is the stable population of oryx, in the absence of hunting. It is proportional to the area of the preserve. With our 1/4 kilo-oryx per year hunt, then, we have

$$y' = g(y) = (a-y)y - 1/4 = -(y^2 - ay + 1/4)$$

I invoked the Mathlet <Phase Lines> to visualize what happens.

For small a , the population crashes.

When a is increased to a critical size, a single equilibrium appears, which is "semistable." You can watch the graph of g(y) at the lower left of the applet; it's a parabola, opening downward, and it rises as a increases till at a = 1 it has a single root.

We can solve for where the critical points are, using the quadratic formula:

$$y = a/2 +- (1/2)(\sqrt{a^2 - 1})$$

There is just one exactly when the square root is zero, ie a = 1 (or a = -1).

So the Kenyan Chamber of Commerce recommends a preserve of this size, a = 1.

Is this wise? if the population falls every so slightly below the critical population (which seems to be y = 1/2), it crashes.

When you increase a further, things get better. There are two critical points. The farther they are from each other, the more stable the population system is.

I returned a to zero and opened the Bifurcation Diagram. and watched what happened as a increased. The bifurcation diagram puts all the phase lines, for various a , together.

[3] In an autonomous equation y' = g(y) , the conditions represented by the ODE are constant in time. Direction fields are constant in the horizontal direction. Consequently, any horizontal (time) translate of a solution is another solution. A "time translate" of a function y(t) is a function y(t-t_0): the graph gets shifted horizontally (to the right) by t_0 units.

Ex. Solutions of y' = k_0 y exhibit three different behaviors, illustrated by

$$y = e^{k_0 t} , \quad y = 0 , \quad y = - e^{k_0 t}$$

Any solution is a horizontal translate of one of them: any solution is either

$$y = e^{k_0 (t - t_0)} ,$$

y = 0 (whose only time-translate is itself) , or

$$y = - e^{{k_0 (t - t_0)}}$$

Question 8.2. Solutions of autonomous equations can have a

strict local maximum.

(A strict local maximum for $f(t)$ is a time $t = a$ such that $f(a) > f(t)$ for all t near but not equal to a .)

1. True
2. False

Extreme points occur where $y' = 0$, i.e. where $g(y) = 0$. These are the constant solutions, and they don't have strict maxima or minima. So it's true, solutions can't have strict extrema.

Question 8.3. Nonconstant solutions of the autonomous ODE $y' = g(y)$ have inflection points at y for which:

1. $g(y)=0$
2. $g'(y)=0$
3. $g''(y)=0$

$y'' = g'(y)\, y'$ by the chain rule. So if $y'' = 0$ at $y = c$ then either $g'(c) = 0$ (constant solutions) or $g'(c) = 0$ there. So (2) is the case. You can see it on the S curve.

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18.03 Class 9, Feb 22, 2010

Linear vs Nonlinear: a debate

[1] Linearization near equilibrium
[2] Exponential Response Formula (first order)
[3] Potential blow-up of solutions to a nonlinear equation

Hour Exam I Wednesday. Locations for both lectures:

10-250 A -- G
Walker H -- Z

Office hours as usual on Wednesday

Review: Nonlinear vs Linear

$y'(t) = F(t,y(t))$ vs $r(t)\,x'(t) + p(t)\,x(t) = q(t)$

This is in the form of a debate, between Linn E. R. (on the right) and Chao S. (on the left). I'll take a vote at the end.

Linn: I'd like to begin by making the point that there is a solution procedure for linear equations, which reduces solution of any linear equation to integration: Multiply the equation through by a factor so that the two terms become the two terms in $(d/dx)(ux)$. Then integrate.

Sometimes you can just *see* this: $t^2 x' + 2t x = (d/dt)(t^2 x)$ for example.

If we are in *reduced* standard form, so $r = 1$, then this can be done systematically:

We seek $u(t)$ such that $u(x' + px) = (d/dt)(ux)$
i.e. $pu = u'$: separable, with solution $u = e^{\int p(t)\,dt}$
(any constant of integration will do here). Then integrate both sides of

$(d/dt)(ux) = uq$: $x = u^{-t} \int uq\,dt$

The constant of integration is in this integral, so the general solution has the form

$x = x_p + c\,u^{-1}$.

Another lovely feature of linear equations is that the constant of integration in the solution of a linear equation always appears right there.

The "associated homogeneous equation" is $x' + px = 0$: separable, with solution $x_h = e^{-\int p(t)\,dt}$. Look! this is the reciprocal of the integrating factor! Wonderful!

In most applications, u^{-1} falls off to zero as t gets large; the term $c\,u^{-1}$ is a "transient."

Chao: That's a lot of integration .... I'm more interested in the general

behavior of solutions, rather than an incomprehensible expression of them as integrals or a boring expression of them in terms of sin, cos, exponential.

I prefer arguments like this: take an equation like $y' = y^2 - x$. This doesn't have a single solution which you, Linn, in your linear cave, have anything to say about. But I can look at the direction field, recognize that there is a funnel along $y = -\sqrt{x}$, so all solutions near there are trapped and are asymptotic to $- \sqrt{x}$ .... and even argue that they are all ultimately a bit larger than $- \sqrt{x}$. No integration involved, but very good information.

Or take an equation like the logistic equation

$$y' = k_0 (1 - y/p) y$$

This is an auotomous equation, and remains so even if I allow a harvest rate, even one depending up on y :

$$y' = k_0 (1 - y/p) y - a(y)$$

This equation gives genuine insight into real population dynamics. By looking at the phase line it is easy to analyze the behavior of solutions, in a way useful for policy makers.

Linn: Well, now, most of the time a system is near equilibrium. Engineers get very anxious when their systems get too far from equilibrium. Let's look at your nice nonlinear logistic equation. As you said, there's a critical point at $y = p$, and so an equilibrium solution. Just how does the system relax to this equilibrium?

Let's write $y = p + u$ and change variables:

$$1 - y/p = - u/p$$

$$u' = - k_0 u/p (p-u) = - k_0 u + k_0 u^2/p$$

For small u the second term is *very* small, and can be ignored. This is called LINEARIZING the equation near equilibrium. Near equilbrium solutions to the nonlinear equation behave a lot like $p + u$ where u is a solution to the *linear* equation

$$u' = - k_0 u$$

so we can say that the population relaxes to equilibrium exponentially, as $e^{-k_0 t}$ approaches 0.

Chao: There you go again with your fancy exponentials. You think you know all about them, but your computer has to compute their values, after all, and the methods it uses are no different than the methods used to compute the values of linear equations: Euler's method, or its fancier variants.

Linn: Speaking of fancy exponentials, I'd like to point out that smart people almost never use integrating factors to integrate linear

equations with constant coefficients:

$$x' + k x = q(t)$$

Yeah, the integrating factor is e^{kt} , but even I don't like the integrals that come out. But there are these great tricks, Chao! Suppose $q(t) = B e^{rt}$. Then, be optimistic! maybe there's an exponential solution!

$$x_p = A e^{rt}$$

$$x_p' = A r e^{rt}$$

$$B e^{rt} = A (k+r) e^{rt}$$

so $A = B / (k+r)$:

ERF $x_p = B e^{rt} / (k+r)$ is a solution to $x' + kx = B e^{rt}$

This is the "Exponential Response Formula." It works as long as $k+r$ is not 0.

Chao: Bravo.

Linn: And what's better, did I ever tell you about the complex exponential? r can be a complex number! Euler told us that

$$e^{i \theta} = \cos(\theta) + i \sin(\theta)$$

-- it's a point on the unit circle in the complex plane --- so trig functions are incorporated into the complex exponential!!

So to solve $x' - x = 3 \cos(t)$

I replace it by a different equation

$$z' - z = 3 e^{it}$$

of which it is the real part. Then use optimism or the ERF (with $k = -1$, $r = i$, $B = 3$):

$$z_p = 3 e^{it} / (-1+i)$$

and just find the real part $x_p = Re(z_p)$

$$z_p = 3 (-1-i)/2 (\cos(t) + i \sin(t))$$

so $x_p = -(3/2) \cos(t) + (3/2) \sin(t)$

For the general solution we just add in the general sol to the homogeneous equation, which is $c e^t$. It's a little funny to call this a "transient," and I don't, but it does give you the general solution.

Chao: You know, this method results in these sums of sines and cosines, which is very nice but I want to know what they look like. The "nonlinear"

view of sines and cosines writes

$$a \cos(\omega t) + b \sin(\omega t) = A \cos(\omega t - \phi)$$

where A, ϕ are the polar coordinates of the point (a,b) .

In your example, $A = 3\sqrt{2}/2$ and $\phi = 3\pi/4$: so the solution is easy to draw and compare with the input signal.

Linn, I want to point out another charming feature of solutions to many nonlinear equations. ... Take a simple one like $y' = y^2$
for example. This is separable: $y^{-2}\, dy = dx$, $-y^{-1} = x + c$,

$$y = 1 / (c - x)$$

So the IVP with $y(0) = 1$ has $c = 1$, $y = 1/(1-x)$. Its graph is asymptotic to the vertical line at $x = 1$: it goes off to infinity in finite time. It ENDS. The equation $y = 1/(1-x)$ actually represents TWO solutions: one for $x < 1$, and another for $x > 1$. If we are to say that a solution of a differential equation is determined by an initial value, we have to require that the graph be connected.

Linn: You call that a feature? That never happens to solutions to my equations. If they are going to go south on me, I know it from the coefficients or the input signal. As long as $p(t)$ and $q(t)$ are nice and finite (and $r(t)$ is nonzero) so are all solutions.
They live as long as I do!

Chao: Well, the world really is nonlinear. Newton's law of gravitation is highly nonlinear. This kind of explosion actually happens in the case of Newton's laws: Jeff Xia showed that in a certain 5-planet system two of the planets behaves more or less like $(1/t)\sin(1/t)$, oscillating with increasing amplitude and increasing frequency as $t \dashrightarrow 0$ (from $t < 0$) .

Solutions to linear equations are not nearly as diverse and exciting!

Who wins? Linn E. R. or Chao S.?

In both classes, Linear won but Nonlinear was more enthusiastic.

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18.03 Differential Equations

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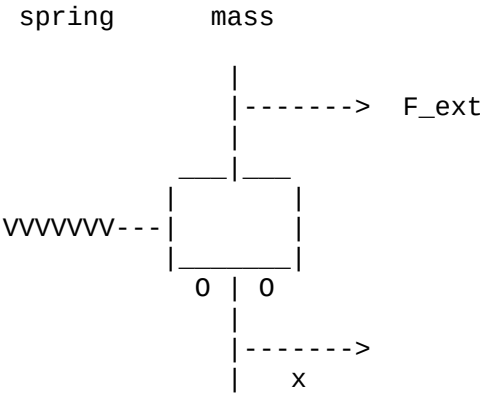
18.03 Class 11, Feb 26, 2010

Second order linear equations:
Physical model, solutions in homogeneous case.
Characteristic polynomial, distinct real roots.

[1] Springs and masses
[2] Dashpots
[3] Second order linear equations
[4] Solutions in homogeneous case: Superposition I
[5] Exponential solutions: characteristic polynomial

[1] Second order equations are the basis of analysis of mechanical and electrical systems. We'll build this up slowly.

A spring is attached to a wall and a cart:



Set up the coordinate system so that at $x = 0$ the spring is relaxed, which means that it is exerting no force.

In addition to the spring, suppose that there is another force acting on the cart -- an "external force," maybe wind blowing on a sail attached to it, maybe gravity, or some other force. Then

$$mx'' = F_{spr} + F_{ext}$$

The spring force is characterized by the fact that it depends only on position. In fact:

$$\text{If } x > 0 \text{ , } F_{spr}(x) < 0$$
$$\text{If } x = 0 \text{ , } F_{spr}(x) = 0$$
$$\text{If } x < 0 \text{ , } F_{spr}(x) > 0$$

I sketched a graph of $F_{spr}(x)$ as a function of x.
The simplest way to model this behavior (and one which is valid in general for small x, by the tangent line approximation) is

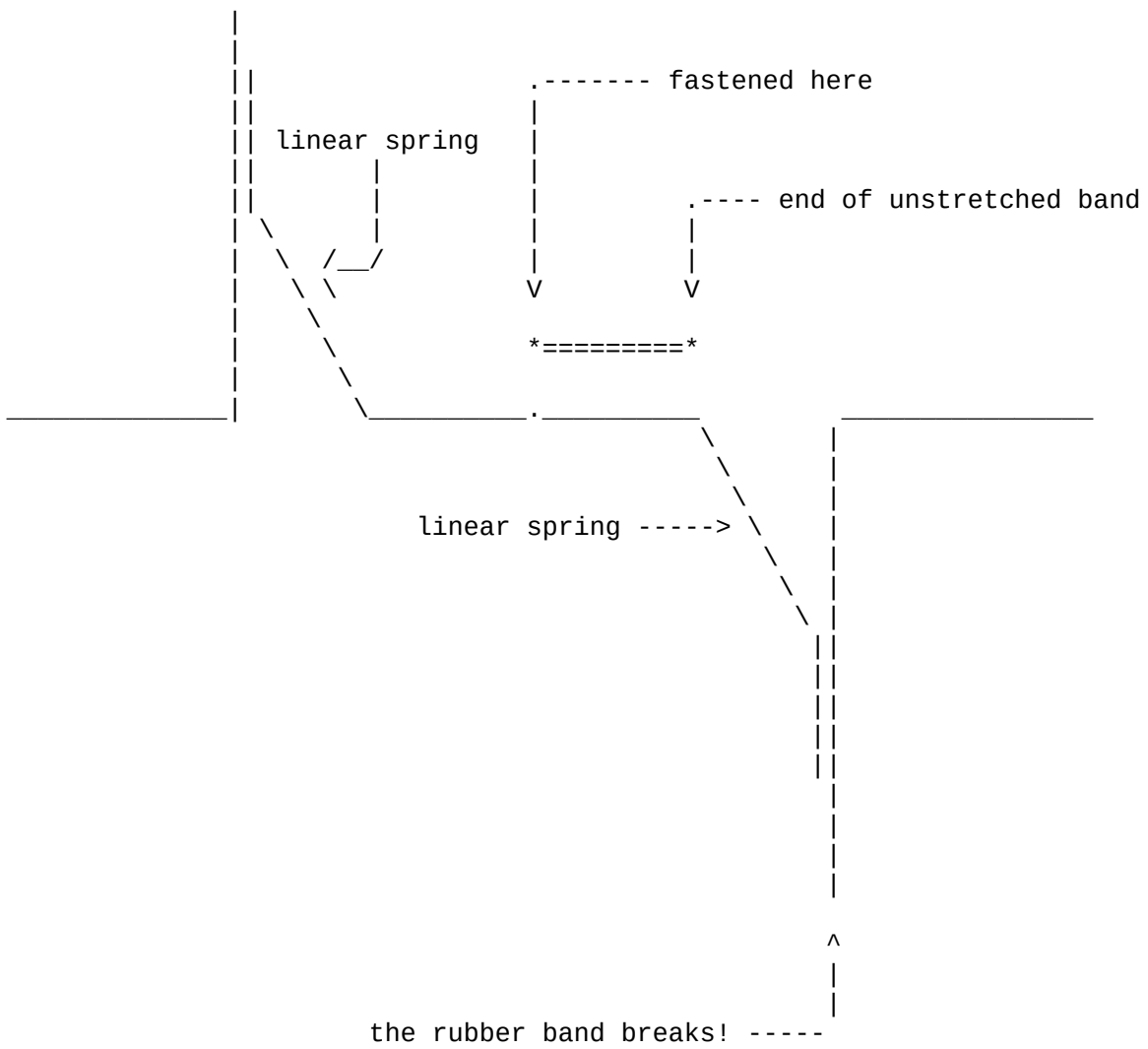
$$F_{spr}(x) = -kx \qquad k > 0 \text{ the "spring constant."} \qquad \text{"Hooke's Law"}$$

This is another example of the linear approximation that Linn was

discussing on Monday. So we get

$$mx'' + kx = F_{ext} .$$

I displayed a weight on a rubber band. This is not a spring, as you usually think of one, but it behaves like one, at least in a range. Lay a rubber band laid out on a table. Fix the right end of it and set $x = 0$ where the left end is in a relaxed state, then the graph of the force exerted by the rubber band looks something like this ...



[2] Any real mechanical system has friction. Friction takes many forms. It is characterized by the fact that it depends on the motion of the mass. We will suppose that it depends only on the velocity of the mass and not on its position. Often the damping is controlled by a "dashpot." This is a cylinder filled with oil, that a piston moves through. Door dampers and car shock absorbers often actually work this way. Write $F_{dash}(x')$.

It acts against the velocity:

```
      If  x' > 0 ,  F_Dash(x') < 0
      If  x' = 0 ,  F_Dash(x') = 0
      If  x' < 0 ,  F_Dash(x') > 0
```

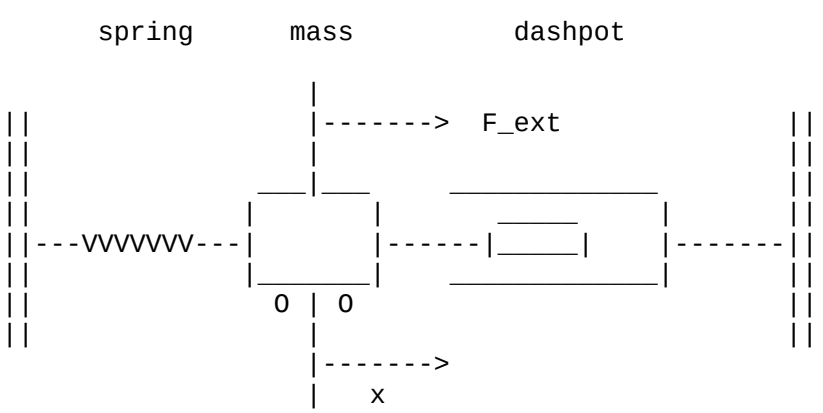
The simplest way to model this behavior (and one which is valid in general for small x' , by the tangent line approximation) is

```
      F_fric(x)  =  -bx    b > 0  the "damping constant."
```

This is "linear damping." Altogether the equation is

```
      mx" + bx' + kx  =  F_ext
```

Diagrammatically:



[3] A linear differential equation is one of the following form:

```
      a_n x^(n) + a_{n-1} x^(n-1) + ... + a_1 x' + a_0 = q(t)
```

(no constant term on the left hand side!) The a_k's are the "coefficients." They may depend upon t (but not on x). If a_n is not zero, this is "of order n."

Our spring system is an example of a *second order* linear equation. (Two springs in series will give a fourth order equation.)

The left hand side represents the SYSTEM, the spring/mass/dashpot system. The right hand side encodes the INPUT SIGNAL, an external force at work. The coefficients represent parameters of the system -- for example, mass, damping constant, spring constant. In general they may depend upon time: maybe the force is actually a rocket, and the fuel burns so m decreases. Maybe the spring gets softer as it ages. Maybe the honey in the dashpot gets stiffer with time.

Most of the time we will assume that the coefficients are CONSTANT: the timescale of their variation is much longer than the timescale of the dynamical variable x .
But the external force F_ext(t) can certainly vary (maybe sinusoidally).

We can see physically that there are many solutions with fixed

initial value of x , x(t_0) = x_0 . So many different solution
curves of a second order equation pass through a given point.
But, also on physical grounds, the initial position together with
the initial vecocity determine the solution: "initial position" data
for a second order equation consist of both position and velocity
at a give instant. This is a version of the existence/uniqueness theorem.
In an n-th order equation, you need to know n numbers:
x(t_0), x'(t_0), ... x^(n-1)(t_0) .

[4] We'll begin with F_ext = 0 : the system is allowed to
evolve on its own, without outside interference.

     mx" + bx' + kx = 0   ,   m nonzero   (*)

Think of a saloon door swinging. This is a *homogeneous* equation.

Special case (from recitation): the harmonic oscillator: b = 0, k > 0 :

     x" + (k/m) x = 0

Solutions come from simple facts about sines and cosines:

      cos(omega t) ---> - omega sin(omega t) ---> - omega^2 cos(omega t)
      sin(omega t) --->   omega cos(omega t) ---> - omega^2 sin(omega t)

So if omega^2 = k/m , so that our equation is   x" + omega^2 x = 0 ,

then   x = cos(omega t)   and   x = sin(omega t)   are solutions:

ANOTHER FUNDAMENTAL FACT TO MEMORIZE!

In fact, you showed that any sinusoid of circular frequency omega,

     x = a cos(omega t) + b sin(omega t) = A cos(omega t - phi)   (*)

is also a solution. In fact these are the only solutions, because

     x(0) = a

     x'(0) = omega b

and so you can solve (uniquely) for a and b to give any desired
initial condition: you don't need more.

Q11.1. What is the period of a nonzero solution of x" + 4x = 0 ?

1. Depends upon the solution
2. 2
3. pi
4. 4
5. 2pi
6. pi/2
Blank. Don't know.

Ans: : think of what t has to do to take (2t) from 0 to 2pi.
Or use P = 2pi/omega, with omega = 2 . Ans: 3: pi.

The fact that cos(omega t) and sin(omega t) are solutions implies that (*) is also a solution, via

Superposition I: If x_1 and x_2 are two solutions of the homogeneous linear equation, then so is

$$x = c_1 x_1 + c_2 x_2$$

Check (in second order case $mx'' + bx' + kx = 0$):

$$m(c_1x_1 + c_2x_2)'' + b(c_1x_1 + c_2x_2) + k(c_1x_1 + c_2x_2)$$

$$= m(c_1x_1'' + c_2x_2'') + b(c_1x_1' + c_2x_2') + k(c_1x_1 + c_2x_2)$$

$$= c_1(mx_1'' + bx_1' + kx_1) + c_2(mx_2'' + bx_2' + kx_2)$$

$$= c_1 (0) + c_2 (0) = 0 .$$

[5] The equation $mx'' + bx' + kx = 0$, for m, b, k constant, is a lot like $x' + kx = 0$, which has as solution $x = e^{-kt}$ (and more generally multiples of this). It makes sense to try for exponential solutions of (*): e^{rt} for some as yet undetermined constant and r .

To see which r might work, plug $x = e^{rt}$ into (*). Organize the calculation: the k] , b] , m] are flags indicating that I should multiply the corresponding line by this number.

$$\begin{array}{lll} k] & x = c & e^{rt} \\ b] & x' = c\,r & e^{rt} \\ m] & x'' = c\,r^2 & e^{rt} \end{array}$$

$$0 = mx'' + bx' + kx = c (mr^2 + br + k) e^{rt}$$

An exponential is never zero, so we can cancel to see that $c e^{rt}$ is a solution to (*) for any c exactly when r is a root of the "characteristic polynomial"

$$p(s) = ms^2 + bs + k$$

Example. $x'' + 5x' + 4x = 0$

The characteristic polynomial $s^2 + 5s + 4$. We want the roots. One reason I wanted to write out the polynomial was to remember that you can find roots by factoring it. This one factors as $(s + 1)(s + 4)$ so the roots are $r = -1$ and $r = -4$. The corresponding exponential solutions are e^{-t} and e^{-4t} .

By superposition, the "linear combination"

$$x(t) = c_1 e^{-t} + c_2 e^{-4t}$$

is a solution as well. This is the general solution.

Suppose we know also that $x(0) = 2$ and $x'(0) = -5$. To use the second condition we'll need to know

$$x(t) = -c_1 e^{-t} - 4 c_2 e^{-4t}$$

Then

$$\begin{aligned} 2 &= c_1 + c_2 \\ -5 &= -c_1 - 4c_2 \\ \hline -3 &= -3c_2 \end{aligned}$$

so $c_2 = 1$ and then $c_1 = 1$:

$$x(t) = e^{-t} + e^{-4t}$$

General picture: A linear equation of degree n

$$a_n x^{(n)} + \ldots + a_1 x' + a_0 x = q(t)$$

with constant coefficients has a characteristic polynomial,

$$p(s) = a_n s^n + \ldots + a_1 s + a_0$$

e^{rt} is a solution if and only if r is a root of $p(s)$: $p(r) = 0$.

By superposition, any linear combination of these exponentials is also a solution.

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18.03 Class 12, March 1, 2010

Good Vibrations

[0] Summary table [on the blackboard]
[1] Overdamping
[2] Underdamping
[3] Real solution theorem
[4] Natural and damped circular frequencies
[5] Critical damping
[6] Transience
[7] Root diagram

[0] Summary table of unforced system responses.

Name*	m,b,k relation	Char. roots	Exp. sol's	Basic real solns
Overdamped	b^2/4m > k	Two diff. real roots, r1, r2	e^{r1 t}, e^{r2 t}	same
Critically damped	b^2/4m = k	Repeated root r = - b/2m	e^{rt}	e^{rt}, te^{rt}
Underdamped	b^2/4m < k	-b/2m +- iw	e^{r1 t}, e^{r2 t}	e^{-bt/2m}cos(w t) e^{-bt/2m}sin(w t)

(w = omega_d = sqrt{ (k/m) - (b/2m)^2 })

* The names here are appropriate under the assumption that b and k are both non-negative. The rest of the table makes sense in general, but it doesn't have a good interpretation in terms of a mechanical system.

We are studying equations of the form

$$mx'' + bx' + kx = 0 \quad (*)$$

which model a mass, dashpot, spring system without external forcing term. Homogeneous constant coefficient linear second order.

We found that (*) has an exponential solution e^{rt} exactly when r is a root of the "characteristic polynomial" $p(s) = ms^2 + bs + k$. These are called the MODES of the system.

[1] Example. [Overdamped: Distinct real roots] $x'' + 5x' + 4x = 0$.
We did this on Friday:
The characteristic polynomial $s^2 + 5s + 4$ factors as $(s + 1)(s + 4)$ so the roots are $r = -1$ and $r = -4$. The corresponding exponential solutions are e^{-t} and e^{-4t}. These are called *modes* of the system. They represent pure states. The general solution is mixture of the two states, $x = c_1 e^{-t} + c_2 e^{-4t}$.

But where's the vibration?

[2] Example. [Underdamped: Nonreal roots] Make the spring a little stronger and the dashpot a little weaker: x" + 4x' + 5x = 0

A good way to find the roots of a quadratic polynomial is to complete the square:

$$s^2 + 4s + 5 = (s + 2)^2 + 1 .$$

Setting this equal to zero, $(s + 2)^2 = -1$ or $s + 2 = +- i$ or $s = -2 +- i$. So our exponential solutions are

$$e^{(-2+i)t} , \quad e^{(-2-i)t}$$

The general solution is a linear combination of these two "basic" solutions. But I guess we were expecting REAL valued solutions.

For this we have:

[3] [Slide] Real Solution Theorem:
If z is a complex-valued solution to $mz" + bz' + kz = 0$, where m , b , and k are real, then the real and imaginary parts of z are also solutions.

Proof: Write $z = u + iv$ and build the table.

k]	z	= u	+ iv
b]	z'	= u'	+ iv'
m]	z"	= u"	+ iv"

$$0 = (mu" + bu' + ku) + i(mv" + bv' + kv)$$

Both things in parentheses are real, so the only way this can happen is for both of them to be zero.

In our situation,

$e^{(-2+i)t}$	has real part	$e^{-2t} \cos(t)$
	and imaginary part	$e^{-2t} \sin(t)$

so we have those two solutions.

If I had chosen the other exponential solution,

$e^{(-2-i)t}$	has real part	$e^{-2t} \cos(-t) = e^{-2t} \cos(t)$
	and imaginary part	$e^{-2t} \sin(-t) = - e^{-2t} \sin(t)$

(This also follows from the fact that the roots are complex conjugates of each other and hence the two exponential functions are too.)

So you get the same (up to sign) basic real solutions using either solution.

Taking linear combinations, we get the general solution

x = e^{-2t} (a cos(t) + b sin(t))

or A e^{-2t} cos(t - phi)

This is a "damped sinusoid," with "circular pseudo-frequency" 1.

I demonstrated equations like this using the Mathlet Damped Vibration.

In this underdamped case, you should go straight to the answer and not pass through the complex exponential, which was just a convenient way to find the answer. If the roots are a +- omega i , the solutions are e^{at} cos(omega t) and e^{at} sin(omega t) .

[4] Let's get formulas for the roots. The quadratic formula tells you, or complete the square:

0 = s^2 + (b/m) s + k/m = (s + (b/2m))^2 - ((b/2m)^2 - k/m)

So r = -b/2m +- sqrt{(b/2m)^2 - k/m} .

Note that sqrt(-d) = i sqrt(d) , so if (b/2m)^2 < k/m then the roots are

r = -b/2m +- omega_d i , omega_d = sqrt(k/m-(b/2m)^2)

It is called the "damped circular frequency" of the system represented by mx" + bx' + kx .

Example. [Undamped, special case of Underdamped]
mx" + kx = 0 , k/m > 0 : no damping : "Harmonic Oscillator."

Roots are +- omega_n i , omega_n = sqrt{k/m} so we have exponential solutions e^{i omega t} and e^{-i omega t} with real and imaginary parts cos(omega_n t) and sin(omega_n t) . This recovers our earlier calculation.

Even if b is not zero, sqrt{k/m} is denoted omega_n and called the "natural circular frequency of the system.

Question 12.1. As b increases from b = 0 (while m and k remain fixed) what happens to the period of oscillation of nonzero solutions of mx" + bx' + kx = 0 ?

1. It stays the same

2. It gets shorter

3. It gets longer

4. It depends on the other parameters

5. It depends on the initial conditions

Blank. Don't know

Ans: The quantity inside the square root decreases, so the circular frequency decreases and the period increases. As b increases towards critical damping, the period goes from $2pi/omega_n$ to infinity.

[5] In between overdamped and underdamped there is a marginal case, when the roots are equal.

Example. [Critically damped] $x" + 4x' + 4 = 0$.
Then $p(s) = (s+2)^2$ has $r = -2$ as a repeated root. The only exponential solution is e^{-2t} . Another solution, not a constant multiple of $e^{-2t}$$, is given by te^{-t} . I will not check this for you, you know how to do it: plug in and use the product rule.

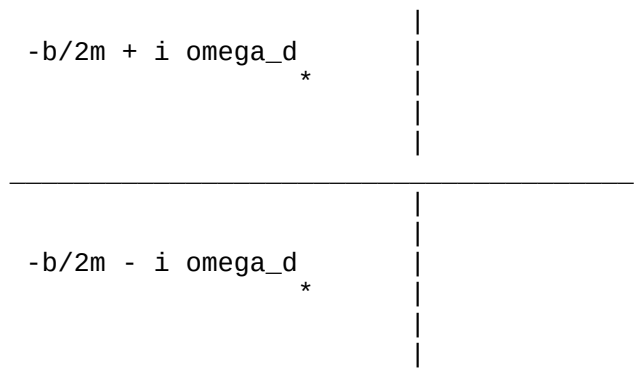
So the general solution is $e^{-2t} (a+ct)$ in this case.

[6] Transience: If m, b, and k are all positive, then every solution dies away for large t.

Look at the roots! $r = -b/2m +- sqrt{(b/2m)^2 - k/m}$.

In the underdamped case, the amplitude is $e^{-bt/2m}$, which dies off.
In the overdamped case, the square root is smaller than $b/2m$, so both roots are negative, and all solutions die off.
In the critically damped case, the root is negative so (even though you get to multiply by t) all solutions die off.

[7] In the complex plane, the roots look like this:



I used the Damped Vibrations Mathlet to illustrate that as the roots move to the left, the solutions decay faster; when they move onto the imaginary axis, there is no damping. If we could envision anti-damping, then we really get into Beach Boy territory and the solutions grow exponentially in amplitude.

If the roots are pushed towards the real axis (by increasing the spring constant, for example, then the frequency decreases; the period increases. This can be a bit hard to see on the applet, but we did see it analytically.

When the two roots coalesce, we have critical damping, and no more vibration. In fact no solution crosses the t axis more than once. This behavior continues when the roots separate on the real axis, in the overdamped regime.

Muddy cards.

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18.03 Differential Equations

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18.03 Class 13, March 3, 2010

Forced linear equations

[1] Superposition II
[2] Harmonic sinusoidal response
[3] Exponential response formula
[4] Sinusoidal response using complex replacement

I drew the spring/mass/dashpot system and added a force to it:
the little sail comes back into play.

$$mx'' + bx' + kx = F_{ext} \quad (*)$$

Notice by the way that I can put the damper on the left in parallel with the spring: it still opposes velocity. If $x' < 0$, $F_{dash} > 0$ and so on: so you get exactly the same equation.

Also important will be the "associated homogeneous equation"

$$mx'' + bx' + kx = 0 \qquad (*)_h$$

which we know all about after Lecture 12. Final comment on this:
We can "reduce" this by dividing by m:

$$x'' + (b/m)x' + (k/m) = 0$$

If $b = 0$ we get solutions with circular frequency $\sqrt{k/m}$.
If $b > 0$, you get exponentially damped sinusoids, with smaller circular frequency ω_d (or not oscillating at all, if b is big enough).
In general, even if $b > 0$, we call $\sqrt{k/m}$ the "natural circular frequency" of the system, and write ω_n for it. So in the underdamped case, when there is an ω_d, $\omega_d < \omega_n$. So the reduced homogeneous equation is $x'' + (b/m)\,x' + \omega_n^2 = 0$.

[1] The general strategy in finding solutions is based on "superposition."

[Slide:]
Superposition I: If $x1$ and $x2$ are solutions of a homogeneous linear equation, then so is any linear combination $c1\,x1 + c2\,x2$.
If the equation is of second order and neither of $x1$, $x2$ is a multiple of the other, then $c1\,x1 + c2\,x2$ is the general solution.

Now we have:

Superposition II: If xp is any solution to $(*)$ and xh is a solution to $(*)_h$, then $xp + xh$ is again a solution to $(*)$.

Proof: Plug x into $(*)$:

k] $x = xp + xh$
b] $x' = xp' + xh'$
m] $x'' = xp'' + xh''$
_____

$$mx'' + bx' + kx = (m\,xp'' + b\,xp' + k\,xp) + (m\,xh'' + b\,xh' + k\,xh)$$
$$= F_{ext} + 0$$

as we wanted.

In fact, if xh is the general solution to $(*)_h$ then $xp + xh$ is the general solution to $(*)$.

This proof works for linear equations of any order; for example first order, where we already saw it as a consequence of the method of integrating factors.

Superposition II splits the problem of finding the general solution to $(*)$ into two parts:

(1) find SOME solution to $(*)$, a "particular solution," and then

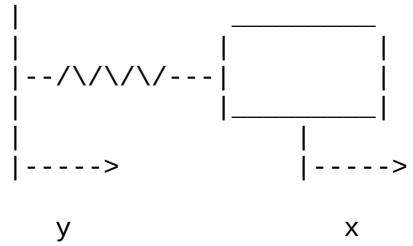
(2) find the general solution of $(*)_h$.

We have just been studying solutions too $(*)_h$, so let's focus on finding a particular solution.

[2] First case: $b = 0$, sinusoidal driving term.
The model is the weight on the rubber band. I showed its natural frequency, and then drove it by moving my hand sinusoidally, first slowly and then faster. At first the mass just follows the motion of my hand, but as the frequency picks up the rubber band starts to stretch at the lows and compress at the highs. At a certain point things get really wild, but then settle down - but now the mass is moving UP when my hand goes down, and vice versa.

Can we explain all that mathematically?

I drew a spring/mass system, forced through the spring:



Set up the parameters x and y so that when $x = y$ the spring is relaxed. So when $x > y$ the spring is stretched and the spring force is negative:

$$m\,x'' = k(y - x)$$

or $$mx'' + kx = ky$$

Think of y as the input signal - the position of my hand - and x as the system response or output signal.

So in the sinusoidal case, set the clock so that the input signal has the

$$y = A \cos(\omega t)$$

So : $x'' + \omega_n^2 x = A \omega_n^2 \cos(\omega t)$ (**)

There are two frequencies here: the natural frequency ω_n of the system and the frequency ω of the input signal $A \cos(\omega t)$.

The experiment with the weight and rubber and encourages us to use the Method of Optimism, and hope for a solution of the form

$$x_p = B \cos(\omega t)$$

To see what B must be, plug this into (**) :

ω_n^2] $x_p = B \cos(\omega t)$
1] $x_p'' = -B \omega^2 \cos(\omega t)$

$$A \omega_n^2 \cos(\omega t) = B(\omega_n^2 - \omega^2) \cos(\omega t)$$

This works out if we take

$$B = \left(\omega_n^2 / (\omega_n^2 - \omega^2) \right) A$$

The output amplitude is a multiple of the input amplitude, and the ratio is the GAIN:

$$H(\omega) = |B/A| = \omega_n^2 / |\omega_n^2 - \omega^2|$$

Imagine the natural frequency of the oscillator fixed, and we slowly increase the frequency of the input signal. The graph of the gain starts when $\omega = 0$ at $H = 1$ - consistent with what we saw - and then increases to a vertical asymptote at $\omega = \omega_n$. This is RESONANCE, and then no such sinusoidal solution exists. There are solutions, of course, and we will come back to this case later. What happens with the weight and rubber band is that the nonlinear character of the spring asserts itself for large amplitude.

When $\omega > \omega_n$, the gain falls back towards zero.

Also: when $\omega < \omega_n$ the denominator is positive, and the output is a positive multiple of the input. When $\omega > \omega_n$ the denominator is negative, and the output signal is a negative multiple of the input: this is PHASE REVERSAL.

We have nicely accounted for all the things we observed.
I demonstrated this effect using the Mathlet Harmonic Frequency Response: Variable Input Frequency.

[3] Exponential signals

$$mx'' + bx' + kx = A e^{rt} \quad (*)$$

We want to find some solution, x_p . Again as in the first order case,

try for a solution of the form $x_p = B\, e^{rt}$:

$$k] \quad x_p = B\, e^{rt}$$

$$b] \quad x_p' = B\, r\, e^{rt}$$

$$m] \quad x_p'' = B\, r^2\, e^{rt}$$

$$A\, e^{rt} = B\,(mr^2 + br + k)\, e^{rt} = B\, p(r)\, e^{rt}$$

We saw the same thing in the first order case. It's a general fact about linear constant coefficient equations with exponential right hand side:

$$a_n x^{(n)} + \ldots + a_1 x' + a_0 x = A\, e^{rt} \qquad (*)$$

with characteristic polynomial $p(s) = a_n s^n + \ldots + a_1 s + a_0$:

Exponential Response Formula: A (particular) solution to (*) is given by:

$$x_p = A\, e^{rt} / p(r)$$

as long as $p(r)$ is not zero.

- i.e., if r is *not* a root of the characteristic polynomial. If $p(r) = 0$, the system is "in resonance" with the input signal, and there is no solution of the form $B\, e^{rt}$.

Eg $x'' + x' + 2x = 4\, e^{3t}$ $p(3) = 3^2 + 3 + 2 = 14$ so

$$x_p = (4 / 14)\, e^{3t} .$$

[4] Now let's try this: $x'' + x' + 2x = \cos(t)$

by *complex replacement*: This is the real part of

$$z'' + z' + 2z = e^{it}$$

$p(s) = s^2 + s + 2$, $p(i) = (i)^2 + i + 2 = 1 + i$,

$$z_p = (1/(1+i))\, e^{it}$$

We want the real part.

$$z_p = ((1-i)/2)\,(\cos(t) + i \sin(t))$$

$$x_p = (1/2)\,(\cos(t) + \sin(t))$$

To understand this, sketch the triangle: sides 1/2 and 1/2 so hypotenuse $A = \mathrm{sqrt}(2)/2$ and angle $\phi = \pi/4$:

$$x_p = (\mathrm{sqrt}(2)/2) \cos(t - \pi/4) .$$

Lesson: the damping term, b, creates a nonzero imaginary part in the denominator of the ERF, and that causes phase lag in the solution. It also causes the magnitude of the denominator to increase, and hence the amplitude of the sinusoidal solution to decrease.

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18.03 Differential Equations

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18.03 Class 14, March 5, 2010

Complex gain

1. Recap
2. Phase lag
3. Driving via the dashpot
4. Complex gain

[1] The story so far: We're studying solutions of linear constant coefficient equations

$$a_n x^{(n)} + \ldots + a_1 x + a_0 = q(t) \quad (*)$$

A key is the characteristic polynomial

$$p(s) = a_n s^n + \ldots + a_1 s + a_0$$

For the homogeneous case,

$$a_n x^{(n)} + \ldots + a_1 x + a_0 = 0 \quad (*)_h$$

we found that the roots of $p(s)$ give the exponents in exponential solutions, and that the general solution is a linear combination of these or (these times a power of t in case there are repeated roots).
Euler's formula shows that

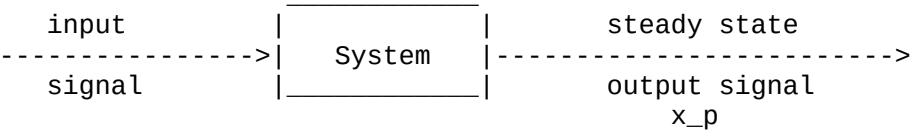
$$|e^{(a+bi)t}| = e^{at}$$

so: [Slide]

Transience Theorem:
All homogeneous solutions of $(*)_h$ decay to zero provided that all the roots of $p(s)$ have negative real parts.

In this case the solutions to $(*)_h$ are called "transients,"
By superposition, all solutions to (*) converge together as t gets large, and we say that the equation is "stable."

If we have a system modeled by a stable equation, and we are only interested in what it looks like after the transients have died down, we can eliminate the initial condition:



So we look for a particular solution x_p. Sinusoidal input signals are of particular importance. Experiments indicate that sinusoidal in gives sinusoidal out. We decide to set our clock so that the input signal is

$$\text{input} = A \cos(\omega t)$$

Experiments indicate that the steady state output signal is again sinusoidal, of the same circular frequency:

output = x = B cos (omega t - phi)

A consequence of linearity of the system is that B is proportional to A:

x = g A cos (omega t - phi)

So there are just two numbers I need to know, in understanding this kind of system:

the "gain" g and
the phase lag phi.

Both of them will depend upon the input circular frequency omega .

[2] Polar treatment of example from Wednesday:

x" + x' + 2x = cos(t) p(s) = s^2 + s + 2

The default is to regard the right hand side as the input signal, and x as the output. We are looking for the steady state solution. Make the complex replacement:

z" + z' + 2z = e^{it} p(i) = - 1 + i + 2 = 1 + i

z_p = e^{it}/(1+i) from ERF, [Slide]

Rectangular solution: 1/(1+i) = (1-i)/2

x_p = Re(z_p) = (1/2) cos(t) + (1/2) sin(t)

We used the triangle to rewrite this in polar form:

x_p = (sqrt(2)/2) cos (t - pi/4)

This expression gives more insight: the amplitude is sqrt(2)/2 ~ 0.707 times the amplitude of the input signal - the gain is sqrt(2)/2 - and the steady state system response lags pi/4 radians or 1/8 cycle behind the input signal.

I want to show you how you can get to this information directly, by passing to polar coordinates earlier. So we start from

z_p = e^{it}/p(i)

and calculate

p(i) = 1+i = sqrt(2) e^{i pi/4)

z_p = e^{it}/p(i) = (1/sqrt(2)) e^{-i pi/4) e^{it}

= (sqrt(2)/2) e^{i(t - pi/4)}

$$x_p = \mathrm{Re}(z_p) = (\sqrt{2}/2)\cos(t - \pi/4)$$

-- much more efficient.

Question 14.1. In this equation, if m and k are left alone
and the damping constant b is increased from 1, the phase lag

1. increases and I can see why from the mathematics
2. increases but I only see this from physical reasoning
3. decreases and I can see why from the mathematics
4. decreases but I only see this from physical reasoning
5. stays the same and I can see why from the mathematics
6. stays the same but I only see this from physical reasoning
7. don't know

Ans: the only effect of b is to produce the imaginary part of $p(i)$.
If it increases, then the argument of the complex number $p(i)$ increases,
The argument of $p(i)$ is the phase lag in this example, and that increases.

[The class was on board with this one.]

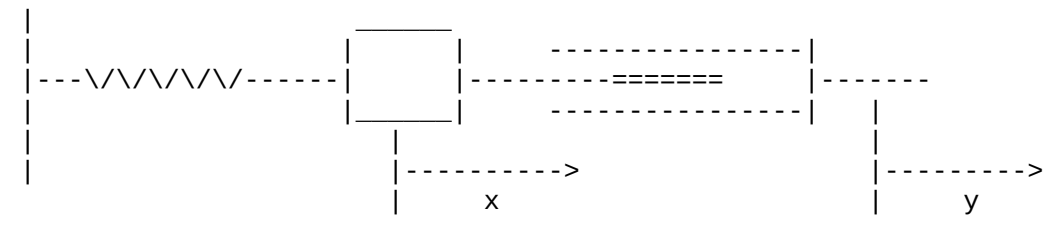
Question 14.2. In this equation, if m and k are left alone and
the damping constant b is increased from 1, the amplitude of the solution

1. increases and I can see why from the mathematics
2. increases but I only see this from physical reasoning
3. decreases and I can see why from the mathematics
4. decreases but I only see this from physical reasoning
5. stays the same and I can see why from the mathematics
6. stays the same but I only see this from physical reasoning
7. don't know

The amplitude of the solution is $1/|p(i)|$. $p(i)$ increases if b increases,
so $1/|p(i)|$ decreases.

[This was harder. Both classes discussed it. I think the mistake was
forgetting that you *divide* by p(i omega).]

[3] Another way to drive the spring system: though the dashpot:



Now the force on the mass exerted by the dashpot is $b(y-x)'$:

$$m\,x'' + bx' + kx = by' \qquad (*)$$

Input signal: y
System response: x

Notice! the right hand side is not the input signal; it's not even a multiple of the input signal.

Again let's think about driving this system sinusoidally;

$$y = A \cos(\omega t) .$$

We know we will analyze this by making a complex replacement.
Let's take the next step, push the complex replacement back even farther, and replace the input signal itself with a complex exponential signal:

$$y_{cx} = A e^{i \omega t}$$

Now solve (*) with y_{cx} in place of y :

$$m z'' + b z' + k z = b y_{cx}' = b A i \omega e^{i \omega t}$$

ERF $z_p = b A i \omega e^{i \omega t} / p(i \omega)$

where $p(i \omega) = (k - m \omega^2) + b i \omega$

[4] Define the *complex gain* as the complex number you multiply the complex exponential input by in order to get the complex exponential system response:

$$z_p = H(\omega) y_{cx}$$

In this case it is

$$H(\omega) = b i \omega / p(i \omega)$$

Now, to return to original equation we pass to real parts:

$$x_p = \mathrm{Re} (H(\omega) e^{i \omega t})$$

Let's compute the real part using the polar approach as in [1].
The following calculation works in general, not just for this particular case.

$$H(\omega) = |H(\omega)| e^{- i \phi}$$

so $-\phi$ is the argument of $H(\omega)$. Then

$$z_p = A |H(\omega)| e^{- i \phi} e^{i \omega t}$$

$$= A |H(\omega)| e^{i(\omega t - \phi)}$$

Now when I take real parts,

$$x_p = A |H(\omega)| \cos(\omega t - \phi)$$

So: $|H(\omega)|$ is the gain of the system

$- \mathrm{Arg}(H(\omega))$ is the phase lag of the system.

(and that accounts for my choice to write - phi for Arg(H(omega)) .)

This last conclusion is not special to this particular system; it is a general fact.

I demonstrated the Mathlet Amplitude and Phase: Second Order II.

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18.03 Differential Equations

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18.03 Class 15, March 8, 2010

Operators, polynomial signals, resonance

[1] Operators
[2] Resonance
[3] Polynomial signals

Several different topics today, and a respite from the gain game.

[1] Operators

Just as number --function--> number

function --operator--> function

The *differentiation operator D takes x to x' : Dx = x' .

For example, D sin(t) = cos(t) , D x^n = n x^{n-1} , D8 = 0 .

We can iterate: D^2 = x" .

There's also the "identity operator": Ix = x

And we can take linear combinations of operators:

(D^2 + 2D + 2I) x = x" + 2x' + 2x .

The characteristic polynomial here is p(s) = s^2 + 2s + 2 , and it's irresistible to write

D^2 + 2D + 2I = p(D)

so x" + 2x' + 2x = p(D) x

This formalism lets us discuss linear equations of higher order with no extra work. Such an equation has the form

an x^{(n)} + ... + a1 x' + a0 x = q(t) (*)

It has a characteristic polynomial

p(s) = a_n s^n + ... + a_1 s + a_0

and so it can be written

p(D) x = q(t)

Now we can say that the *operator*

p(D) = an D^n + ... + a_1 D + a_0 I

represents the system. This is a "linear, time-invariant differential

operator." In the systems and signals yoga, an input signal determines a function $q(t)$, and the system response x satisfies $Lx = q$. We could write

$$x = L^{-1} q$$

and indeed most of this course is about finding ways to "invert" these operators.

The Exponential Response Formula lets us find a solution of

$$p(D)\, x = B\, e^{rt}$$

very efficiently, as long as
(1) r is not a root of $p(s)$ and
(2) B is constant

Today we'll see how to deal with the first limitation.
The second will come on Wednesday.

[2] For example suppose we want to solve $x'' - 4x = e^{-2t}$. $p(s) = s^2-4$ vanishes at $r = -2$. We can't apply ERF. To see what we CAN do, I want to recall where ERF came from: [Slide]

$$p(s) = a_n s^n + \ldots + a_1 s + a_0$$

$$\begin{aligned}
a_0] \quad & e^{rt} = e^{rt} \\
a_1] \quad D \; & e^{rt} = r \; e^{rt} \\
a_2] \quad D^2 \; & e^{rt} = r^2 \; e^{rt} \\
& \ldots \qquad\qquad \ldots \\
a_n] \quad D^n \; & e^{rt} = r^n \; e^{rt} \\
\hline
& p(D)\, e^{rt} = p(r)\, e^{rt} \qquad (*)
\end{aligned}$$

To get the ERF, you divide by $p(r)$ to see that $e^{rt}/p(r)$ is a solution to $p(D)\, x = e^{rt}$.

This formula (*) is true even if $p(r) = 0$. I will exploit that fact by thinking of t as a constant and r as a variable, and differentiate with respect to r. There are two independent variables now. D is differentiation with respect to t. In 18.02 you learn that at least when applied to nice functions (eg e^{rt}), mixed partials are equal:

(partial/partial r)(partial/partial d) = (partial\partial d)(partial/partial r)

So (partial/partial r) D = D (partial/partial r) and

$$(\partial/\partial r)\, p(D)\, e^{rt} = p(D)\, (\partial/\partial r)\, e^{rt}$$

$$= p(D)\, (t\, e^{rt})$$

By (*), the LHS equals

(partial/partial r) (p(r) e^{rt}) = p'(r) e^{rt} + p(r) t e^{rt}

so

p(D) (t e^{rt}) = p'(r) e^{rt} + p(r) t e^{rt}

Now the hypothesis that p(r) = 0 becomes a virtue! - the second term goes away. As long as p'(r) is not zero we can divide by it:

Resonant Response Formula: If p(r) = 0 and p'(r) is not zero, then

x_p = B t e^{rt} / p'(r)

is a solution to p(D) x = B e^{rt} .

Example: If p(s) = s^2 - 4 , p'(s) = 2s , so a solution to x" - 4x = e^{-2t} is given by

x_p = t e^{-2t} / 4

When r is a root of p(s) we say that the system is "in resonance" with the input signal. Here's why:

Example: x" + 4x = cos(2t)

z" + 4z = e^{2it}

p(s) = s^2 + 4 , p(2i) = 0 . p'(s) = 2s , p'(2i) = 4i .

z_p = t e^{2it} / 4i

x_p = Re z_p = (1/4) t sin(2t) .

So it's like a swing - you push in synch with the natural frequency and the amplitude increases, linearly. You're in resonance.
Our solution in the first example doesn't increase in size - it decays to zero, but slower than the e^{-2t} decay that ERF would lead us to expect.

Speaking of swings, with the economy the way it is, the CIA is about the only group hiring academics these days. One day a biologist, a physicist, and a mathematician were staking out an empty house. Two people went in, and a little while later three people came out. The biologist started talking about reproduction, the physicist about tunneling, and the mathematician said, "If one more person goes into that house, it will be empty again."

[3] Polynomial signals are next.

Theorem (Undetermined coefficients) Suppose that

$$q(t) = b_k t^k + \ldots + b_1 t + b_0 .$$

Then $p(D)x = q(t)$ has exactly one solution which is polynomial of degree less than or equal to k, provided that $p(0)$ is not zero.

Notice that if $p(s)$ is the polynomial

$$a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0$$

then $p(0) = a_0$.

Proof by example:

$$3x'' + 2x' + x = t^2 + 1$$

The theorem applies since 3 is not 0 : there is exactly one solution of the form

$$x = at^2 + bt + c$$

To find a , b , c , plug in:

$$\begin{array}{lll} 1] & x = at^2 + bt + c \\ 2] & x' = 2at + b \\ 3] & x'' = 2a \\ \hline & t^2 + 1 = at^2 + (b+4a)t + (c+2b+6a) \end{array}$$

The coefficients must be equal. $a = 1$. Then $b+4a = 0$ implies $b = -4$. Finally $c+2b+6a = 1$ reads $c = 1 - 2b - 6a = 1 - 2(-4) - 6 = 3$.

$$x_p = t^2 - 4t + 3$$

If $a_0 = 0$, the theorem doesn't apply, but we can use "reduction of order":

Eg $x'' + x' = t$

Substitute $u = x'$ so $u' + u = t$

$$\begin{array}{ll} u = & at + b \\ u' = & a \\ \hline t = & at + (a+b) \end{array}$$

$a = 1$, $b = -1$, $u_p = t - 1$. Now integrate: $x_p = t^2/2 - t$.

For the general solution, the roots of $p(s) = s(s+1)$ are 0 and -1, so $x_h = c_1 + c_2 e^{-t}$.

Q15.1. The differential equation $3x^{(4)} + 2x^{(3)} + x^{(2)} = t^2 + 1$

(1) has no polynomial solutions.

(2) has exactly one polynomial solution.
(3) has a polynomial solution of degree at most 3 .
(4) has exactly one polynomial solution of the form $at^5 + bt^4 + ct^3$.
(5) has only polynomial solutions.
(6) None of the above
(7) Several of the above
(Blank) Don't know

Ans: First of all, if I plug 1 or t into the left hand side, I get zero. These are solutions to the associated homogeneous equation, so I can add them to any solution and get new solutions. So if there is one polynomial solution then there are many.

To find one, set $u = x''$, so the equation is $3u'' + 2u' + u = t^2 + 1$.

We solved this above, but in any case the Undetermined Coefficients Theorem shows that it has a solution $u = at^2 + bt + c$. Integrating twice to get x gives a polynomial involving t^4, t^3, t^2 (and, if you like, t and 1 as well, by different choice of constant of integration).

So I think the answer is (6).

Incidentally using the calculation we did, you get

$$x_p = (1/12)\, t^4 - (2/3)\, t^3 + (3/2)\, t^2 .$$

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18.03 Differential Equations

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18.03 Class 16, March 10, 2010

Frequency response

[1] Variation of parameters
[2] Summary of complex gain
[3] Second order frequency response
[4: Supplement] First order frequency response
[5: Supplement] Other systems [not done in lecture]

[1] I promised on Monday to show you what you can do if A is not constant in $p(D)x = A e^{rt}$.

Example: $3x'' + 8x' + 6x = (t^2 + 1) e^{-t}$.
Now B is not constant.

Try for a solution of the form $x_p = u e^{-t}$ for some u .
This is what led us to the ERF; but now u is allowed to be nonconstant.
This is called *variation of parameters*.

$$\begin{array}{rl} 6\,] & x_p = u e^{-t} \\ 8\,] & x_p' = (u' - u) e^{-t} \\ 3\,] & x_p'' = (u'' - u' - u' + u) e^{-t} \\ \hline (t^2 + 1) e^{-t} = & (3u'' + 2u' + u) e^{-t} \end{array}$$

Cancel the e^{-t} : $3u'' + 2u' + u = t^2 + 1$

This is solvable by undetermined coefficients. In fact, by an incredible stroke of luck, we have already solved it!

$u_p = t^2 - 4t + 3$

so $x_p = u_p e^{-t} = (t^2 - 4t + 3) e^{-t}$

This method will always replace the given equation with another on in which the right hand side is the same as before but without the e^{rt} .

[2] Complex gain summary:

General stuff (independent of order!) Input: $y = A \cos(\omega t)$.

Complex input: $y_{cx} = A e^{i \omega t}$

z_p = exponential system response to input y_{cx}

Complex gain: $z_p = H(\omega) y_{cx}$

Polar: $H(\omega) = |H(\omega)| e^{-i \phi}$

$$z_p = |H(\omega)| e^{-i \phi} A e^{i \omega t}$$

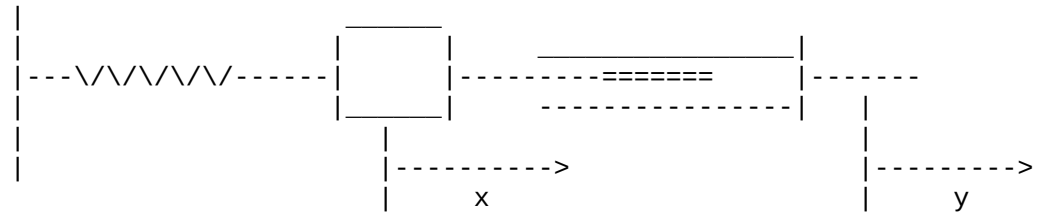
$$= |H(\omega)| A e^{i(\omega t - \phi)}$$

Real part: $x_p = |H(\omega)| A \cos(\omega t - \phi)$

Gain: g(omega) = |H(omega)|

Phase lag: phi(omega) = \phi = - Arg(H(omega))

[3] I want to go back to the spring/mass/dashpot system and think some more about what we can learn about the system response.



m x" + bx' + kx = by' (*)

Input signal: y = A cos(omega t)
System response: x

I demonstrated the Mathlet Amplitude and Phase: Second Order II.
It has m = 1 but we will work this out for general m .
It's interesting. What regularities can we see? Get responses from the class.

It seems that:

- When omega is small, g(omega) is small.

- The maximal amplitude of the system response is 1;
i.e. the maximal gain is 1. Write omega_r for that circular frequency;
it's the "resonant frequency."

- When omega is large, g(omega) is small.

and

- When omega is small, phi(omega) is about - pi/2 .

- phi(omega_r) = 0 (so the dashpot is "locked").

- When omega is large, phi(omega) is about + pi/2 .

Let's invoke the Bode plots: they support these observations. The hump is called the "resonant peak." As long as b is not zero, this hump does not go all the way to infinity; the maximal gain is finite.
I'll call the resonance we studied on Monday "true resonance."

From them you can see what happens when I fiddle with the system parameters.
Let's fix k but vary b . It seems that:

- omega_r is independent of b .

- If I make b small, the resonant peak gets sharper.

- and the phase lag switches from - pi/2 to + pi/2 more abruptly.

This equation is the SAME as the equation modeling an AM radio receiver. Think of the input signal as an incoming radio wave, and the system response as the induced current in the receiver. The environment is full of radio waves of different frequencies. But only waves of frequency near to omega_r excite the receiver. This is "tuning." The sharper the spike, the better the reception.

Analysis: $y = A \cos(\omega t)$

Complex replacement: $y_{cx} = A e^{i \omega t}$

$$m z'' + b z' + kz = b (y_{cx})' = b A i \omega e^{i \omega t}$$

Steady state solution: $z_p = b A i \omega e^{i \omega t} / p(i \omega)$

$$= (b i \omega / p(i \omega)) y_{cx}$$

$$H(\omega) = b i \omega / p(i \omega)$$

$$p(i \omega) = (k - m \omega^2) + b i \omega$$

$$g(\omega) = |H(\omega)| = b \omega / | p(i \omega) |$$

$$= b \omega / \sqrt{ (k - m \omega^2)^2 + b^2 \omega^2 }$$

When omega is small, this is small.

You can see that the denominator is always bigger than the numerator: the maximal gain is $g = 1$, and this happens just when the extra term in the denominator, $(k - m \omega^2)^2$, vanishes: that is,

$$k = m \omega_r^2$$

or $$\omega_r = \sqrt{k/m}$$

which is the natural frequency of the system! This is a surprise!

$$\omega_r = \omega_n$$

for this system. It is *independent of b* .

What is the phase lag at that frequency?

$$H(\omega_r) = b i \omega / b i \omega = 1$$

so the gain is 1 , as predicted, and the phase lag is 0 , as predicted.

All this is also clear from the Nyquist plot, which I invoked on the Mathlet. The Nyquist plot seems to be independent of the system parameters

(as it was in the first order case). It looks to be a circle of radius 1/2 centered at 1/2 . Almost exactly the same calculation we did before shows that this is true.

[Here it is: Write $x = b\,\omega$, $a = k - m\,\omega^2$

and compute $\left| (1/2) - ix/(a + ix) \right|$

$$= (1/2) \left| 1 - 2ix/(a + ix) \right|$$

$$= (1/2) \left| (a + ix - 2ix) / (a + ix) \right|$$

$$= (1/2) \left| a - ix \right| / \left| a + ix \right| = 1/2$$]

[4. Supplement] First order gain/phase lag analysis.

A bay on Cape Cod communicates with the ocean via a narrow channel. The ocean experiences tides. What happens to the water level in the bay?

Linear model for this: y = ocean water level
x = bay water level

both with respect to the same mark, so that $x = y$ means they are at the same level.

$x < y \implies x' > 0$
$x > y \implies x' < 0$

so the linear model is $x' = k(y-x)$ or $x' + kx = ky$

k is the coupling constant. This is just like Newtonian cooling.

Start with sinusoidal input: $y = A \cos(\omega t)$:

$$x' + ky = k A \cos(\omega t) .$$

I illustrated this using the Mathlet Amplitude and Phase: First Order.

Question: the cyan curve seems to cross the yellow curve right at the maxima and minima. Can you explain that?
Work with your neighbor for a minute and I'll take proposals.

Ans: the yellow curve is x . At extrema, $x' = 0$, which means that $ky = kx$, or $x = y$.

Complex replacement: $y_{cx} = A e^{i \omega t}$

$$z' + kz = k A e^{i \omega t}$$

Steady state solution: $z_p = k A e^{i \omega t} / p(i \omega)$

$$= (k / (i \omega + k)) A e^{i \omega t}$$

In terms of complex gain: $H(\omega) = k / (i \omega + k)$

$$g(\omega) = k / \sqrt{k^2 + \omega^2}$$

$$\tan(\phi(\omega)) = \omega / k$$

Also ϕ is between 0 and $\pi/2$, so

$$\phi(\omega) = \arctan(\omega/k)$$

So now we have these formulas: the amplitude of the steady state solution of

$$x' + ky = k A \cos(\omega t)$$

is $g(\omega) A = k A / \sqrt{k^2 + \omega^2}$

and we understand the phase lag as well.

This computation gives equations for the Bode plots in the Mathlet.
A Bode plot is just a graph of the gain or phase lag against the
input frequency.

The "Nyquist plot" is the trajectory of the complex gain as a complex-valued function of ω. In this case it looks like it sweeps out a semi-circle which is independent of k. Let's check this:

$$1/2 - H(\omega) = 1/2 - k/(i\omega + k) =$$

$$= ((i\omega + k) - 2k) / 2 (i\omega + k)$$

$$= (1/2) ((i\omega - k) / (i\omega + k)$$

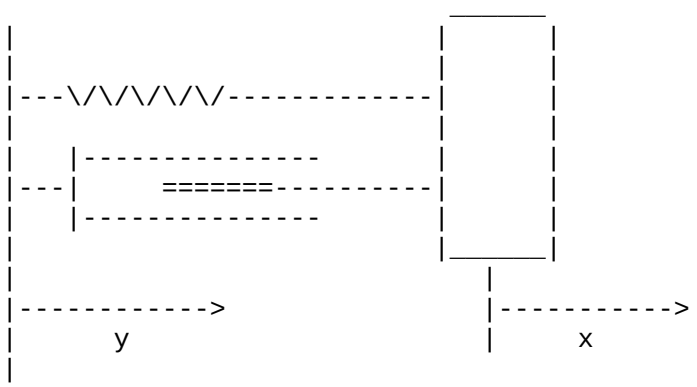
Since $|z/w| = |z|/|w|$ (because magnitudes multiply),

$$1/2 - H(\omega) = (1/2) | i\omega - k | / | i\omega + k |$$

But k is the same distance from $i\omega$ as $-k$ is : so this ratio is 1.

[5. Supplement] Some other systems:

Drive through both spring and dashpot:



Now m x" + b x' + k x = k y + b y'

-- both position and velocity occur on the right hand side.

Still Input signal = y
       Output   = x

This is illustrated by Amplitude and Phase: Second Order III.

This is a standard model of an automobile suspension.
Now the gain can be well larger than 1, Nyquist plot depends heavily
on the system parameters.

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18.03 Differential Equations

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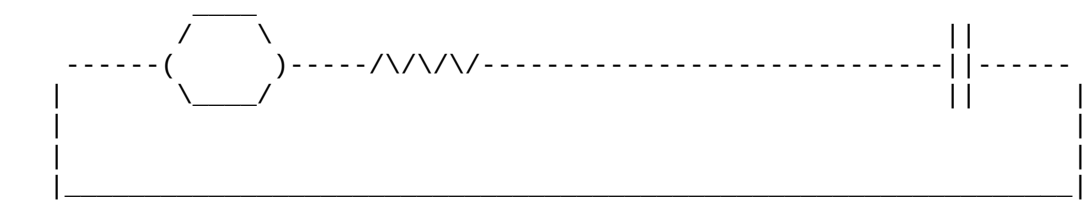
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18.03 Class 17, March 12, 2010

Linearity and time invariance

[1] RLC
[2] Superposition III
[3] Time invariance
[4] Review of solution methods

[1] We've spent a lot of time with mx" + bx' + cx = q(t) . There are many other systems modeled by this equation. For example here is a series RLC circuit:



I   = current; the same everywhere
V   = voltage increase across the power source
V_R = voltage drop across the resistor
V_C = voltage drop across the capacitor

Input signal: V
System response: I

KVL: V_R + V_C = V

Component behavior:

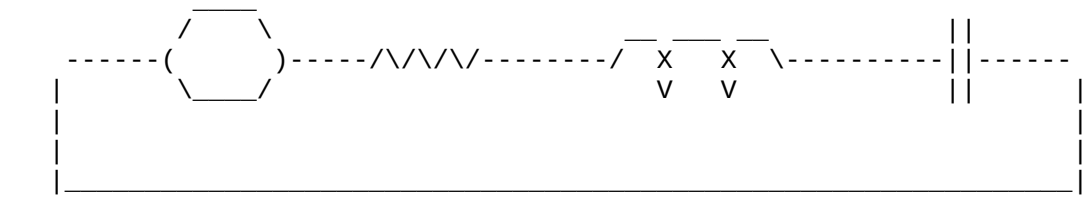
V_R = R I V_C' = (1/C) I

Differentiate KVL : V_R' + V_C' = V'

so R I' + (1/C) I = V'

In Lecture 3 we offered this as an example of a first order LTI system.

Now let's add another component, an inductor.



The voltage drop across an inductor depends not on the current but rather on the derivative of the current:

$$V_L = L\, I' \qquad \text{so} \quad V_L' = L\, I'' \ .$$

KVL now says $V_L + V_R + V_C = V$

so $$L\, I'' + R\, I' + (1/C)\, I = V' \qquad (*)$$

The system serves as an OPERATOR:

$$L\, D^2 + R\, D + (1/C)\, Id$$

takes the current I, as a function of time, and gives the derivative of the impressed voltage. (I have to use a different symbol for the identity operator here, so it doesn't get confused with current. I chose "Id.")

[2] Suppose you want to solve $u'' - 4u = \cosh(2t)$

Remember, I can write the left hand side as $p(D)\, u$ where $p(s) = s^2 - 4$.

First you have to know what $\cosh(t)$ is :

$$\cosh(x) = (e^x + e^{-x}) / 2$$
$$\sinh(x) = (e^x - e^{-x}) / 2$$

The right hand side is a linear combination of exponentials, each of which we know very well how to deal with.

Superposition III: $p(D) (c_1 x_1 + c_2 x_2) = c_1\, p(D)\, x_1 + c_2\, p(D)\, x_2$ [Slide]

Therefore, if $p(D)\, x_1 = q_1(t)$ and $p(D)\, x_2 = q_2(t)$, then

$x = c_1 x_1 + c_2 x_2$ solves $p(D)\, x = c_1\, q_1(t) + c_2\, q_2(t)$

Proof of Super III: [Slide]

$$p(D)\, x = a_n x^{(n)} + \ldots + a_1 x' + a_0 x$$

a_0] $x = c_1 x_1 + c_2 x_2$
a_1] $x' = c_1 x_1' + c_2 x_2'$
a_2] $x'' = c_1 x_1'' + c_2 x_2''$
. . .

$$p(D)\, x = c_1\, p(D)\, x_1 + c_2\, p(D)\, x_2$$

So we should solve separately: $u_1'' - 4u_1 = e^{2t}$, $u_2'' - 4u_2 = e^{-2t}$

Try to apply ERF: $p(s) = s^2 - 4$, $p(2) = 0$ and $p(-2) = 0$:
so we must use ERF/R : $p'(s) = 2s$, $p'(2) = 4$, $p'(-2) = -4$,

u_1 = te^{2t}/4 solves u_1" - 4u_1 = e^{2t}
u_2 = te^{-2t}/(-4) solves u_2" - 4u_2 = e^{-2t}

so u = (t/4) (e^{2t} - e^{-2t}) / 2 = (t/4) sinh(2t) is a solution.

Super III contains the earlier episodes.
With q_2 = 0 , this gives us "Superposition II,"
With q_1 = 0 and q_2 = 0 , we get" Superposition I."

The property Superposition III says that p(D) is a LINEAR OPERATOR.

[3] You may be worried that we always say that we are interested in sinusoidal input, but then we always consider A cos(omega t) , with no phase lag. Isn't that kind of restrictive?

Ans: No. Here's why.

First, if the right hand side is a sine, there is a special thing you can do:

Example: x" + 9x = sin(2t)

This is the IMAGINARY part of z" + 9z = e^{2t} : p(s) = s^2 + 9 , p(2i) = 9 - 4 , z_p = e^{2it} / (9-4) , so
x_p = Im (e^{2it}/(9-4)) = (1/(9-4) sin(2t)

This is worth remembering, by the way: if you drive a harmonic oscillator, there is no phase lag:

x" + omega_n^2 x = cos(omega t) : x_p = cos(omega t) / (omega_n^2 - omega^2)

x" + omega_n^2 x = sin(omega t) : x_p = sin(omega t) / (omega_n^2 - omega^2)

But there is a better and more general reason:
We are studying the response of a system which is not changing: the coefficients are constant. So if I start the signal a little bit later, all that happens is that the system response is correspondingly delayed.

If we shift the graph of a function x(t) to the right by a units, we get the graph of the function x(t-a) .

TIME INVARIANCE: If p(D) y(t) = q(t) , then p(D) y(t-a) = q(t-a)
[Slide]

To solve p(D) x = q(t-a) , first solve p(D) y = q(t) . Then x(t) = y(t-a).

It's very important that the coefficients are *constant* for this to work.

Example: x" + x' + 6x = cos(2t - pi/3)

Method: (1) Rewrite the right hand side as f(t-a):

cos(2t - pi/3) = cos(2(t - pi/6)) so f(t) = cos(2t) and a = pi/6 .

(2) Solve y" + y' + 6y = f(t) = cos(2t) ;

z" + z' + 6z = e^{2it} ,

p(s) = s^2 + s + 6 ,   p(2i) = -4 + 2i + 6 = 2 + 2i = 2sqrt(2) e^{i pi/4},

z_p = e^{2it} / 2(1+i) = (1/2sqrt(2)) e^{- i pi/4} e^{2it}
    = (1/2sqrt 2) e^{i(2t - pi/4)} ,

y_p = Re(z_p) = (1/2sqrt 2) cos(2t - pi/4)

(3) Then a solution to the original equation is

x(t) = y(t-a) = y_p(t - pi/6) = (1/2sqrt 2) cos(2(t - pi/6) - pi/4) .
                              = (1/2sqrt 2) cos((2t - pi/3) - pi/4) .

Be careful here: the (t - pi/6) goes in place of t , so gets multiplied by the circular frequency.

In any case: [Slide] The amplitude and the phase lag of the solution to

p(D) x = A cos(omega t - phi_0)

depend only on omega and A , and not on phi_0 .

If there is a system you are studying, it has a gain and a phase lag, functions of the circular frequency of the input signal. These are the same for any sinusoidal input with the given frequency. They don't depend on amplitude or phase of input.

[4] Brief summary of solution methods

We are looking at LTI operators. Solutions x_h of p(D) x = 0 are linear combinations of e^{rt} where r is a root of the characteristic polynomial: "modes." If r is a double root you have to add te^{rt} (and so on). If r = a+bi is a non-real root (and p(s) has real coefficients) the r-conjugate is also a root, and these two conjugate modes combine to give e^{at} cos(bt) and e^{at} sin(bt) .

If all roots have negative real part, then all these solutions decay to zero as t ---> infinity : they are transients.

To solve p(D) x = q(t) , find some solution x_p ; then the general solution is x_p + x_h : all solutions converge as t ---> infinity.

q(t) = e^{rt} : ERF or ERF/R

q(t) = cos : complex replacement reduces to exponential input signal.

q(t) = exp x cos : ditto.

q(t) = sinusoidal : use time invariance to shift to cosine.

q(t) = polynomial : undetermined coefficients, preceded by reduction of order if necessary

q(t) = (anything) x exponential : variation of parameters leads to a new differential equation in which the exponential has been eliminated from the right hand side.

q(t) = linear combination: Superposition III .

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18.03 Class 18, March 14, 2010

Applications in Engineering: A visit by Professor Kim Vandiver

[1] Damping ratio
[2] Measuring the damping ratio
[3] Cell phones on vibrate
[4] Extracting energy from a river

[1] HM: In this unit, we've been studying the equation controlling a spring system

$$m x'' + b x' + k x = F_{ext}$$

We began by thinking about the homogeneous case, the unforced or free system. If the system has any damping, then all these solutions die off; they are transients. This equation is surely too simple to be of any use in an engineering context, isn't it?

KV: I use this every day. This is the oscillator equation. If it makes noise, it vibrates.

How do you describe the solutions?

HM: We factored the characteristic polynomial $p(s) = ms^2 + bs + k$ to analyse this. Factor out the m and complete the square:

$$p(s) = m(s^2 + (b/m)\, s + (k/m)) = m((s + k/2m)^2 + (k/m - (b/2m)^2)$$

If $k/m > (b/2m)^2$ the roots are imaginary, the system is *underdamped*. Is that an engineering term too?

KV: Yep, and that's the only situation in which you get vibrations. So lets study that case today.

HM: OK. So the root in that case are $-b/2m \pm i\, \omega_d$

$$\omega_d = \sqrt{ k/m - (b/2m)^2 } .$$

So the general solution to this homogeneous equation is

x_p = A e^{-bt/2m} cos(omega_d t - phi)

Does that look familiar?

KV: Well it's familiar, but now I begin to realize why students look at me and say hunh? I use different notation.

When there's no damping you get the undamped, natural frequency

omega_n = sqrt(k/m)

Let's take omega_d and factor out the quantity omega_n :

omega_d = omega_n (1 - (b/(2 omega_n m))^2)

which I could write as

= omega_n (1 - zeta^2)

where zeta is called the *damping ratio*.

HM: Let's try to re-express the original equation in terms of this zeta . Where's your expression for zeta? Ah, it's there; zeta satisfies

b/(2 m) = omega_n zeta.

So plugging back into the differential equation gives

x" + 2 zeta omega_n x' + omega_n^2 x = 0 .

Now you can see that critical damping occurs when zeta = 1 , underdamped when zeta < 1 . And we can write the solutions in terms of zeta :

x_h = A e^{-omega_n zeta t} cos(omega_d t - phi)

By the way, what are the units of this zeta thing? I mean, omega_n has units 1/sec . What about zeta ?

KV: Well, I'd look at the exponent, which must be dimensionless. Since t is in seconds, omega_n t is already dimensionless, so zeta must be too.

HM: You mean if I switched from metric to feet and hours, the number

zeta wouldn't change?

KV: Yep, that's what it means.

[2] Let's try to make a measurement of the damping ratio here. We'll hang a spring, draw the neutral position, draw the spring back, let it go,

and see how long it takes to execute 10 cycles.

The stopwatch reads 14.2 sec . So the period, which I write tau and which is written P in this class, is 1.42 sec/cycle.

So 1/P = damped natural frequency in Hertz = cycles/sec , or about 0.7 Hz.

HM: We've always been talking about circular fequency.

KV: Yes: omega_d = 2 pi (1/P) which comes out to about 4.3 radians/sec.

HM: OK! So we've figured out omega_d by observation. Can we get the system constants omega_n and zeta ?

KV: As a vibration engineer, I have a quick and dirty formula:

$$\zeta \sim 0.11 / n_{50\%}$$

where $n_{50\%}$ is the number of cycles it takes for the amplitude to decay by 50%

HM: OK.... well let's measure that.

KV: Let's pull the weight down to here, and count how many cycles it takes till it comes down only half way. By the way, rubber bands are very nonlinear.

HM: Shhh.

KV: Looks like about 4 cycles.

HM: ... so zeta ~ .11/4 ~ .025 ...

KV: Yes. As engineers say, that's two and a half percent damping, 2.5% of critical damping.

HM: So where did this .11 come from? First lets graph this solution. First I'll draw the exponential decay and its negative: +- e^{-zeta omega_n t}.

This gets multiplied by a delayed cosine, cos(omega_d t - phi) .
I don't want to try to control what the delay is. The time gap between places where it becomes zero is half the period. But were measuring the bottoms. They don't occur quite where the damped oscillation touches the exponential - the slope isn't zero there. We have to be a little more careful. Think about what happens when I differentiate

$$x_p = A e^{-\zeta \omega_n t} \cos(\omega_d t - \phi)$$

using the product rule. You get two terms. Both will contain $e^{-bt/2m}$.
One will contain $\cos(\omega_d t - \phi)$, the other $\sin(\omega_d t - \phi)$.
But any linear combination of those two sinusoids is another sinusoid of the same circular frequency! so

$$x_p' = A' e^{-\zeta \omega_n t} \cos(\omega_d t - \phi')$$

The distance between places where this is zero is AGAIN half the period, so the time gap between successive minima is the period P .

When t increases by P , what happens to the amplitude $A e^{-\zeta \omega_n t}$?
Ans: it gets multiplied by

$$e^{-\zeta \omega_n P}$$

So after n cycles it gets multipled by $e^{-n \zeta \omega_n P}$. So:

$$1/2 = e^{- \zeta \omega_n P n_{50\%} }$$

Take natural logs:

$$\ln(2) = \zeta \omega_n P n_{50\%} .$$

$$P = 2\pi/\omega_d = 2\pi/(\omega_n \sqrt{1-\zeta^2})$$

When zeta is small, $1 - \zeta^2$ is very close to 1; ω_d is very close to ω_n , so, to good approximation, we can replace ω_d by ω_n.
Put this in:

$$\ln(2) \sim \zeta \omega_n (2\pi/\omega_n) n_{50\%} = 2\pi \zeta n_{50\%}$$

or $\zeta \sim (\ln(2)/2\pi) / n_{50\%}$

And now ln(2)/2pi = 0.1103178 ....

This is the mysterious .11 .

KV: This oscillator equation applies to many things. This string when it vibrates has many modes. The fundamental looks like half a sine wave, which is controlled by the same equation as before, and we can measure its damping ratio.

HM: So the string is controlled by a much more complicated equation, it has infinitely many degrees of freedom and is described by a PDE, but nevertheless its properties can be understood using the simple spring system we've been studying.

KV: So if I pull this back and count the number of cycles to half amplitude .... I count about n_50% ~ 5 ... so what's the damping? zeta ~ .11/5 ~ .02 .

HM: Amazing - so this spring system and this string have something in common - they both have damping of 2% or so.

[3] Now, how about forced vibrations?

KV: How does a cell phone vibrate? Inside there's a little motor, and on the shaft there's an off-center mass. Here's a DC motor with a mass on the shaft. You can see it in the document camera.

A "squiggle pen" works the same way.

Here is a beam, a steel ruler. It has its own natrual frequency, which is determined by its length - it goes like one over length squared - so when you make the length just right the natural frequency would be the same as the frequency of this squigglepen which is attached to the end of it. We can see this with the help of a strobe light. So dim the lights, and I'll keep changing the length of the beam ... till it appears to be permanently bent. You can see the weight, stop action, too.

Now, what if you didn't have a strobe handy, but you wanted to measure the frequency of the squiggle pen.

You can measure it with the ruler, if you can predict the natural frequency of the ruler. There's a formuala for this in terms of the shape of the beam and property of the steel it's made of. It turns out that

omega_n is proportional to 1/L^2 where L is the length. For this beam,

omega_n = 3.706/L^2 .

HM: So it's a ruler.

KV: Yes, it's a frequency ruler. In the old days they had these things marked out in Hertz.

Checking the squiggle pen again, with the strobe, we find resonance - the ruler looks very bent.

HM: Can you bend spoons too?

KV: When you measure the length, you find a frequency.

HM: So if you had the strobe going half as fast, it would look just the same, would't it?

KV: Well you should double the rate of the strobe: you should then see two fixed images.

HM: We've talked a lot about what happens to a fixed system when you vary the frequency of the input signal: you get one of these amplitude response curves. Here what's happening is a little bit different: the input signal (from the squiggle pen) is fixed, and you are varying the system parameters (the length of the ruler).

But still, when the two frequencies get close you see near resonance.

Thank you, Professor Vandiveer. Your research involves vibration, doesn't it?

Yes. I study structures in the ocean. Imagine you have a cylinder (3 feet long, D = 3 inches in diameter), mounted on springs so it can move up and down. Water is flowing past the cylinder with velocity U . In the wake of the cylinder vortices are shed, eddies. The frequency of the vortex shedding is approximately

$$f_S = .2\, U/D$$

If the cylinder were an oil well pipe, D would be about 1 meter;

1 meter per second is a pretty normal speed of water moving in the ocean; so then you expect to see oscillations with frequency .2 Hz.

The vortex shedding causes a force on the cyclinder transverse to the direction of the water flow.

Using this one can extract power from a river. Here is a film that shows a way to do it. Instead of a dashpot, you have a generator sucking energy out of the system.

HM: You wouldn't want to be under one of those things, they look like they have a lot of power.

KV: Yeah, it looks like you could chop carrots with them.

HM: I was thinking of stamping wine ...

Thank you very much, Professor Vandiver.

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18.03 Differential Equations

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18.03 Class 20, March 19, 2010

Periodic signals, Fourier series

[1] Periodic functions
[2] Sines and cosines
[3] Parity
[4] Integrals

[1] Periodic functions: for example the heartbeat, or the sound of a violin, or innumerable electronic signals. I showed an example of somewhat simplified waveforms of a violin and a flute.

A function $f(t)$ is "periodic" if there is $L > 0$ such that $f(t+2L) = f(t)$ for every t. $2L$ is a "period."

So strictly speaking the examples given are not periodic, but rather they coincide with periodic functions for some period of time. Our methods will accept this approximation, and yield results which merely approxmimate real life behavior, as usual.

The constant function is periodic of every period. Otherwise, all the periodic functions we'll encounter have a minimal period, which is often called THE period.

Any "window" (interval) of length $2L$ determines the function. You can choose the window as convenient. We'll often use the window $[-L,L]$. (This is one reason why chose to write the period as $2L$.)

[2] Sine and cosines are basic periodic functions. For this reason a natural period to start with is $2L = 2\pi$: $L = \pi$.

We'll use the basic window $[-\pi,\pi]$.

Question: what other cosines have period 2π ?

1. cos (t), cos(t/2), cos(t/3), ...

2. cos(pi t), cos(2 pi t), ...

3. cos(t), cos(2t), cos(3t), ...

Answer: $\cos(nt)$ for $n = 1, 2, 3, \ldots$ has minimal period $2\pi/n$.

I sketched graphs of $\cos(t)$, $\cos(2t)$, and $\cos(3t)$.

Among the sines, we have $\sin(t), \sin(2t), \ldots$.

These are "harmonics" of the "fundamental" sinusoids with $n = 1$.

If $f(t)$ and $g(t)$ are periodic of period $2L$ then so is $af(t) + bg(t)$.

So we can form linear combinations:

```
f(t) = a0/2 + a1 cos(t) + a2 cos(2t) +... + b1 sin(t) + b2 sin(2t) +... (*)
       \                                /  \                          /
         ------- cosine series ---------      ----- sine series ------
```

This is a "Fourier Series." The an and bn are the "Fourier Coefficients."

We'll see why the odd choice of a0/2 for the constant term shortly.

Theorem. Any reasonable [piecewise continuous] function of period 2pi has exactly one expression as a Fourier series.

The *definition* of the Fourier coefficients of a function f(t) is this: they are the coefficients that make (*) true.

I set up the Mathlet FourierCoefficients and showed multiples of cos(2t) , and then added multiples of cos(t) - you can see the addition of functions.

For example, the standard squarewave is periodic of period 2pi and

```
Sq(t)  =   1  for    0 < t < pi
       =  -1  for  -pi < t < 0
       =   0  for    t = 0 , pi
```

The applet actually shows (pi/4) Sq(t) . I tried to approximate it using cosines; but they are even and Sq is odd, so it doesn't look so good. Working with sines, it seems that the even ones are not useful - they break the symmetry round pi/2 exhibited by Sq(t) . I got pretty good results with b_1 = 1.00 , b_3 = .33 , b_5 = .20 .

[3] Parity. A function f(t) is "even" if f(-t) = f(t) , odd if f(-t) = -f(t).

Even + even is even, Odd + odd is odd; Even + Odd can be anything.

Even x even is even. 5 times 7 is odd, so I ask:

Q2. The product of two odd functions is
1. Even
2. Odd
3. Could be either or neither

Well, if f(t) and g(t) are odd then

$$f(-t)g(-t) = (-f(t))(-g(t)) = f(t)g(t)$$

so the product is even, in contrast to products of odd numbers. Also even times odd is odd.

Linear combinations of evens are even, of odds are odd.

cos(nt) is even, sin(nt) is odd.

The only function which is both even and odd is the zero function. For f(t) = f(-t) and f(t) = -f(-t) together imply that f(t) = -f(t) .

If a periodic function f(t) is even, then f(t) - (cosine series) is a linear combination of evens and hence even, but it's also (sine series) and so odd, so it's zero, so:

The Fourier series of an even function is a cosine series: bn = 0 .

The Fourier series of an odd function is a sine series: an = 0

[The same argument shows that if a polynomial is even then it's a sum of even powers of t ; if it's odd then it's a sum of odd powers of t .]

[4] Average. The average of a function of period 2pi is

Ave(f) = (1/2pi) integral_{-pi}^pi f(t) dt

Ave(f(t)+g(t)) = Ave(f(t)) + Ave(g(t))

Ave(cos(nt)) = 0 for n > 0 and Ave(sin(nt)) = 0 ,

so applying Ave to (*) :

Ave(f(t)) = a0/2 or

a0 = (1/pi) integral_{-pi}^pi f(t) dt.

Other integral expressions: This will use the trigonometric integrals [Slide]

integral_{-pi}^pi cos(mt) sin(nt) dt = 0

integral_{-pi}^pi cos(mt) cos(nt) dt = 2pi if m = n = 0
= pi if m = n > 0
= 0 if m is not equal to n

integral_{-pi}^pi sin(mt) sin(nt) dt = pi if m = n
= 0 if m is not equal to n

The first of these is easy, since the product is odd and the interval you are integrating over is symmetric. The others require some trig identity which you can find in Edwards and Penney.

Application: Substitute (*) into integral_{-pi}^pi f(t) cos(nt) dt (for n > 0)

Compute this integral term by term:

integral_{-pi}^pi (a0/2) cos(nt) dt = 0 (since n > 0)

Then we have a bunch of cosines. The m-th one gives:

integral_{-pi}^pi am cos(mt) cos(nt) dt = am pi if m = n
                                        =   0   if m is not equal to n

And then a bunch of sines. The m-th of them gives:

integral_{-pi}^pi am sin(mt) cos(nt) dt = 0

Only one of all these terms is nonzero: the cosine term with m = n ,
and since then am = an , we discover

integral_{-pi}^pi f(t) cos(nt) dt = am pi ,   or

an = (1/pi) integral_{-pi}^pi f(t) cos(nt) dt

We did this calculation assuming n > 0 , but since cos(0t) = 1
the formula is true for n = 0 (by our comment about averages above).

Exactly the same method shows:

bn = (1/pi) integral_{-pi}^pi f(t) sin(nt) dt

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18.03 Class 21, March 29

Fourier series II

[1] Review
[2] Square wave
[3] Piecewise continuity
[4] Tricks

[1] Recall from before break: A function $f(t)$ is periodic of period $2L$ if $f(t+2L) = f(t)$.

Theorem: Any decent periodic function $f(t)$ of period $2pi$ has can be written in exactly one way as a *Fourier series*:

$$f(t) = a_0/2 + a_1 \cos(t) + a_2 \cos(2t) + \ldots + b_1 \sin(t) + b_2 \sin(2t) + \ldots$$

If the need arises, the "Fourier coefficients" can be computed as integrals:

$$a_n = (1/pi) \int_{-pi}^{pi} f(t) \cos(nt)\, dt , \quad n \geq 0$$
$$b_n = (1/pi) \int_{-pi}^{pi} f(t) \sin(nt)\, dt , \quad n > 0$$

[2] Squarewave: A basic example is given by the "standard squarewave," which I denote by $sq(t)$: it has period $2pi$ and

$$sq(t) = 1 \text{ for } 0 < t < pi$$
$$= -1 \text{ for } -pi < t < 0$$
$$= 0 \text{ for } t = 0 , t = pi$$

This is a standard building block for all sorts of "on/off" periodic signals.

It's odd, so $a_n = \int_{-¥pi}^{pi}$ odd . even $dt = 0$ for all n .

If $f(t)$ is an odd function of period $2pi$, we can simplify the integral for bn a little bit. The integrand $f(t) \sin(nt)$ is even, so the integral is twice the integral from 0 to pi:

$$bn = (2/pi) \int_0^{pi} f(t) \sin(nt)\, dt$$

Similarly, if $f(t)$ is even then

$$an = (2/pi) \int_0^{pi} f(t) \cos(nt)\, dt$$

In our case this is particularly convenient, since $sq(t)$ itself needs different definitions depending on the sign of t. We have:

$$bn = (2/pi) \int_0^{pi} \sin(nt)\, dt$$

$$= (2/pi) \left[- \cos(nt) / n \right]_0^{pi}$$

$$= (2/pi\, n) \left[- \cos(n\, pi) - (-1) \right]$$

= (2/pi n) [1 - cos(n pi)]

This depends upon n :

n	cos(n pi)	1 - cos(n pi)
1	-1	2
2	1	0
3	-1	2

and so on. Thus: bn = 0 for n even
= 4pi/n for n odd and

sq(t) = (4/pi) [sin(t) + (1/3) sin(3t) + (1/5) sin(5t) + ...]

This is the Fourier series for the standard squarewave.

I used the Mathlet FourierCoefficients to illustrate this. Actually, I built up the function

(pi/4) sq(t) = sin(t) + (1/3) sin(3t) + (1/5) sin(5t) + ... (**)

and observed the fit.

[3] What is "decent"?

This is quite amazing: the entire function is recovered from a *discrete* sequence of slider settings. They record the strength of the harmonics above the fundamental tone. The sequence of Fourier coefficients is a "transform" of the function, one which only applies (in this form at least) to periodic functions. We'll see another example of a transform later, the Laplace transform.

Let's be more precise about decency. First, a function is *piecewise continuous* if it is broken into continuous segments and such that at each point t = a of discontinuity,

f(a-) = lim_{t ---> a from below} f(t) and

f(a+) = lim_{t ---> a from above} f(t)

exist. They exist at points t = a where f(t) is continuous, too, and there they are equal. So f(t) = 1/t is NOT piecewise continuous, but sq(t) is .

A function is "decent" if it is piecewise continuous and is such that at each point of discontinuity, t = a , the value at a is the average of the left and right limits:

f(a) = (1/2) (f(a+) + f(a-))

So the square wave is decent, and any continuous function is decent.

Addendum to the theorem:

At points of discontinuity, the Fourier series can't make up its mind, so it converges to the average of f(a+) and f(a-) .

For example, evaluate the Fourier series for sq(t) at t = pi/2:

sin(pi/2) = +1
sin(3pi/2) = -1
sin(5pi/2) = +1
... so

1 = (4/pi) (1 - 1/3 + 1/5 - 1/7 + ...) or

1 - 1/3 + 1/5 - 1/7 + ... = pi/4

Did you know this? It's due to Newton and Leibnitz.

[4] Tricks: Any way to get an expression (*) will give the same answer!

Example [trig id]: cos(t - pi/4) .

How to write it like (*) ? Well, there's a trig identity we can use:

a cos(t) + b sin(t) = A cos(t - phi)
if (a,b) has polar coord's (A,phi)

a = A cos(phi), b = A sin(phi) :

For us, A = 1, phi = pi/4, so a = b = 1/sqrt(2) and

cos(t - pi/4) = (1/sqrt(2)) cos(t) + (1/sqrt(2) sin(t) .

That's it: that's the Fourier series. This means a1 = b1 = sqrt(2) and all the others are zero.

Example [linear combinations]:

1 + 2 sq(t) = 1 + (8/pi) (sin(t) + (1/3) sin(3t) + ...)

Example [shifts]: f(t) = sq(t + pi/2)

= (1/2) (4/pi) (sin(t + pi/2) + (1/3) sin(3(t + pi/2)) + ...)

sin(theta + pi/2) = cos(theta) , sin(theta - pi/2) = - cos(theta) so

f(t) = (4/pi) (cos(t) - (1/3) cos(3t) + (1/5) cos(5t) - ...)

Example [Stretching]: What about functions of other periods? Suppose g(x) has period 2L .

Building blocks: cos(n(pi/L)x)
and sin(n(pi/L)x) are periodic of period 2L .

Then the Fourier series for g(x) is:

$$g(x) = a_0/2 + a_1 \cos((\pi/L)\, x) + a_2 \cos((2\pi/L)\, x) + \ldots + b_2 \sin((\pi/L)\, x) + b_2 \sin((2\pi/L)\, x) + \ldots$$

Example: sq((pi/2) x) has period 4 , not 2pi : L = 2. But we can still write (using the *substitution* t = (pi/2) x) :

$$sq(2\pi x) = (4/\pi)\,(\sin((\pi/2)x) + (1/3) \sin(3(\pi/2)\, x) + \ldots)$$

There are integral formulas as well. [Slide]

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18.03 Class 22, March 31, 2010

Fourier Series III

[1] Differentiation and integration
[2] Harmonic oscillator with periodic input
[3] What you hear

[1] You can differentiate and integrate Fourier series.

Example: Consider the function f(t) which is periodic of period 2pi and is given by f(t) = |t| between -pi and pi.

We could calculate the coefficients, using the fact that f(t) is even and integration by parts. For a start, a0/2 is the average value, which is pi/2.

Or we could realize that

f'(t) = sq(t) (except where f'(t) doesn't exist)

or what is the same

f(t) = integral_0^t sq(u) du

and integrate the Fourier series of the squarewave.

NB: it is not true in general that the integral of a periodic function is periodic; think of integrating the constant function 1 for example. But the integral IS periodic if the average value of the function is zero. If you think of this one term at a time, the point is that the integral of cos(nt) is periodic unless n = 0 and the integral of sin(nt) is always periodic.

Let's compute:

f(t) = integral_0^t (4/pi) sum_{n odd} sin(nx)/n dx

= (4/pi) sum_{n odd} integral_0^t sin(nx)/n dx

= (4/pi) sum_{n odd} [- cos(nx) / n^2]_0^t

= (4/pi) sum_{n odd} (1/n^2) - (4/pi) sum_{n odd} cos(nt) / n^2

That's it, that's the Fourier series for f(t). The constant term is a little odd. It's a specific number, but not a sum you can find by the geometric series or by telescoping. In fact the only way to evaluate it is this way, using Fourier series. Because we know that the constant term in the Fourier series for f(t) is the average value of f(t), which is pi/2:

(4/pi) sum_{n odd} (1/n^2) = pi/2 or

sum_{n odd} (1/n^2) = pi^2 / 8 .

That is,

(odd)^2: 1 + 1/3^2 + 1/5^2 + 1/7^2 + ... = pi^2 / 8.

Just to carry this one step further: Try to sum all the reciprocal squares.

(2 x odd)^2: 1/2^2 + 1/6^2 + 1/(10)^2 + ... = (1/4) pi^2 / 8

(4 x odd)^2: 1/4^2 + 1/(12)^2 + 1/(20)^2 + ... = (1/4)^2 pi^2 / 8

...

so sum 1/n^2 = (1 + 1/4 + (1/4)^2 + (1/4)^3 + ...) pi^2 / 8

The first factor is a geometric series:

1 + 1/4 + (1/4)^2 + ... = 1/(1 - (1/4) = 4/3 so

1 + 11/2^2 + 1/3^2 + 1/4^2 + ... = pi^2/6

This is one of the most famous equations in all of mathematics.
It made Euler's reputation when he discovered it in 1736.

So we learn two things from this calculation: this interesting mathematical formula, and the calculation of the Fourier series for |t| extended periodically:

f(t) = pi/2 - (4/pi) sum_{n odd} cos(nt) / n^2 .

[2] Harmonic oscillator with periodic forcing.

Now we come to the relationship with differential equations:

We have a complicated wave, perhaps a square wave, f(t) .
We drive a harmonic oscillator with it:

x" + omega_n^2 x = f(t)

What is the system response? We might imagine the system as a radio tuner; f(t) represents the radio wave, and x represents the output of the receiver.

Remember [Slide]: x" + omega_n^2 x = cos(omega t)

has sinusoidal solution x_p = A cos(omega t)/(omega_n^2 - omega^2)

and x" + omega_n^2 x = sin(omega t)

has sinusoidal solution x_p = A sin(omega t)/(omega_n^2 - omega^2)

When the denominator vanishes we have resonance and no periodic solution.

Well, by Superposition III we can now handle ANY periodic input signal.
For example, suppose

f(t) = sq(t) = (4/pi) (sin(t) + (1/3) sin(3t) + ...)

Then we will have a particular solution

$$x_p = (4/\pi)\,(\sin(t)/(\omega_n^2 - 1) + \sin(3t)/(\omega_n^2 - 9) + \ldots)$$

I showed the Harmonic Frequency Response applet. This applet actually shows the system response of a spring system driven through the spring, so it is

$$x'' + \omega_n^2 = \omega_n^2 f(t)$$

so when $f(t) = \sin(t)$, $x_n = \omega_n^2 \sin(t) / (\omega_n^2 - 1)$

with amplitude $\omega_n^2 / (\omega_n^2 - 1)$. This is the "RMS" graph.

When $f(t) = sq(t)$,

$$x_p = \omega_n^2 (4/\pi)\,(\sin(t)/(\omega_n^2 - 1) + \sin(3t)/(\omega_n^2 - 3^2) + \ldots)$$

There is resonance when

$$\omega_n = 1, 3, 5, \ldots$$

but NOT when $\omega_n = 2, 4, 6, \ldots$

When ω_n is very near to but less than k^2, k odd, the term

$$\sin(kt)/(\omega_n^2 - k^2)$$

is a large negative multiple of $\sin(kt)$. This appears on the applet.

Then when ω_n passes k^2 the dominant term flips sign and becomes a large positive multiple of $\sin(kt)$.

[3] Let's write $\omega = \pi/L$ for the fundamental circular frequency of the periodic function $f(t)$. The Fourier series

$$f(t) = a0/2 + a1 \cos(\omega t) + a2 \cos(2\omega t) + \ldots + b1 \sin(\omega t) + b2 \sin(2\omega t) + \ldots$$

can be rewritten in polar form as

$$f(t) = A0 + A1 \cos(\omega t - \phi 1) + A2 \cos(2\omega t - \phi 2) + \ldots$$

If you think of this as the pressure variation at your eardrum, the $A0$ is atmospheric pressure. What you hear is the rest.

I showed the Fourier Coefficients: Complex with Sound Mathlet.

Notice how dramatically the phase alters the waveform.

It turns out that your ear hears only the amplitudes of the various fourier components, or harmonics, not their relative phases. Just listen ....

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18.03 Class 23, April 2, 2010

Step and delta

[1] Step function $u(t)$
[2] Rates and delta(t)
[3] Regular, singular, and generalized functions
[4] Generalized derivative
[5] Heaviside and Dirac

Two additions to your mathematical modeling toolkit.
- Step functions [Heaviside]
- Delta functions [Dirac]

[1] Model of on/off process: a light turns on; first it is dark, then it is light. The basic model is the Heaviside unit step function

$$u(t) = \begin{cases} 0 & \text{for } t < 0 \\ 1 & \text{for } t > 0 \end{cases}$$

Of course a light doesn't reach its steady state instantaneously; it takes a small amount of time. If we use a finer time scale, you can see what happens. It might move up smoothly; it might overshoot; it might move up in fits and starts as different elements come on line. At the longer time scale, we don't care about these details. Modeling the process by $u(t)$ lets us just ignore those details. We simplify the model by supposing that $u(t) = 0$ for all $t < 0$ no matter how near to zero,
$u(t) = 1$ for all $t > 9$ no matter how near to zero,
and $u(0)$ is left undefined.

$u(t-a)$ turns on at $t = a$.

If $a < b$, $u(t-a) - u(t-b)$ turns on at $t = a$ and off again at $t = b$: it's a "window."

We can use $u(t-a)$ to turn on another function:
$u(t-a)\, f(t)$ is zero when $t < a$ and agrees with $f(t)$ when $t > a$.

Q1: What is the equation for the function which agrees with $f(t)$ between a and b ($a < b$) and is zero outside this window?

(1) $(u(t-b) - u(t-a))\, f(t)$
(2) $(u(t-a) - u(t-b))\, f(t-a)$
(3) $(u(t-a) - u(t-b))\, f(t)$
(4) $u(t-a)\, f(t-a) - u(t-b)\, f(t-b)$
(5) none of these

Ans: (3).

[2] From bank accounts to delta functions.

Bank account equation: $x' + Ix = q(t)$
$x = x(t)$ = balance (K$)
I = interest rate ($(yr)^{-1}$)
$q(t)$ = rate of savings (K$/yr)

I am happily saving at the rate K$1/yr. The concept of rate can be clarified] by thinking about the cumulative total, $Q(t)$ (from some starting time, perhaps $t = 0$);

$$Q'(t) = q(t)$$

or $Q(t) = \int_0^t q(u)\, du$

At $t = 1$ I won $2K at the race track! I deposit this into the account. I can model the cumulative total deposit using the step function:

$$Q(t) = t + 2\, u(t-1)$$

What about the rate? For this we would need to be able to talk about the derivative of $u(t)$, in such a way that its integral recovers $u(t)$.

Of course there is no such function.

But let's approximate $u(t)$ as we did before. I can differentiate each of them, to form a rate (varying with time).

Once again, I don't care about the details. What matters is the two properties:

- The integral is always 1
- Concentrated very near to zero.

These functions approximate *something*, written $\delta(t)$; and

$$\delta(t) = u'(t)$$

Using this we can write down a formula for the new rate of savings:

$$q(t) = 1 + 2\, \delta(t-1)$$

We can graph this using a "harpoon" at $t = 1$ with the number 2 next to it; the area under the harpoon is 2.

Not too long after this, at $t = 2$, I bought a camera for $3K.

$$q(t) = 1 + 2\, \delta(t-1) - 3\, \delta(t-2) .$$

The negative multiple of delta can be represented using a harpoon pointing up with -3 next to it, or by a harpoon pointing down with $+3$ next to it.

[3] Let me go back and be more precise about the kind of functions we have been using before these delta things.

A function is "piecewise smooth" if it is broken up into segments so that:
- each segment has all higher derivatives
- at each endpoint, left and right limits of all derivatives exist.

We'll call piecewise smooth functions "regular." We can now
add in combinations of delta functions, called "singularity functions."
A sum of a regular function and a linear combination of delta functions
is a "generalized function":

$$f(t) = f_r(t) + f_s(t)$$

For example, the $q(t)$ above has

$$q_r(t) = u(t)$$

$$q_s(t) = 2\,\delta(t-1) - 3\,\delta(t-2)$$

I can differentiate a regular function, but if there are breaks in
the graph the result will be a generalized function:
Whenever $Q(a+)$ is different from $Q(a-)$,
the derivative will have a delta contribution:

$$(Q(a+) - Q(a-))\,\delta(t-a)$$

Keeping these terms in the derivative lets us reconstruct $f(t)$ up to a
constant. With the singular terms in place this is called the
"generalized derivative."

****
*
* In this Unit whenever we differentiate *
* a discontinuous function we will mean   *
* the generalized derivative.   *
*
****

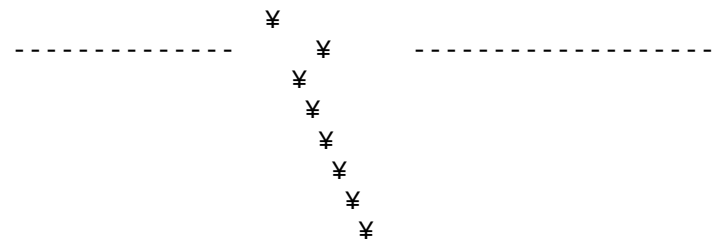
[4] When you fire a gun, you exert a very large force on the bullet
over a very short period of time. If we integrate $F = ma = mx''$
we see that a large force over a short time creates a sudden change in
the momentum, mx'. This is called an "impulse."

I fire straight up.
The graph of the elevation of the bullet, plotted against t,
starts at zero, then abruptly rises in an inverted parabola, and then
when it hits the ground it stops again.

The derivative is zero for $t < 0$; then it rises abruptly to v_0;
then it falls at constant slope (the acceleration of gravity) till
the instant when it hits the ground, when it returns abruptly to zero.

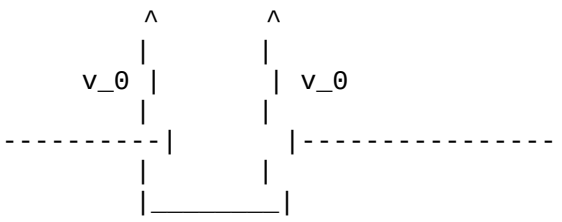
$v(t)$ has a graph like this:



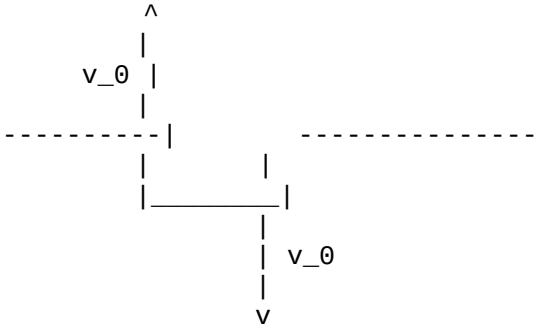


Q2: What does the graph of the generalized derivative of v(t) look like?

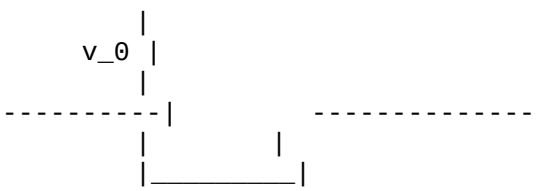
(1)



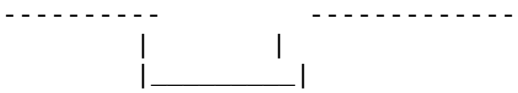
(2)



(3)



(4)



Ans: (1).

[5] Oliver Heaviside, 1850--1925, British mathematical engineer
``... whose profound researches into electro-magnetic waves
have penetrated further than anyone yet understands.''

He was the one who wrote down Maxwell's equations in the compact vector
form you see now on ``Let there be light'' T-shirts.]

Paul A. M. Dirac, 1902--1984, Swiss/British theoretical physicist.
Nobel prize 1933, for the relativistic theory of the electron.

Lucasian Chair, Cambridge University:

Isaac Newton, 1669
...
P.A.M. Dirac, 1932--1969
...
Stephen Hawking, 1980

Quotes:

``I consider that I understand an equation when I can predict
the properties of its solutions without actually solving it.''
(Quoted in Frank Wilczek and Betsy Devine, "Longing for the Harmonies")

``God used beautiful mathematics in creating the world.'']

[6: Supplement] People often want to know what the delta function REALLY IS. One answer is that it is a symbol, representing a certain approximation to reality and obeying certain rules.

There are other answers.

One is this: one measures the value of a function by means of a piece of equipment of some sort. This equipment gathers light, for example, over a period of time, and reports an integrated value. The time interval may be short but it is not of width zero. Each measuring device has a sensitivity profile, $m(t)$, which rises to a peak and then falls again, and which mathematically is assumed to be smooth.

If the light profile is $f(t)$, what this instrument actually measures is

$$M(f;m) = \text{integral } f(t)\, m(t)\, dt$$

The most we can ever know about the function $f(t)$ is the collection of all these measurements, $M(f;m)$ as m varies over all measuring devices.

So f determines a new "function," sending each m to a number. This is what is called a "distribution." Notice that changing the value of $f(t)$ at a single point doesn't change $M(f;m)$: the distribution determined by a function is independent of the value of the function at any single point.

There are other ways to assign a number to each measuring device. For example, we can send m to $m(0)$. That is what the "delta function" does:

$$\text{integral } \delta(t)\, m(t)\, dt = m(0)$$

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18.03 Class 24, April 5, 2010

Unit impulse and step responses

1. Generalized derivative
2. Rest initial conditions
3. First order unit step/impulse response
4. Second order unit step/impulse response

[1] Generalized derivative

The unit step and delta functions help deal with events happening on a time scale which is very short relative to our interest.

u(t) can be thought of as a function which, except for t very near zero, is given by u(t) = 0 for t < 0 and u(t) = 1 for t > 0 .

delta(t) can be thought of as a function which is zero except for t very near zero, and has area under its graph equal to 1 .

u'(t) = delta(t) .

A generalized function is by definition a sum g(t) = g_r(t) + g_s(t) , where its *regular part* g_r(t) is piecewise smooth, and its *singular part* g_s(t) is a linear combination of shifted delta functions.

Any regular f(t) function has a "generalized derivative" f'(t) which is a generalized function: f'(t) = f'_r(t) + f'_s(t) . The regular part is the ordinary derivative of f(t) (except at the break points, where it is undefined). The singular part is a sum of delta functions, one for each break in the graph:

(f(a+)-f(a-)) delta(t-a)

There's no separate notation for the generalized derivative to distinguish it from the ordinary derivative, and we will just write f'(t) or dot-x (t).

For example, if f(t) = t + 2 u(t) ,
f'(t) = 1 + 2 delta(t)
f'_r(t) = 1
f'_s(t) = 2 delta(t)

Conversely, if g(t) is a generalized function then

f(t) = int_a^t g(t) dt

makes sense: take the cumulative total of the regular part by integrating as usual, and add in whatever contributions the delta functions make.

Then FTC is still true: f(b) - f(a) = int_a^b f'(t) dt

[2] Let's solve some differential equations with discontinuous input signals. In this part of the course we will make the following standing assumptions (which are covered by the general rubric "rest initial conditions":

(1) the input signal is zero for $t < 0$
(2) the desired system response is zero for $t < 0$
(3) the solution has as many derivatives as possible

[3] First order

Ex 1. $x' + kx = r\,u(t)$.

Interpretation: I add uranium to the reactor at a constant rate of r kg/year, say, starting at $t = 0$.

Look for a continuous solution. Thus $x(0) = 0$, and for $t > 0$ we must solve

$$x' + kx = r \quad .$$

The general solution is $x(t) = (r/k) + ce^{-kt}$. To find c , we use $x(0) = 0$: $0 = x(0) = (r/k) + c$, $c = -r/k$,

$$x(t) = (r/k)\,(\,1 - e^{-kt}\,) \qquad t > 0 \ .$$

With $r = 1$, this is the *unit step response,* sometimes written $v(t)$. To be more precise, we could write

$$v(t) = u(t)\,(1/k)\,(\,1 - e^{-kt}\,)$$

Question 24.1. For the solution to $v' + kv = u(t)$ with rest initial conditions, $v(0-) = 0$. What is $v'(0+)$?

1. $v'(0+) = 0$
2. $v'(0+) = 1/k$
3. $v'(0+) = 1$
4. $v'(0+) = k$
5. Don't know

Ans: (3): $v(0) = 0$, so $v'(0+) = u(0+) = 1$.

Ex 2. $x' + kx = \mathrm{delta}(t)$.

Interpretation: I insert one kilogram of uranium into the reactor at $t = 0$ and leave it alone thereafter. The corresponding rate is given by the delta function. The solution can't be continuous; $x(0-) = 0$ but $x(0+) = 1$. Thereafter, though, the equation is

$$x' + kx = 0$$

--- homogeneous, with solution e^{-kt} , $t > 0$.

This is the *unit impulse response*, which we will write $w(t)$. In some sense it is the simplest nontrivial solution; you just give the system a unit kick at $t = 0$, stand back, and watch the result. Since we began with rest initial conditions, the full solution

is

$$w(t) = u(t)\, e^{-kt}$$

Question 24.2. For solution $w(t)$ to $x' + kx = \delta(t)$ with rest initial conditions, what is $w'(0+)$?

1. $w'(0+) = 0$
2. $w'(0+) = -1/k$
3. $w'(0+) = -1$
4. $w'(0+) = -k$
5. Don't know

Ans: (4) : $w(0+) = 1$, while $\delta(0+) = 0$, so $w'(0+) = -k$.

[4] Second order

Ex 3. $mx'' + kx = u(t)$

Interpretation: at $t = 0$ a steady force starts to act on the mass. The mass does not change position abruptly, nor does it change velocity instantaneously. Therefore we should expect a solution which is continuous with continuous derivative; only the acceleration experiences a discontinuity.

There is a constant solution: $x = 1/k$. So the general solution is

$$x(t) = (1/k) + a \cos(\omega_n t) + b \sin(\omega_n t) .$$

where $\omega_n = \sqrt{k/m}$

$$0 = x(0) = (1/k) + a$$

$$0 = x'(0) = - a \omega_n \sin(\omega_n t) + b \omega_n \cos(\omega_n t)$$

so $a = -1/k$, $b = 0$,

$$v(t) = u(t)\, (1/k)\, (1 - \cos(\omega_n t)$$

Ex 4. $mx'' + kx = \delta(t)$

Interpretation: There is a sudden very brief very intense force, rather like getting hit on the head by a hammer. The effect is to increase the momentum instantaneously, without changing the position.

Since $x(0) = 0$ and x is continuous, $x(t)$ small for small t , so for small t we have approximately $mv' = \delta(t)$, or $v' = (1/m)\, \delta(t)$, so v increases abruptly by $(1/m)$ at $t = 0$. The momentum increases by 1 . This is a "unit *impulse*."

So for $t > 0$ the equation is equivalent to

$$mx'' + kx = 0 \ , \ x(0+) = 0 \ , \ x'(0+) = 1/m$$

--- homogeneous. General solution is sinusoidal with circular frequency

omega_n = sqrt{k/m} . x(0+) = 0 implies sine:

    x = c sin(omega_n t) ,   x' = c omega_n cos(omega_n t)

so 1/m = c omega_n   or c = 1/(m omega_n)

and the unit impulse response is

    w = u(t) (1/(m omega_n) sin(omega_n t)

Question 24.3. For solution w(t) to mx" + kx = delta(t)
with rest initial conditions, what is w'(0+) ?

1. w'(0+) = 0
2. w'(0+) = omega_n
3. w'(0+) = k
4. w'(0+) = k/m
5. w'(0+) = 1/m
6. Don't know

Ans: (5): As we saw.

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18.03 Class 25, April 7, 2010

Convolution, or, the superposition of impulse response

1. General equivalent initial conditions for delta response
2. Two implications of LTI
3. Superposition of unit impulse responses
4. Convolution product

1. Equivalent for general delta-response (t > 0) :

p(D) x = a_n x^(n) + ... + a_0 x (a_n not zero)

p(D) w = delta(t) with rest initial conditions specifies the
unit impulse response.

For t > 0 , this equation is equivalent (for t > 0) to

a_n x^(n) + ... + a_0 x = 0
with initial conditions x(0) = ... = x^(n-2)(0) = 0 , x^(n-1)(0) = 1/a_n .

Ex: n = 2 : mx" + bx' + kx = delta(t) , rest initial conditions,
is equivalent to mx" + bx' + kx = 0 , x(0) = 0 , x'(0) = 1/m .

[2] We work with a system modeled by an LTI operator p(D) = a_n D^n + ...
The time invariance has two important consequences:

(a) If p(D) x = q(t) , then p(D) x' = q'(t) .

This is because D p(D) = p(D) D (because the coefficients are constant)
so p(D) x' = p(D) Dx = D p(D) x = D q = q' .

In particular, since u'(t) = delta(t), v'(t) = w(t) :
the derivative of the unit step response is the unit impulse response.

e.g. D v = D (u(t) (1/k) (1 - e^{-kt})) = e^{-kt} = w .
(Notice that v is continuous: there are no delta terms in its
generalized derivative.)

(b) It doesn't really matter when you start the clock.

So, for example, we found that
the unit impulse response for D+kI is w(t) = u(t) e^{-kt}
and the unit step response is v(t) = u(t) (1/k) (1 - e^{-kt}) .

By time invariance the solution to x' + kx = delta(t-a) with x(t) = 0
for t < a is x = u(t-a) e^{-k(t-a)}) .

[3] We learn about a system by studying how it responds to various input
signals.

I claim that the unit impulse response $w(t)$ contains complete information about the LTI operator $p(D)$ (and so about the system it represents).

In fact there is a formula which gives the system response (with rest initial conditions) to any input signal $q(t)$ (with $q(t) = 0$ for $t < 0$) as a kind of "product" of $w(t)$ with $q(t)$, namely $w(t)*q(t)$.

Definition. Let $f(t)$ and $g(t)$ be such that $f(t) = 0 = g(t)$ for $t < 0$. The *convolution product* of $f(t)$ and $g(t)$ is

$$f(t) * g(t) = \int_0^t f(t-\tau)\, g(\tau)\, d\tau$$

I hope this crazy looking formula becomes sensible to you by the end of the lecture!

Story:

Suppose phosphates from a farm run off fields into a lake, at a rate $q(t)$ which varies with the seasons. For definiteness let's say

$$q(t) = u(t)\,(1 + \cos(bt))$$

Once in the lake, the phosphate load decays -- it's carried out of the stream at a rate proportional to the amount in the lake:

$$x' + ax = q(t), \quad x(0) = 0$$

We will find the solution to this differential equation in a clever way.

The unit impulse response for this system is

$$w(t) = u(t)\, e^{-at}$$

This tells you how much of each pound is left after t units of time have elapsed. If c pounds go in at time 0, then $w(t)\,c$ pounds remain at time $t > 0$. More generally (by time invariance) if c pounds go in at time τ, then $w(t-\tau)\,c$ pounds remain at time $t > \tau$.

More realistically, let h be a small time interval. If $q(\tau)\,h$ pounds go in between time τ and time $\tau + h$, then $w(t-\tau)\,q(\tau)\,h$ pounds remain at time $t > \tau$.

Now fix a time t. We'll think about how $x(t)$ gets built up from the contributions made during the time between 0 and t.

Break time up into small intervals each of width h, the "step size."

In the first interval of time, between $t = 0$ and $t = h$, approximately

$$q(0)\,h$$

pounds of phosphate gets dumped into the lake.

By time t this has decayed to the amount

$$w(t)\, q(0)\, h$$

In the next time interval, between t = h and t = 2h , approximately

$$q(h)\, h$$

pounds of phosphate gets added, and by time t it has decayed to

$$w(t-h)\, q(h)\, h$$

In the next interval we have

$$q(2h)\, h$$

which at time t has been reduced to

$$w(t-2h)\, q(2h)\, h$$

This continues (assuming t is a multiple of h) up to time t - h,
which yields at time t :

$$w(h)\, q(t-h)\, h$$

Add these up:

$$w(t)\, q(0)\, h + w(t-h)\, q(h)\, h + w(t-2h)\, q(2h)\, h + \ldots + w(h)\, q(t-h)\, h$$

Take the limit as h --> 0 : this is a Riemann sum approximating the
integral

$$x(t) = \int_0^t w(t-\tau)\, q(\tau)\, d\tau \qquad (\#)$$

That's the formula. For us,

$$x(t) = \int_0^t e^{-a(t-\tau)} (1 + \cos(b\tau))\, d\tau$$

The unit impulse response serves as a weight here - it tells you what
the long term impact of an action is.
"Weight function" = "Unit impulse response."

I used the "Convolution: Accumulation" tool with signal f(t) = 1 + cos(bt)
and weight function (written g(t) in the tool) $u(t)e^{-at}$.

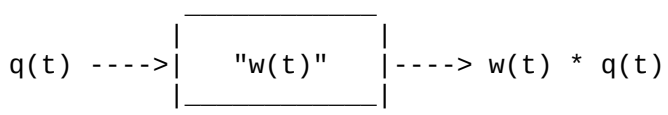
Restatement: Suppose w(t) is the weight function of p(D) . Then the
solution of p(D)x = q(t) satisfying rest initial conditions is

$$x(t) = \int_0^t w(t-\tau)\, q(\tau)\, d\tau$$

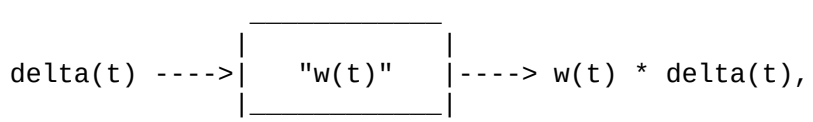
This works for any order. In the second order case, you think of the
input signal as made up of many little blows, each producing a decaying
ringing into the future. They get added up by superposition, and this is
the convolution integral. It is sometimes called the superposition
integral. It's the superposition of unit impulse responses.

You learn about a system by studying how it responds to input signals.
The interesting thing is that just one single system response suffices
to determine the system response to any signal at all. This is based on
two major assumptions: Linear, and Time Invariant.

[4] This means we can represent the system by its unit impulse response.
In terms of an input/output diagram (where rest initial conditions are assumed)



In particular,



but also by definition the output is w(t) : w(t) * delta(t) = w(t).

This * is NOT just the product w(t)f(t) . Nevertheless it deserves
to be called a "product." For one thing, it is associative and commutative:

$$(f(t) * g(t)) * h(t) = f(t) * (g(t) * h(t))$$

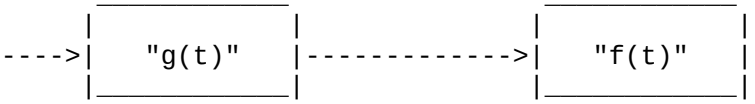
$$f(t) * g(t) = g(t) * f(t)$$

an distributive:

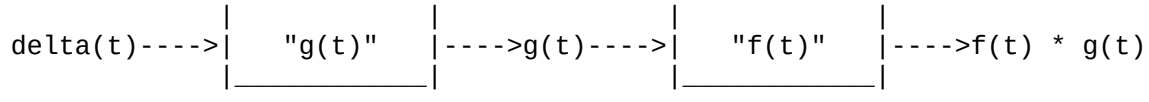
$$f(t)\,(c_1\, g_1(t) + c_2\, g_2(t)) = c_1\, f(t) * g_1(t) + c_2\, f(t) * g_2(t)$$

Also $f(t) * \text{delta}(t) = f(t)$ shows that the delta function serves as a
"unit" for this "multiplication."

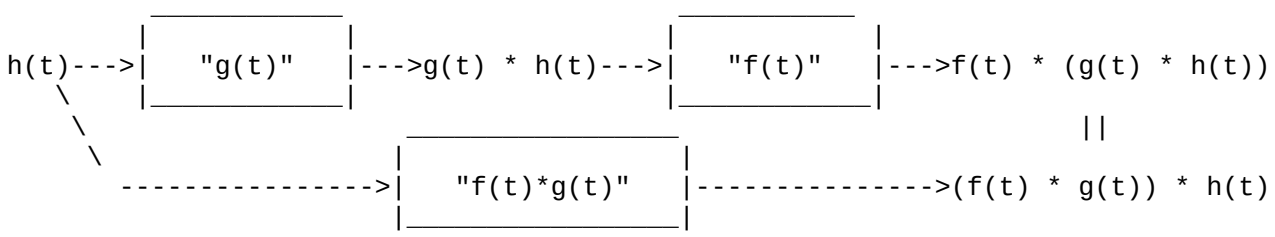
The book carries out the integration manipulation you need to do to see
these facts. Here's a proof of associativity using systems and signals:
Think about feeding the output of the "g(t)" system into the "f(t)" system:



This is a new system, a "composite" system. What is its unit impulse response?



so an input of h(t) gives an output of (f(t) * g(t)) * h(t) .
On the other hand,



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18.03 Lecture 26, April 9, 2010

Laplace Transform: definition and basic properties

1. Definition of LT; L[1]
2. Region of convergence
3. Powers
4. Linearity
5. s-shift rule
6. sines and cosines
7. t-domain and s-domain

[1] Laplace Transform

We continue to consider functions $f(t)$ which are zero for $t < 0$.
(I may forget to multiply by $u(t)$ now and then.)

The Laplace transform takes a function $f(t)$ (of "time") and uses it to manufacture another function $F(s)$ (where s can be complex). It: [Slide]

(1) makes explicit long term behavior of $f(t)$.

(2) answers the question: if I know $w(t)$, how can I compute $p(s)$?

(3) converts differential equations into algebraic equations.

But we won't see these virtues right away.

Definition: The Laplace transform of $f(t)$ is the improper integral

$$F(s) = \int_0^\infty e^{-st} f(t)\, dt$$

(formula subject to two refinements).

We will often write $f(t) \dashrightarrow F(s)$

and $\mathcal{L}[f(t)] = F(s)$

(This notation isn't so good, because there's no room for "s" on the left.)

For each value of s, $F(s)$ is a weighted integral of $f(t)$.
Wnen $s = 0$, for example, the Laplace integral is just the integral of $f(t)$. When $s > 0$, the values of $f(t)$ for large t are given less weight. Each value of $F(s)$ contains information about the whole of $f(t)$.

Example: $f(t) = 1$:

$$F(s) = \int_0^\infty e^{-st}\, dt$$

$$= \lim_{T \to \infty} e^{-st}/(-s) \,|_0^T$$

$$= (-1/s)\left(\lim_{T \to \infty} e^{-st} - 1\right).$$

If $s > 0$, $e^{-st} \to 0$ as $T \to \infty$, and we get $1/s$.

If $s < 0$, $e^{-st} \to \infty$ as $T \to \infty$, and the improper integral diverges.

Actually, I'll want s to be a complex number, not just a real number. With $s = a+bi$, $|e^{st}| = |e^{at}e^{ibt}| = e^{at}$, and the same argument shows that the integral converges for $\mathrm{Re}(s) > 0$.

So this is our first calculation:

1 has Laplace transform $1/s$

[2] For a more general function $f(t)$, the integral will converge for a given value of s provided that $|f(t)e^{-st}|$ can be integrated. This will happen if

$$|f(t)| < |e^{st}| = e^{(\mathrm{Re}\, s)t}$$

at least for t large. So to make the integral converge *somewhere*, we make the assumption that

> $f(t)$ is "of exponential order" if there is a constant M such that for large t , $|f(t)| < e^{at}$.

In that case, the integral for $F(s)$ converges for $\mathrm{Re}(s) \geq a$.

Eg e^{t^2} has no Laplace transform.

So in the definition I really meant:

Refinement #1: "for $\mathrm{Re}(s)$ large." -- the "Region of Convergence."

We'll see that the region of convergence contains important information and it's good practice to declare it when you state a Laplace transform:

> 1 has LT $1/s$ with region of convergence $\mathrm{Re}(s) > 0$.

The expression obtained by means of the integration makes sense everywhere in C except for a few points -- like $s = 0$ here -- and this is how we define the Laplace transform for values of s whose real part is too small. This is called "analytic continuation."

[3] Powers

$$L[t^n] = \int_0^\infty t^n e^{-st}\, dt$$

$$u = t^n \qquad dv = e^{-st}\, dt$$
$$du = n t^{n-1}\, dt \qquad v = (-1/s)\, e^{-st}$$

$$= t^n (-1/s) e^{-st} \Big|_0^\infty + (n/s) \int_0^\infty t^{n-1} e^{-st}\, dt$$

$$= n/s\, L[t^{n-1}] .$$

Start with $n = 1$: $L[t] = (1/s)(1/s) = 1/s^2$.

$n = 2$: $L[t^2] = (2/s)(1/s^2) = 2/s^3$

n = 3 : L[t^3] = (3/s)(2/s^3) = 3!/s^4

... L[t^n] = n!/s^{n+1} .

The region of convergence is Re(s) > 0 .

[4] These first computations can be exploited using general properties of the
Laplace Transform. We'll develop quite a few of these rules, and in fact normally you will not be using the integral definition to compute Laplace transforms.

Rule 1 (Linearity): af(t) + bg(t) ------> aF(s) + bG(s).

This is clear from the linearity of the integral.

Question 26.1. What is L[(1+t)^2] ?

1. (1/s + 1/s^2)^2 = 1/s^2 + 2/s^3 + 1/s^4
2. 1/s + 2/s^2 + 2/s^3
3. (1+t)(1/s + 1/s^2)
4. Don't know

[5] Rule 2 (s-shift): If z is any complex number,

L[e^{zt}f(t)] = F(s-z) with region of convergence Re(s) > Re(z)

Here's the calculation:

L[e^{zt}f(t)] = integral_0^infinity e^{zt} f(t) e^{-st} dt

= integral_0^infinity f(t) e^{-(s-z)t} dt

= F(s-z).

Using f(t) = 1 and our calculation of its Laplace transform we find

L[e^{zt}] = 1/(s-z). (*)

[6] As usual we can get sinusoids out from the complex exponential.

Using linearity and Euler's identity

cos(omega t) = (1/2)(e^{i omega t} + e^{-i omega t})

we find

L[cos(omega t)] = (1/2)((1/(s - i omega) + 1/(s + i omega))

= s/(s^2 + omega^2)

Using

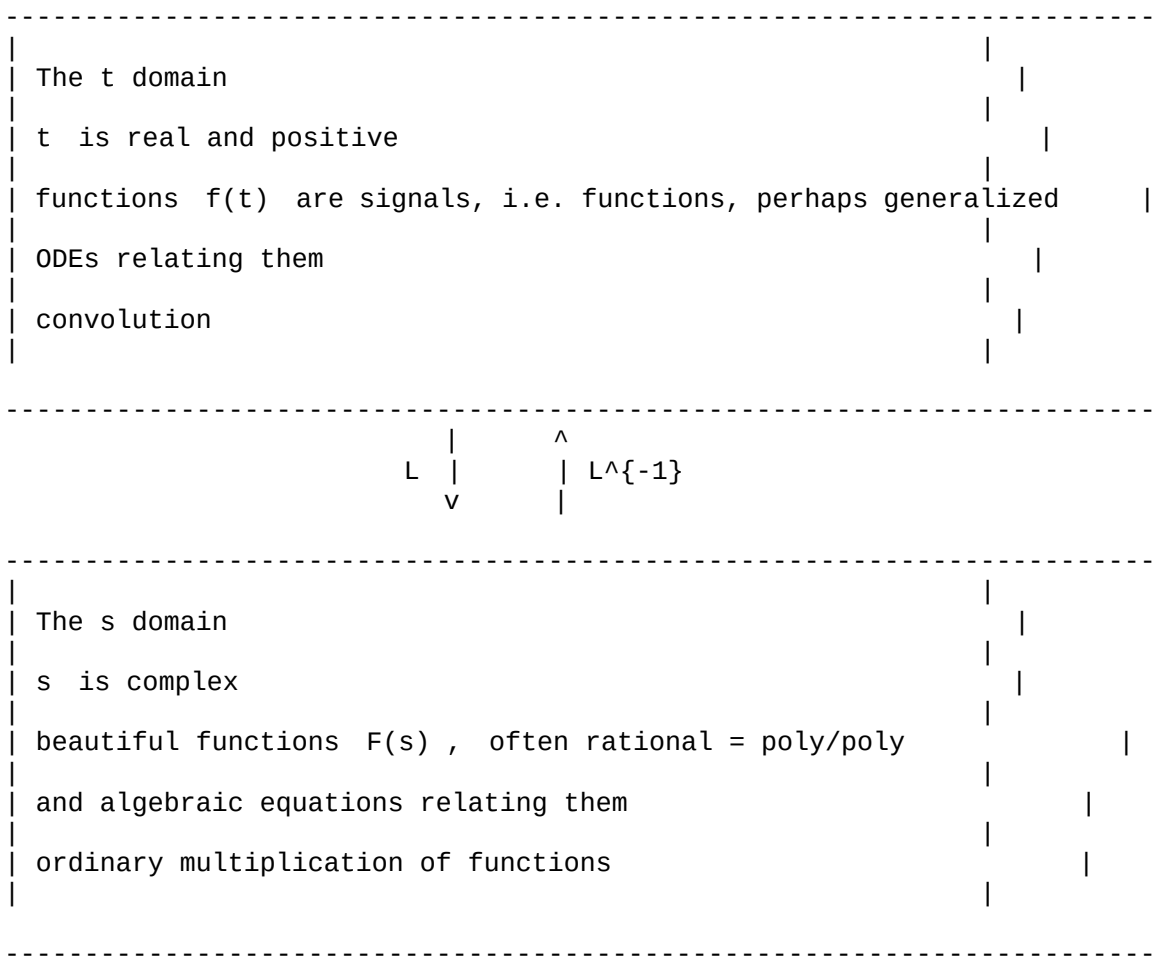
$$\sin(\omega t) = (1/2i)(e^{i\omega} - e^{-i\omega})$$

we find

$$L[\sin(\omega t)] = \omega/(s^2 + \omega^2).$$

Both are convergent for $\mathrm{Re}(s) > 0$, since $\pm i\omega t$ lie on the imaginary axis.

[7] Two worlds ... [Slide]



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18.03 Class 27, April 12, 2010

Laplace Transform II

1. Delta signal
2. t-derivative rule
3. Inverse transform
4. Unit impulse response
5. Partial fractions
6. L[f'_r]

Laplace Transform: F(s) = int_0^\infty f(t) e^{-st} dt

We saw that this improper integral may only converge for Re(s) > a, for some a depending upon f(t) . The smallest such a gives the "region of convergence."

Computations so far:

L[1] = 1/s

L[t^n] = n!/s^{n+1}

L[e^{rt}] = 1/(s-r)

L[cos(t)] = s / (s^2 + omega^2)

L[sin(t)] = omega / (s^2 + omega^2)

We also have

Rule 1 (Linearity): af(t) + bg(t) -----> aF(s) + bG(s).

Rule 2 (s-shift): L[e^{rt}f(t)] = F(s-r)

Warm-up: What is the Laplace transform of f(t) = e^{-t} cos(3t) ?

We could do this by writing it as (1/2)(e^{(-1+3i)t} + e^{(-1-3i)t})

but it's a bit easier to use the s-shift : cos(3t) ----> s / (s^2 + 9)

so e^{-t} cos(3t) ----> (s+1)/((s+1)^2 + 9)

If you like you can "uncomplete the square" and write this as

= (s+1) / (s^2 + 2s + 10)

[1] The delta function: For b > 0 ,

L[delta(t-b)] = integral_0^infty delta(t-b) e^{-st} dt

What could this mean?

If f(t) is continuous at t = b , delta(t-b) f(t) = f(b) delta(t-b) .

so integral delta(t-b) f(t) dt = f(b) u(t-b) + const

or, if a < b < c , integral_a^c delta(t-b) f(t) dt = f(b) .

In some accounts, this is the DEFINITION of the delta function.

In our situation, you get

L[delta(t-b)] = e^{-bs}

In this case, the region of convergence is the entire complex plane: the limit you take to get the improper integral is constant as soon as t > b.

There is a problem when b = 0 . delta(t) e^{-st} = delta(t) for any s , but int_0^infty delta(t) dt = u(infty) - u(0) and u(0) is indeterminate.

We want the formula that worked for b > 0 to work for b = 0 as well:

L[delta(t)] = 1

To be sure this happens, we should refine the definition of the LT integral so the lower limit (as well as the upper limit) occurs as a limit:

Refinement #2:

L[f(t)] = lim_{c increasing to infty, a increasing to 0}

int_a^c f(t) e^{-st} dt

and then we have the new computation

L[delta(t-b)] = e^{-bs} for b greater or equal to 0
region of convergence: the whole plane.

[2] To use LT in understanding differential equations, we will need:

L[f'(t)] = integral_{0-}^infty f'(t) e^{-st} dt

Parts: u = e^{-st} du = -s e^{-st} dt

dv = f'(t) dt v = f(t)

... = e^{-st} f(t) |_{0-}^infty + s integral f(t) e^{-st} dt

= s F(s)

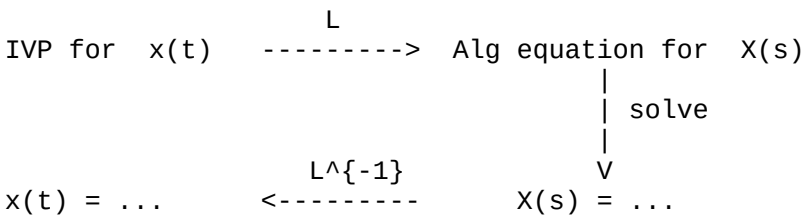
Now, what is f'(t) ? If f(t) has discontinuities, we must mean the generalized derivative; that's the only way to make the integral of the derivative work right. Even if f(t) has no breaks in its graph for t > 0 , it probably will have one when t = 0 since we are assuming that f(0-) = 0 but have not assumed that f(0+) = 0 . We have to expect a discontinuity at t = 0 , and so a delta function in f'(t) at t = 0 .

For example, u(t) = f(t) ------> 1/s
and   delta(t) = f'(t) ------> s (1/s) = 1

or   t^n = f(t) ------> n!/s^{n+1}
and   n t^{n-1} = f'(t) ------> s n!/s^{n+1} = n (n-1)!/s^n

or   cos(t) = f(t) ------> s/(s^2+1)
 delta(t) - sin(t) = f'(t) ------> s^2/(s^2+1) = 1 - 1/(s^2+1)

[3] In summary the use of Laplace transform in solving initial value problems goes like this:



For this to work we have to recover information about f(t) from F(s).
There isn't a formula for L^{-1}; what one does is look for parts of
F(s) in our table of computations. It's an art, like integration.
There is no free lunch.

We can't expect to recover f(t) exactly, if f(t) isn't required
to be continuous, since F(s) is defined by an integral, which is left
unchanged if we alter any individual value of f(t) . What we have is:

Theorem: If f(t) and g(t) are generalized functions with the same
Laplace transform, then for every a greater or equal to 0
f(a+) = g(a+), f(a-) = g(a-) , and any
occurrences of delta functions are the same in f(t) as in g(t).

So if f(t) and g(t) are continuous at t = a, then f(a) = g(a).

[4] Example: Find the unit impulse response for the operator D + 3I

ie solve   w' + 3w = delta(t) with rest initial conditions.

Step 1: Apply L : sW + 3W = 1

Step 2: Solve for W :   W = 1/(s+3)

Step 3: Find w(t) with this as Laplace transform: e^{-3t}

or more precisely, u(t) e^{-3t} .

Laplace transform is a good way to find unit impulse responses.

"Unit impulse response" = "weight function"

Its Laplace transform is called the "transfer function."

[5] Example: Solve $x' + 3x = e^{-t}$, with initial condition $x(0+) = 0$.

Step 1: Apply L : $sX + 3X = 1/(s+1)$, using linearity, the s-shift rule, and the t-derivative rule.

Step 2: Solve for X: $(s+3)X = 1/(s+1)$

so $X = 1/((s+1)(s+3))$

Step 3: Massage the result into a linear combination of recognizable forms.

Here the technique is:

Partial Fractions: $1/((s+1)(s+3)) = a/(s+1) + b/(s+3)$.

Old method: cross multiply and identify coefficients.

This works fine, but for excitement let me offer:

The Cover-up Method: Step (i) Multiply through by $(s+1)$:

$$1/(s+3) = a + (s+1)(a/(s+3))$$

Step (ii) Set $s + 1 = 0$, or $s = -1$:

$$1/(3-1) = a + 0 : \quad a = 1/2 .$$

This process "covers up" occurrences of the factor $(s+1)$, and also all unwanted unknown coefficients. The same method gives b :

$$1/(-3+1) = 0 + b : \quad b = -1/2.$$

So $X = (1/2)/(s+1) - (1/2)/(s+3)$

Step 4: Apply L^{-1}: we can now recognize both terms:

$$x = (1/2)\, e^{-t} - (1/2)\, e^{-3t} . \qquad \text{(times } u(t)\text{)}$$

Of course, this is very easy to do by our earlier methods: The ERF gives the first term, the general solution to the homogeneous equation is ce^{-3t} , and the transient needed for initial condition $x(0) = 0$ is $c = -1/2$.

[5] Consider the equation

$$x' + 3x = e^{-t} \quad , \quad x(0) = 5$$

Since we have the standing agreement that $x(t) = 0$ for $t < 0$, $x(t)$ has a jump, apparently, at $t = 0$, and perhaps what is intended is

$$x' + 3x = e^{-t} \quad , \quad x(0+) = 5$$

But this equation does not have a solution! Since $x(0-) = 0$, x' contains

the singular part 5 delta(t) ; but there's no 5 delta(t) on the right hand side.

What is really intended in a problem like this is, in connection with LT is:

$$x'_r + 3x = e^{-t} , \quad x(0+) = 0 .$$

Just to keep the notation in bounds, let's suppose that f(t) is continuous for t > 0. Then the only singular part of the generalized derivative occurs at t = 0 :

$$(f')_s(t) = f(0+) \text{ delta } (t)$$

The generalized derivative is the sum of this and the ordinary derivative (f')_r(t) . By linearity and our value L[delta(t)] = 1 ,

$$L[f'(t)] = f(0+) + L[f'_r(t)]$$

and so

$$L[f'_r(t)] = s F(s) - f(0+) .$$

This is what you always see in books and most of the time it's what is used in practice. Let's use it:

$$x'_r + 3x = e^{-t}$$

Then $(sX - 5) + 3X = 1/(s+1)$

$$X = 1 / (s+1)(s+3) + 5/(s+3)$$

so we add $L^{-1}[5/(s+3)] = 5 e^{-3t}$ to the earlier solution -- this corrects the transient.

Current list of Rules:

L is linear: L(af+bg) = aF + bG

s-shift: $e^{at}f(t)$ -----> F(s-a)

t-derivative: f'(t) -----> s F(s)

f'_r(t) -----> s F(s) - f(0+)
if f(t) has continuous derivative for t > 0 .

Computations:

1 ----> 1/s

e^{as} ----> 1/(s-a)

```
cos(omega t) ----> s/(s^2+omega^2)

sin(omega t) ----> omega/(s^2+omega^2)

delta(t-a)   ----> e^{as}
```

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18.03 Class 28, Apr 14, 2010

Laplace Transform III: Second order equations; completing the square.

1. Question period: f', FTC
2. L[f'_r(t)]
3. s-derivative rule
4. Second order equations

Question period:

[1] There were good questions from the class about finding the generalized derivative of

f(t) = u(t) cos(omega t) .

Let's do this again. The graph shows that f(t) is continuous except at t = 0 . The regular part of the derivative is

f'_r(t) = - omega u(t) sin(omega t)

and the singular part is

f'_s(t) = delta(t)

You can also apply the product rule:

f'(t) = - omega u(t) sin(omega t) + delta(t) cos(omega t)

= - omega u(t) sin(omega t) + delta(t)

since cos(omega 0) = 1 .

In order for the "fundamental theorem of calculus"

integral_a^c f'(t) dt = f(c) - f(a)

to be true, you MUST use the generalized derivative.

For example, take a = -1, b = 1, f(t) = u(t) .
If you say u'(t) = 0 , then

integral_a^c u'(t) dt = 0 , not 1

... while

integral_a^c delta(t) dt = 1

[2] At the end of class on Monday I took a regular function f(t) (with f(0) = 0 for t < 0) whose only jump in value is at t = 0 , and found L[f'_r(t)] :

f'(t) = f'_r(t) + f'_s(t)

$= f'_r(t) + (f(0+) - f(0-))\ \delta(t)$

$= f'_r(t) + f(0+)\ \delta(t)$

So:

$s\,F(s) = L[f'(t)] = L[f'_r(t)] + f(0+)$

or

$L[f'_r(t)] = s\,F(s) - f(0+)$

Example: Solve $x' + 2x = 4t$ with initial condition $x(0) = 1$ using Laplace transform.

Some interpretation is needed: Since we are always assuming $x(0) = 0$ for $t < 0$, we must mean $x'(0+) = 0$.

So there is a jump at $t = 0$ in the value of $x(t)$. This must produce a delta function in the derivative. Where is it in the equation? Missing! There's no solution to this equation if x' is interpreted as the generalized derivative. What is intended is the ordinary derivative:

$x'_r + 2x = 4t$ with initial condition $x(0+) = 1$.

Apply LT:

$sX - 1 + 2X = 4/s^2$

$X = 4/s^2(s+2) + 1/(s+2)$

Work on these separately. The first is $L[e^{-2t}]$.

For the second we need a fancier version of partial fractions:

$4/s^2(s+2) = (as + b)/s^2 + c/(s+2)$

$= a/s + b/s^2 + c/(s+2)$

Find b and c by coverup:

$4/(s+2) = b + s\,(...)$, so with $s = 0$ you find $b = 2$.

$4/s^2 = (s+2)\,(...) + c$, so with $s = -2$ you find $c = 1$

To find a you have to do something different. How about setting s = anything else, say 1:

$4/3 = a + 2 + 1/3$ or $a = -1$

$1/s^2(s+2) = -1/s + 2/s^2 + 1/(s+2)$

$x = -1 + 2t + 2e^{-2t}$

We could have gotten this more easily using undetermined coefficients!

[3] Another rule: The s-derivative rule :

$$F'(s) = (d/ds) \int_0^\infty e^{-st} f(t)\, dt$$

$$= \int_0^\infty (-t\, e^{-st})\, f(t)\, dt$$

which is the Laplace transform of $-t\, f(t)$. Thus:

$$L[t\, f(t)] = -F'(s)$$

Compare this with the t-derivative rule $L[f'(t)] = s\, F(s)$.

Example: $f(t) = \sin(\omega t)$, $F(s) = \omega / (s^2 + \omega^2)$.

$$L[t \sin(t)] = -F'(s) = 2\omega s / (s^2 + \omega^2)^2$$

[4] Second order differential equations.

$L[f''(t)] = ?$ Let $g(t) = f'(t)$: $G(s) = sF(s)$

$L[f''(t)] = L[g'(t)] = sG(s) = s^2 F(s)$

We will be careful to use this only when $f(0+) = 0$, so that $f'(t)$ doesn't have a delta function in it; we are not going to differentiate the delta function in this course.

Example: Find the unit impulse response for $D^2 + 2D + 2I$

$$w'' + 2w' + 2w = \delta(t) \text{ , rest initial conditions}$$

$$s^2W + 2sW + 2W = 1$$

$$W = 1/(s^2 + 2s + 2)$$

Technique: complete the square:

$$W = 1/((s+1)^2 + 1)$$

$$\sin(t) \text{ -----> } 1/(s^2 + 1)$$

so by the s-shift rule

$$e^{-t} \sin(t) \text{ -----> } W(s)$$

and $w(t) = u(t)\, e^{-t} \sin(t)$

TERMINOLOGY: unit impulse response = weight function
L[unit impulse response] = transfer function

GENERAL FACT: The transfer function of $p(D)$ is $W(s) = 1/p(s)$:

$$a_n w^{(n)} + \ldots + a_0 w = \delta(t)$$

$$a_n s^n W + \ldots + a_0 W = 1$$

$$p(s) W = 1$$

so $\quad LT[w] = 1/p(s)$

or $\quad LT^{-1}[1/p(s)] = w(t)$

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18.03 Class 29, April 16, 2010

Laplace Transform IV: The pole diagram

1. Another example
2. t-shift rule
3. Poles
4. What the pole diagram of F(s) says about f(t)

[1] We saw that if $p(D)w = \delta(t)$,

i.e. $a_n w^{(n)} + \ldots + a_1 w' + a_0 w = \delta(t)$

with rest initial conditions, then $p(s) W(s) = 1$,

or $W(s) = 1/p(s)$, or $w(t) = \mathcal{L}^{-1}(1/p(s))$

w(t) = weight function
W(s) = transfer function

This is another RULE.

Other input signals?

$x'' + 4x = \cos(2t)$ rest initial conditions

(No delta functions in this signal, so we know that $x(0+) = 0$ and $x'(0+) = 0$, but that will come out automatically.)

$X = W(s)\, \mathcal{L}[\cos(2t)] = 2s/(s^2+4)^2$

From our latest addition to the table, we find that

$$x = u(t)\, 1/2\, t \sin(t)$$

(Resonance!)

GENERAL FACT: $p(D)\, x = f(t)$ with rest initial conditions

has Laplace transformed equation

$$p(s)\, X(s) = F(s)$$

with solution

$$X(s) = W(s)\, F(s)$$

In the s-domain, the system response is obtained by multiplying by the transfer function!

[2] Another rule: Let $a > 0$ and define

$f_a(t) = 0$ if $t < a$

= f(t-a) if t > a

(If f_s(t) contains c delta(t) , I also want f_a(t) to contain
c delta(t-a) . So for example delta_a(t) = delta(t-a) .)

If you know F(s) , what is the LT of f_a(t) ?

integral_{0-}^infty f_a(t) e^{-st} dt = integral_{a-}^\infty f(t-a) e^{-st} dt

Substitution: u = t-a , du = dt ,

... = integral_{0-}^\infty f(u) e^{-s(u+a)} du

= e^{-as} integral_{0-}^infty f(u) e^{-su} du

= e^{-as} F(s)

So we have the t-shift rule: f_a(t) -----> e^{-as} F(s)

Eg u_a(t) ----> e^{-as}/s and delta_a(t) ----> e^{as} , as we know.

[3] F(s) "essentially" determines f(t) , but as we have seen the path
from F(s) to f(t) can be difficult. But there are certain features
of f(t) which can be EASILY read off from F(s) . They have to do with
the long term behavior of f(t) .

Typical LTs are F(s) = 1/s or F(s) = 1/(s^2+2s+5) .

F(s) is complicated because it assigns to each complex number s another
complex number F(s) . Let's content ourselves with understanding the
absolute value of F(s) : |F(s)| . This assigns a real number to each
complex number. In 18.02 you learned how to think of functions on the
plane. It has a graph which is a surface in 3-space.

For example |1/s| = 1/|s| = 1/(distance from the origin). The graph of
this function is radially symmetric: it is a surface of revolution of one
branch of the hyperbola z = 1/x . It rises towards infinity as s goes
to 0, and falls off to zero as s grows large.

The point s = 0 is called a "pole" of F(s) = 1/s . Its presence, and
location, is a simple gross feature of F(s) . In fact, it's the only
thing you see from a distance!

How about F(s) = 1/(s^2+2s+5) ? The poles occur at the roots of the
denominator: s^2+2s+5 = (s+1)^2 + 4 has roots -1 +- 2i .
This is a two-ring circus!

And the sum has three poles.

Have you figured out what functions have those two F's as LT's?

L[1] = 1/s L[(1/2)e^{-t} sin(2t)] = 1/((s+1)^2 + 4) .

Region of convergence again:
The location of the poles explains the region of convergence.

$$F(s) = \int_{0-}^{\infty} e^{-st} f(t)\, dt$$

Since $|e^{-st}|$ depends only on $\mathrm{Re}(s)$, the region of convergence will always be to the right of some line $\mathrm{Re}(s) = a$, or empty, or the whole plane. The value of $F(s)$ at a pole is infinite; this reflects the divergence of the improper integral.

The region of convergence is the half plane to the right of the rightmost pole.

[4] What the poles of $F(s)$ tell us about $f(t)$

Knowing where the poles are is just a small part of knowing the whole of F(s), and it doesn't tell you everything about $f(t)$.

Two examples:

(1) If $f(t) = 0$ for $t > A$, then the improper integral isn't so improper -

$$F(s) = \int_{0-}^{A} e^{-st} f(t)\, dt$$

- so it converges for ALL s : no poles at all.

(2) e^{-as} is never zero and has no poles: so the poles of $e^{-as}F(s)$ are the same as the poles of $F(s)$: time translation of $f(t)$ doesn't affect the pole diagram of $F(s)$. So the pole diagram can't see phase.

But the pole diagram does say a lot about the long term behavior of $f(t)$.

Example 1. $f(t) = \sin(2t)$. $F(s) = 4/(s^2_4)$ has poles at $2i$ and $-2i$. Any $f(t)$ such that $F(s)$ has this pole diagram exhibits (for large t)
(a) oscillation with circular frequency 2, and
(b) neither exponential growth nor exponential decay.

[The Laplace transform of $f(t) = t \sin(2t)$ is $4s/((s^2+4)^2)$, which has the same pole diagram. This function does grow with t, but less than exponentially.]

Example 2: $f(t) = e^{-t} \sin(2t)$. $F(s) = 2s/((s+1)^2+4)$ has poles at $-1+2i$ and $-1-2i$. Any $f(t)$ such that $F(s)$ has this pole diagram exhibits (for large t)
(a) oscillation with circular frequency 2, and
(b) exponential decay on the order of e^{-t}.

Example 3. $f(t) = 1 + e^{-t} \sin(2t)$. $F(t) = 1 + 2s/((s+1)^2+4)$, which has poles at 0, $-1+2i$, and $-1-2i$. Any $f(t)$ such that $F(s)$ has this pole diagram exhibits (for large t)

(a) sub-exponential growth/decay,
i.e. for any $a > 0$, $e^{(-at)} < |f(t)| < e^{(at)}$.

(b) oscillation with circular frequency 2 but with amplitude decaying like e^{-t}.

The rightmost poles give the dominant information.

A major lesson: if all the poles of $F(s)$ have negative real part, then the function $f(t)$ decays to zero exponentially.

The position of the poles of $F(s)$ gives detailed information about the long-term rate of growth and oscillation of $f(t)$, information which it can be hard to glean from the graph of $f(t)$ ... all exponentials look rather similar ...

So when you solve an ODE using LT, it may be good enough to stop with $X(s)$. Find its poles, and you know what the solution behaves like in the long run.

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18.03 Class 30, Apr 21, 2010

Laplace Transform V: Poles and amplitude response

1. Recap on pole diagram
2. Stability
3. Exponential signals
4. Transfer and gain
5. Fourier and Laplace

[1] To recap: The kind of function F(s) that arises as a Laplace transform can be understood, in broad terms, by giving the set of points at which it becomes infinite. This is the "pole diagram."

Examples: The following F(s)'s all have pole diagram {2i,-2i} (a not zero) : [Slide]

F(s)	f(t)
a/(s^2+4)	(a/2) sin(2t)
as/(s^2+4)	a cos(2t)
e^{-bs}/(s^2+4)	cos_b(2t)
1 + a/(s^2+4)	delta(t) + (a/2) sin(2t)
4s/(s^2+4)^2	t cos(2t)

and many other examples. All these functions f(t) share some common features, for sufficiently large t :

- they oscillate with circular frequency 2 .
- they may grow or shink, but less than exponentially.

These features are common to all functions f(t) such that F(s) has this pole diagram.

Pole diagram {2i,-2i,1} : any of the above plus (c not zero)

F(s)	f(t)
c/(s-1)	c e^t + ...
c/(s-1)^2	c t e^t + ...

and many other examples. All these functions f(t) share some common features, for sufficiently large t :

- they grow as t ---> infinity "like e^t ." This means that they grow faster than e^{kt} for any k < 1, and slower than e^{lt} for any l > 1.
- they oscillate but the oscillations become insignificant relative to the size of f(t) as t ---> infinity.

The rightmost poles of F(s) determine the dominant features of f(t) for large t.

In particular: If all poles of $F(s)$ have negative real part - are to the left of the imaginary axis - then $f(t) \to 0$ as $t \to \infty$.

[2] This is a "stability" effect, related to one we have seen before. The unit impulse response $w(t)$ for $p(D)$ has Laplace transform $W(s) = 1/p(s)$. The poles of $W(s)$ are the roots of $p(s)$. If all roots have negative real part, then all solutions to $p(D)x = 0$ die off exponentially as $t \to \infty$. For $t > 0$, $w(t)$ is a solution to $p(D)x = 0$. If all the poles are in the left half plane, then not just the unit impulse response but in fact all solutions of the homogeneous equation die off: they are *transients.*

[3] Exponential signals

The reciprocal of the characteristic polynomial arose long ago, in the Exponential response formula:

Exponential response: $p(D)\,x = e^{rt}$ has an exponential solution

$$x_p = W(r)\, e^{rt}$$

The transfer function is the Laplace transform of the weight function.

This is just what appeared in our study of Sinusoidal response: [Slide]

$$p(D)x = \cos(\omega t)$$

$$p(D)z = e^{i\omega t}$$

$$z_p = W(i\omega)\, e^{i\omega t}$$

$$x_p = \mathrm{Re}[\, W(i\omega)\, e^{i\omega t}\,]$$

$$= \mathrm{gain}(\omega)\cos(\omega t - \phi)$$

where $\mathrm{gain}(\omega) = |W(i\omega)|$

$$-\phi = \mathrm{Arg}(W(i\omega))$$

$W(i\omega)$ is the "complex gain." (Here we are supposing that the input signal, with respect to which we should be measuring the gain and the phase lag, is just $\cos(\omega t)$.)

The gain is the restriction to the imaginary axis of the magnitude of the transfer function.

[4] Here's the vision that unifies most of what we have done in this course so far:

You have a system (a black box, with springs and masses and dashpots, for example) which you wish to understand. This means really that you want to be able to predict its response to various input signals.

We will only be able to analyze systems which are LINEAR and TIME

INVARIANT: so (1) superposition holds, and (2) delaying the input signal results in delaying the system response without otherwise changing it.

We are especially interested in its periodic response to periodic signals. Periodic signals decompose into sinusoidal signals, by Fourier series, so it's enough just to study sinusoidal system responses.
There will be a gain and a phase lag involved. You'll be happy to understand the gain, and leave the phase lag for another day.

So hit the system: feed it delta(t) as input signal.

What comes out is w(t).

Apply L to w(t) to get W(s) .

Graph |W(s)| . This will be a surface lying over the complex plane.
It probably has some poles.

Restrict s to purely imaginary values , s = i omega. This is what is needed to study sinusoidal input response:

p(D) x = e^{i omega t}

has exponential solution

x_p = W(i omega) e^{i omega t}

so |W(i omega)| is the gain.

The intersection of the graph of W(s) with the vertical plane lying over the imaginary axis is the amplitude response curve (extended to an even function, allowing negative omega).

Near resonance occurs because i omega is getting near to one of the poles of W(s) .

If you increase the damping, the poles move deeper into negative real part space, and eventually the two humps in the frequency response curve merge.

If you have a higher order system, you get more poles, and a more complicated amplitude response curve.

I did not talk about:

[5] Laplace and Fourier

A period function neither neither grows nor decays as t ---> infty, so we know that its rightmost poles lie in the imaginary axis.

A periodic function of period 2L has a Fourier series.
Let's write omega = 2pi/2L = pi/L. Then the Fourier series has the form

f(t) = a_0/2 + a_1 cos(omega t) + a_2 cos(2 omega t) + ...
+ b_1 sin(omega t) + b_2 sin(2 omega t) + ...

The poles of L [sin(n omega t)] and of L [cos(n omega t)] occur at

n omega i and - n omega i .

So the poles of u(t)f(t) occur along the imaginary axis, spaced out at multiples of +- i omega .

The location of poles of F(s) gives us some intuitive feeling for the significance of the s-domain. If there are poles to the right of the imaginary axis, then the corresponding function of time grows like e^{at} where a is the largest real part of any pole. The imaginary part of the pole tells us about the circular frequency of (exponentially damped or expanding) oscillations.

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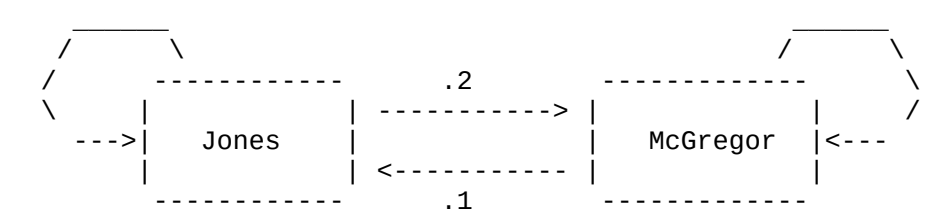
18.03 Class 32, April 26 , 2010

Linear first order systems: Introduction

1. Elimination
2. Matrices
3. Anti-elimination: companion matrices
4. Shakespeare

[1] There are two fields in which rabbits are breeding like rabbits, Farmer Jones's field contains $x(t)$ rabbits, Farmer McGregor's field contains $y(t)$ rabbits. Breeding like rabbits in this case means that Jones's show a net growth rate of .7 rabbits per rabbit per month, and McGregor's show .6. They can also hop over the hedge between the fields, and since the grass is greener in Farmer McGregor's field, Jones's rabbits jump over a the rate of .2 per month, MacGregor's at the rate of .1 per month.

Flow diagram:



So the equations are

$$x' = .5x - .2x + .1y = .3x + .1y \quad (1)$$
$$y' = .5y - .1y + .2x = .2x + .4y \quad (2)$$

This is a linear SYSTEM of equations, homogeneous, and it seems to be impossible to solve, since you need to know y to solve for x and you need to know x to solve for y.

We *can* solve, though: use (1) to express y in terms of x :

$$y = 10\,x' - 3\,x$$

and then plug this into (2):

$$10\,x'' - 3\,x' = .2\,x + 4x' - 1.2\,x$$

$$10\,x'' - 7\,x' + x = 0$$

or

$$x'' - .7\,x' + .1\,x = 0$$

This is a SECOND ORDER ODE , which we can solve:

$$s^2 - .7s + .1 = 0$$

has roots r1 = .5 and r2 = .2 .

(Remember, (s - r1)(s - r2) = s^2 - (r1 + r2) s + r1 r2 .)

So we get two basic solutions,

$$x_1 = e^{.5t}$$
$$x_2 = e^{.2t}$$

Each gives a corresponding solution for y , using y = 10 x' - 3 x :

$$y_1 = 2e^{.5t}$$
$$y_2 = -e^{.1t}$$

Well, some of these solutions are not biologically significant.

We can package this information in a vector: $u(t) = \begin{vmatrix} x(t) \\ y(t) \end{vmatrix}$

This vector traces a curve in the plane, as time increases.
For example, the solution $u_1 = \begin{vmatrix} x_1 \\ x_2 \end{vmatrix}$ traces a ray that passes
through (1,2) at t = 0 and move off towards infinity like e^{.5t} .

[2] Analysis of homogeneous linear equations by matrices.

$$x' = ax + by \qquad (*)$$
$$y' = cx + dy$$

A 2 x 2 matrix is an array of numbers (enclosed by brackets)

$$A = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

In these notes I will use Matlab notation and write this array as

$$[\ a\ b\ ;\ c\ d\ ]$$

There is another matrix in sight, the "column vector"

$$\begin{vmatrix} x \\ y \end{vmatrix}$$

or [x ; y] with entries x and y .

Matrix multiplication is set up so that

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix} = \begin{vmatrix} ax + by \\ cx + dy \end{vmatrix}$$

or [a b ; c d][x ; y] = [ax+by ; cx+dy]

You run along a row of the first matrix and a column of the second,
multiplying and adding up. Another way to look at matrix multiplication
comes from rewriting the right hand side as

```
        | ax + by |  =  x | a |  +  y | b |
        | cx + dy |       | c |       | d |
```

or [ax+by ; cx + dy] = x [a ; c] + y [b ; d]

The ODE (*) can thus be written as

        [x' ; y'] = [a b ; c d][x ; y]

If we write u for the column vector [x ; y] then u' = [x' ; y'] , and

        u' = Au

This compact expression is exactly equivalent to (*) .

In the example we just gave, there are two independent basic solutions:

        u1 = e^{.5t} [1 ; 2]
        u2 = e^{.2t} [1 ; -1]

and the general solution is a linear combination of these two.

[3] Companion matrices.

Any second order linear equation occurs in this way, and in fact the first order system gives an important way to visualize the second order equation.

Suppose we have a second order homogeneous linear equation, say

        x" + b x' + k x = 0   (**)

We can derive a first order linear system from this, by the trick of defining

        y = x'

so then

        y' = x" = - b x' - k x = - kx - b y

The corresponding matrix is A = [0 , 1 ; -k , -b] .
This is the "companion matrix" of the equation (**) .

The solutions to u' = A u are of the form u = [x ; x'] ;
it records both the solution to (**) and its derivative.

For example,

        x" - (1/2)x' + (17/16)x = 0

The companion matrix is [0 , 1 ; -17/16 , 1/2] .

What do solutions of this one look like? The characteristic polynomial of the second order equation is

$$p(s) = s^2 - (1/2)s + (17/16) = (s - (1/4))^2 + 1$$

so a root r satisfies $(r - (1/4))^2 = -1$ or $r = (1/4) +- i$
The general solution is thus

$$x = A e^{t/4} \cos(t - phi)$$

These oscillate under an exponentially growing envelope. The derivative does the same, but is off phase. The result is that the trajectory traced out by (x,y) is an expanding spiral.

[4] It turns out that the same system models the relationship between Romeo and Juliet. The MIT Humanities Department has analyzed the plot of Shakespeare's play and found the following. If R denotes Romeo's love for Juliet, and J denotes Juliet's love for Romeo, then

$$R' = J$$
$$J' = -(17/16)R + (1/2)J$$

Romeo is a puppy dog. He has little selfawareness; the change in his feelings towards Juliet has nothing to do with how he himself feels at the moment; it is completely dependent on how she feels about him. Juliet is more complex. She has a healthy self awareness; if she loves him, that very fact causes her to love him more. On the other hand, if he seems to love her, she gets frightened and starts to love him less.

Let's start the action at (1,0). So Romeo is fond of Juliet but she is neutral towards him. However, she does notice that he is fond of her, and this makes her somewhat hostile. As she becomes more distant, his affection wanes. Eventually he is neutral and she really doesn't like him. This continues; presently he stays away from her, and this very fact makes her more intested. She warms to him, he notices and his rate of increase of disinterest starts to ameliorate. Eventually she is neutral and just as he bottoms out. He then starts to feel better towards her, but still stays away, and now both his attitude and hers cause her to feel progressively more well disposed towards him. This causes him to continue to warm to her.

Following this around, you wind up at $J = 0$ again, but now R has increased. This is a cyclical relationship, but with each cycle the intensity increases. We all know the sad outcome.

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18.03 Class 33, Apr 28, 2010

Eigenvalues and eigenvectors

1. Linear algebra
2. Ray solutions
3. Eigenvalues
4. Eigenvectors
5. Initial values

[1] Prologue on Linear Algebra.

Recall $[a\ b\ ;\ c\ d]\ [x\ ;\ y] = x[a\ ;\ c] + y[b\ ;\ d]$:

A matrix times a column vector is the linear combination of the columns of the matrix weighted by the entries in the column vector.

When is this product zero?

One way is for $x = 0 = y$. If $[a\ ;\ c]$ and $[b\ ;\ d]$ point in different directions, this is the ONLY way. But if they lie along a single line, we can find x and y so that the sum cancels.

Write $A = [a\ b\ ;\ c\ d]$ and $u = [x\ ;\ y]$, so we have been thinking about $A\,u = 0$ as an equation for u . It always has the "trivial" solution $u = 0 = [0\ ;\ 0]$: 0 is a linear combination of the two columns in a "trivial" way, with 0 coefficients, and we are asking when it is a linear combination of them in a different, "nontrivial" way.

We get a nonzero solution $[x\ ;\ y]$ exactly when the slopes of the vectors $[a\ ;\ c]$ and $[b\ ;\ d]$ coincide: $c/a = d/b$, or $ad - bc = 0$. This combination of the entries in A is so important that it's called the "determinant" of the matrix:

$$\det(A) = ad - bc$$

We have found:

Theorem: $Au = 0$ has a nontrivial solution exactly when $\det A = 0$.

If A is a larger *square* matrix the same theorem still holds, with the appropriate definition of the number $\det A$.

[2] Solve $u' = Au$: for example with $A = [-2\ 1\ ;\ -4\ 3]$.

The Mathlet "Linear Phase Portraits: Matrix entry" shows that some trajectories seem to be along rays from the origin. (We saw these also in the rabbits example on Monday. They were not present in the Romeo and Juliet example, though!) That is to say, we are going to look for a solution of the form

$$u(t) = r(t)\,v\ ,\quad v \text{ not } 0$$

One thing for sure: $u'(t)$ also points in the same or reverse direction:

u'(t) = r'(t) v

Use the equation to express u'(t) in terms of A and v :

u'(t) = A u(t) = A r v = r A v

r' v = r A v

So Av points in the same or the reverse direction as v :

A v = lambda v

for some number lambda. This letter is always used in this context. An *eigenvalue* for A is a number lambda such that A v = lambda v for some nonzero v . A vector v such that A v = lambda v is an eigenvector for value lambda. [Thus the zero vector is an eigenvector for every lambda. A number is an eigenvalue for A exactly when it possesses a nonzero eigenvector.]

I showed how this looks on the Mathlet Matrix/Vector.

[3] This is a pure linear algebra problem: A is a square matrix, and we are looking for nonzero vectors v such that A v = lambda v for some number lambda. Let's try to find v . In order to get all the v's together, right the right hand side as

lambda v = (lambda I) v

where I is the identity matrix [1 0 ; 0 1] , and lambda I is the matrix with lambda down the diagonal. Then we can put this on the left:

0 = A v - (lambda I) v = (A - lambda I) v

Don't forget, we are looking for a nonzero v . We have just found an exact condition for such a solution:

det(A - lambda I) = 0

This is an equation in lambda ; we will find lambda first, and then set about solving for v (knowing in advance only that there IS a nonzero solution).

In our example, then, we subtract lambda from both diagonal entries and then take the determinent:

A - lambda I = [-2 - lambda , 1 ; -4 , 3 - lambda]

det (A - lambda I) = (-2-lambda)(3-lambda) + 4

= 1 - lambda + lambda^2 - 2

= lambda^2 - lambda - 2

This is the "characteristic polynomial"

p_A(lambda) = det(A - lambda I)

of A , and its roots are the "characteristic values" or "eigenvalues" of A .

In our case, p_A(lambda) = (lambda + 1)(lambda - 2)

and there are two roots, lambda_1 = -1 and lambda_ 2 = 2 .

(The order is irrelevant.)

[4] Now we can find those special directions. There is one line for lambda_1 and another for lambda_2 . We have to find nonzero solution v to

(A - lambda I) v = 0

eg with lambda = lambda_1 = -1 , A - lambda = [-1 1 ; -4 4]

There is a nontrivial linear relation between the columns:

A [1 ; 1] = 0

All we are claiming is that

A [1 ; 1] = - [1 ; 1]

and you can check this directly. Any such v (even zero) is called an "eigenvector" of A.

It means that there is a ray solution of the form r(t) v where v = [1;1] . We have

r' v = r A v = r lambda v

so (since v is nonzero)

r' = lambda r

and solving this goes straight back to Day One:

r = c e^{lambda t}

so for us r = c e^{-t} and we have found our first straight line solution:

u = e^{-t} [1;1]

In fact we've found all solutions which occur along that line:

u = c e^{-t} [1;1]

Any one of these solutions is called a "normal mode."

General fact: the eigenvalue turns out to play a much more important role than it looked like it would: the ray solutions are *exponential* solutions, e^{lambda t} v , where lambda is an eigenvalue for

the matrix and v is a nonzero eigenvector for this eigenvalue.

The second eigenvalue, lambda_2 = 2 , leads to

A - lambda I = [-4 1 ; -4 1]

and [-4 1 ; -4 1] v = 0 has nonzero solution v = [1;4]

so [1;4] is a nonzero eigenvector for the eigenvalue lambda = 2 ,
and there is another ray solution

e^{2t} [1;4]

[5] The general solution to u' = Au will be a linear combination of
the two eigensolutions (as long as there are two distinct eigenvalues).

In our example, the general solution a linear combination of the normal
modes:

u = c1 e^{-t} [1;1] + c2 e^{2t} [1;4]

We can solve for c1 and c2 using an initial condition: say for example

u(0) = [1 ; 0]. Well,

u(0) = c1 [1 ; 1] + c2 [1 ; 4] = [c1 + c2 ; c1 + 4 c2]

and for this to be [1 ; 0] we must have 3 c2 = -1 : c2 = -1/3 ;
so c1 = - 4 c2 = 4/3 :

u(t) = (4/3) e^{-t} [1 ; 1] - (1/3) e^{2t} [1 ; 4] .

When t is very negative, -10, say, the first term is very big and the
second tiny: the solution is very near the line through [1 ; -1].
As t gets near zero, the two terms become comparable and the solution
curves around. As t gets large, 10, say, the second term is very big
and the first is tiny: the solution becomes asymptotic to the line through
[1 ; 1].

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18.03 Differential Equations

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18.03 Class 34, April 30, 2010

Complex or repeated eigenvalues

1. Eigenvalues and coefficients
2. Complex eigenvalues
3. Repeated eigenvalues
4. Defective and complete

[1] We were solving $u' = Au$, with $A = [\ a\ b\ ;\ c\ d\ ]$:
$A = [\ -2\ 1\ ;\ -4\ 3\ ]$ for example. Or u could be a 3D vector,
and A a 3x3 matrix.

(1) Find the eigenvalues = roots λ_1, λ_2
of the characteristic polynomial

$$p_A(\lambda) = \det(A - \lambda I)$$

(2) For each eigenvalue, find a nonzero eigenvector:

$$(A - \lambda_1 I)\, v_1 = 0 \qquad (A - \lambda_2 I)\, v_2 = 0$$

Normal modes: $u_1 = e^{\lambda_1 t} v_1$, $u_2 = e^{\lambda_2 t} v_2$.

(3) General solution is linear combination of these: $u = c_1 u_1 + c_2 u_2$.

There are a few problems with this, but before pointing them out let me make
three comments:

(a) Any multiple of an eigenvector is another eigenvector for the same
eigenvalue; they form a line, an "eigenline."

(b) The zero vector is an eigenvector for every value lambda, whether
lambda is an eigenvalue or not. Most of the time 0 is the ONLY eigenvector
for value lambda; lambda is an eigenvalue exactly when there is a
nonzero eigenvector for that value.

(c) $$p_A(\lambda) = \det(A - \lambda I)$$

$$= \lambda^2 - (a+d)\lambda - (ad-bc)$$

The sum of the diagonal terms of a square matrix is the "trace" of A,
$\operatorname{tr} A$, so

$$p_A(\lambda) = \lambda^2 - (\operatorname{tr} A)\lambda + (\det A)$$

In our example, $\operatorname{tr} A = 1$ and $\det A = -2$, and

$$p_A(\lambda) = \lambda^2 - \lambda - 2 .$$

[2] There may be no ray solutions. Romeo and Juliet provided one example.
Or, what about

$$A = [\ 1\ 2\ ;\ -2\ 1\ ] .$$

Let's apply the method and see what happens. tr(A) = 2 , det(A) = 5, so

p_A(lambda) = lambda^2 - 2 lambda + 5 = (lambda - 1)^2 + 4

which has roots lambda_1 = 1 + 2i, lambda_2 = 1 - 2i.

We could abandon the effort at this point, but we had so much fun and success with complex numbers earlier that it seems we should carry on.

Find an eigenvector for lambda_1 = 1 + 2i :

A - (1+2i)I : [- 2i 2 ; -2 -2i][? ; ?] = [0 ; 0]

Standard method: use the entries in the top row in reverse order with one sign changed: [2 ; 2i] or, easier, in this case,

v_1 = [1 ; i].

This is set up so the top entry in the product is 0 . We have a chance to check our work (mainly the calculation of the eigenvalues) by seeing that the bottom entry in the product is 0 too:

-2 . 1 - 2i . i = 0

[1 ; i] is a vector with complex entries. OK, so be it. It's hard to visualize, perhaps, and doesn't represent a point on the plane, but we can still compute with it just fine.

Since lambda_2 = conjugate of lambda_1,
an eigenvector for lambda_2 is given by the conjugate of v_1:

v_2 = [1 ; -i]

So the normal modes

$v(t) = e^{(1+2i)t} [1 ; i]$, $\bar{v}(t) = e^{(1-2i)t} [1 ; -i]$.

As in the case of second order equations, the real and imaginary parts of solutions are again solutions,

$u1 = (v + \bar{v})/2 = Re(v)$, $u2 = (v - \bar{v})/(2i)$

and we really only need to write down one of the normal modes.

So these are real solutions:

u = e^{(1+2i)t} [1 ; i]

= e^t (cos(2t) + i sin(2t)) ([1;0] + i[0;1]) so

u1 = Re(u) = e^t (cos(2t) [1;0] - sin(2t) [0;1])

= e^t [cos(2t) ; - sin(2t)] and

u2 = Im(u) = e^t (cos(2t) [0;1] + sin(2t) [1;0])

$= e^t [\sin(2t) ; \cos(2t)]$

These are two independent real solutions. Both spiral around the origin, clockwise, while fleeing away from it exponentially. They satisfy

$u1(0) = [1;0]$, $u2(0) = [0;1]$.

I showed their trajectories on the Mathlet Linear Phase Portraits: Matrix Entry.

The general real solution is

$a\ u1 + b\ u2$, a, b real .

It is very hard for me to visualize the fact that all those spirals are linear combinations of any two of them.

Summary: Nonreal eigenvalues lead to spiral solutions.
Positive real parts lead to solutions going to infinity with t ("unstable")
Negative real parts lead to solutions going to zero with t ("stable")
Zero real parts lead to solutions parametrizing ellipses.

So we discover that the possibility of complex eigenvalues really isn't a failure of the method at all. There are in fact ray solutions, but they are complex and don't show up on our real phase plane.

[3] Second problem with our method: Illustrated by

$A = [-2\ 1 ; -1\ 0]$

$p_A(\lambda) = \lambda^2 + 2\lambda + 1 = (\lambda + 1)^2$

which has only one root, "repeated": $\lambda_1 = \lambda_2 = -1$.

Still, find an eigenvector:

$A - (-1)I = [-1\ 1 ; -1\ 1][? ; ?] = [0 ; 0] : \quad v = [1 ; 1]$

or any nonzero multiple. ALL eigenvectors for A lie on the line containing 0 and $[1 ; 1]$. I showed a picture of the phase portrait, which shows only one pair of opposed ray trajectories.

So there is (up to multiples) only one normal mode:

$u_1 = e^{-t} [1 ; 1]$

But we need another solution. Here is how to find one; I won't go into details, just give you the method.

Write down the same matrix $A - \lambda_1 I$ but now find a vector w such that

$(A - \lambda_1 I)\ w = v$.

Then

u_2 = e^{lambda_1 t} (t v + w)

is a second solution.

In our case:

[-1 1 ; -1 1] [? ; ?] = [1 ; 1]

has solution [0;1] , so

u_2 = e^{-t} (t [1;1] + [0;1])

= e^{-t} [t ; t+1]

[0;1] isn't the only vector that works here; [0;1] + c v does too for any constant c. It doesn't matter which one you pick.

With this choice, u_1(0) = [1;1] , u_2(0) = [0;1] .

The general solution is

u = a u_1 + b u_2 .

To learn more about all this you should take 18.06.

Didn't get to talk about this on Friday:

[4] A matrix with a repeated eigenvalue but only one lineful of eigenvectors is called "defective." A matrix can have a repeated eigenvalue and not be defective:

A = [2 0 ; 0 2]

for example has characteristic polynomial

p(lambda) = lambda^2 - 4 lambda + 4 = (lambda - 2)^2

so lambda_1 = lambda_2 = 2. To find an eigenvector consider

A - lambda_1 I : [0 0 ; 0 0] [? ; ?] = [0 ; 0]

Now ANY vector is an eigenvector! Instead of only one line you get the entire plane. For any vector v ,

e^{2t} v

is a solution, and every solution is a normal mode. This is called the "complete" case.

In the 2x2 case, if the eigenvalue is repeated you are in the defective case unless the matrix is precisely [\lambda_1 , 0 ; 0 , \lambda_1]

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18.03 Differential Equations

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18.03 Class 35, May 3, 2010

Linear Phase Portraits: Eigenvalues Rule (usually)

1. Eigenvalues rule!
2. Trace-determinant plane
3. Marginal cases
4. Stability

[1] Phase portrait: this means the (x,y) plane (the "phase plane") with trajectories of solutions of u' = Au drawn on it (with direction of time indicated).

These homogeneous linear equations exhibit a nice variety of phase portraits, as shown by the Linear Phase Portraits Mathlets. An important fact is this:

EIGENVALUES RULE

We'll classify the linear phase portraits according to the eigenvalues of the matrix A .

Example: those rabbits again: A = [.3 .1 ; .2 .4]

p_A(lambda) = lambda^2 - .7 lambda + .1

has roots lambda_1 = .5 , lambda_2 = .2

so we learn that all solutions flee from the origin: the phase portrait is a "source."

lambda = .5 : [-.2 .1 ; .2 -.1] ==> v_1 = [1 ; 2]

so one normal mode is u_1 = e^{.5 t} [1 ; 2]

If the number of rabbits in MacGregor's field is twice the number in Jones's, it stays that way forever after.

lambda = .2 : [.1 .1 ; .2 .2] ==> v_2 [1 ; -1]

so the other normal mode is u_2 = e^{.2t} [1 ; -1] .

This is not meaningful in itself in our population model, but we can draw it in the phase plane. And the two together provide the general solution.

What to the other trajectories look like?

For large t , the v_1 component is much bigger than the v_2 component.
For small t , the v_1 component is much smaller than the v_2 component.

So near the origin the trajectories become asymototic to the eigenline with smaller eigenvalue.

This phase portrait is a NODE. The same kind of picture occurs whenever the eigenvalues are real, of the same sign, and distinct.

[2] When are the eigenvalues non-real?

[Slide:] p_A(lambda) = det (A - lambda I)

If A = [a b ; c d] then p_A(lambda) = lambda^2 - (tr A) lambda + (det A)

tr(A) = a + d = lambda_1 + lambda_2

det(A) = ad-bc = lambda_1 lambda)2

To find the eigenvalues, complete the square:

p_A(lambda) = (lambda - (tr(A)/2))^2 + (det(A) - (tr(A)/2)^2)

so lambda1,2 = tr(A)/2 +- sqrt(tr(A)^2/4 - det(A)).

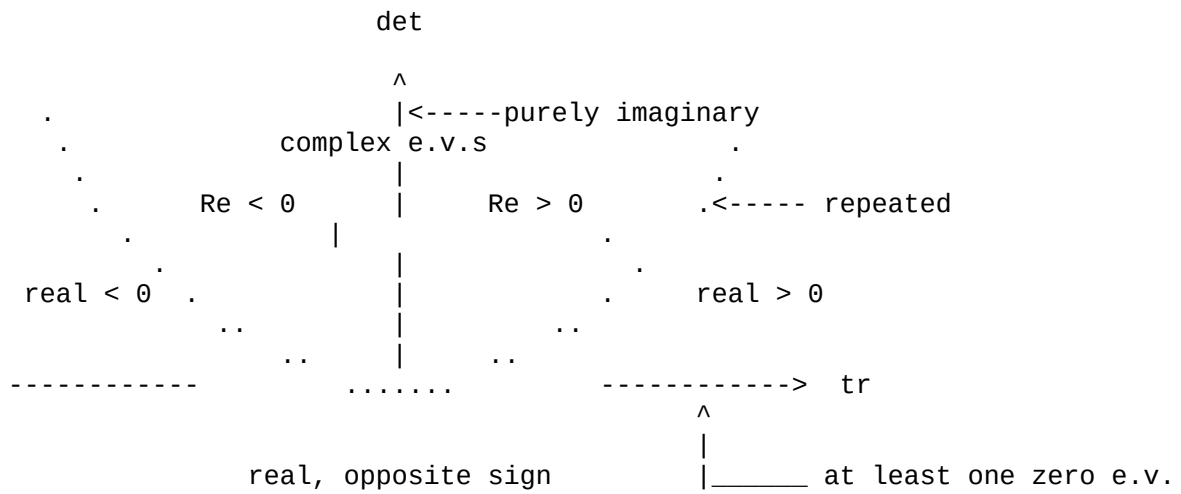
lambda1,2 are not real if det(A) > tr(A)^2/4

are equal if det(A) = tr(A)^2/4

are real and different from each other if det(A) < tr(A)^2/4

The boundary is the "critical parabola," where det(A) = tr(A)^2/4.

Notice that if the eigenvalues are complex, the real part is tr(A)/2.
If the eigenvalues are real, they have the same sign exactly when their product is positive, and that sign is positive if their sum is also positive. Thus:



The eigenvalues determine the general characteristics of the solutions.

NODES: When the eigenvalues are real and of the same sign, but distinct, you have a "node." If the sign is positive, it's "unstable" or a "source." If negative, "stable" or a "sink."

SPIRALS: when the eigenvalues are non-real, we get spirals.
Two more comments on this:

(1) The spirals move IN when Re(lambda) < 0   "stable spiral"
                     OUT when Re(lambda) > 0   "unstable spiral"

When Re(lambda) = 0 it turns out that the trajectories are ellipses.
The technical term for this type of phase portrait is "center."

SADDLES: When the eigenvalues are real and of opposite sign, the phase plane is a "saddle." There are two eigenlines, one with positive eigenvalue and the other with negative. Normal modes along one move out, and along the other move in. The general solution is a combination of these two.

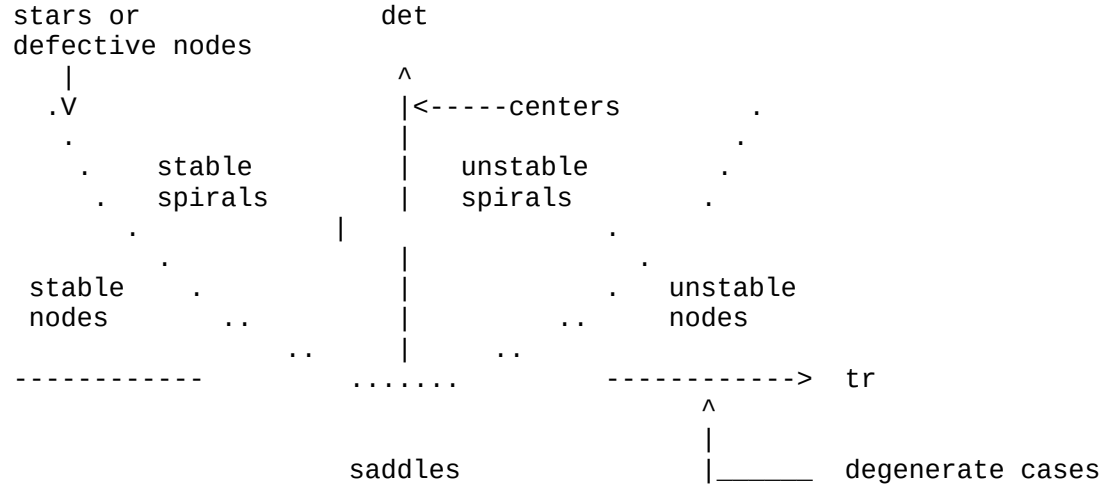
For example A = [1 1 ; 0 -2] has tr(A) = -1 , det(A) = -2
so the eigenvalues are lambda1 = 1 , lambda2 = -2

Eigenvector for lambda1 is again [1;0] , since the first column of the matrix is [1;0] .

For lambda2 we subtract -2 from the diagonal: [3 1 ; 0 0]
so v2 = [-1;3] .

The trajecories of other solutions are asymptotic to the eigenline for value +1 as t ---> infinity and to the eigenline for value -1 as t ---> - infinity.

The corresponding phase portraits exhibit the following behaviors:



[3] Marginal cases: For the most part, the trace and determinant determine the name and essential features of the phase portrait, but there are some things which require inspecting the matrix itself.

-- det = tr^2/4 , along the critical parabola: repeated real eigenvalues.

For example on Friday [Slide] we discussed the matrix $A = \begin{bmatrix} -2 & 1 \\ -1 & 0 \end{bmatrix}$
and saw that it has eigenvalues $\lambda_1 = \lambda_2 = -1$,
eigenvector $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, so normal mode $u_1 = e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
Then we found a solution w to $(A - \lambda - 1) w = v_1$ to be
$w = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and wrote down the extra solution

$$u_2 = e^{\lambda_1 t} (t v_1 + w) = e^{-t} \left[t \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$$

$$= e^{-t} \begin{bmatrix} t+1 \\ t \end{bmatrix}$$

What is this phase portrait like? There are the eigensolutions.
The solution u_2 has $u_2(0) = w = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. When t increases
away from zero, the x-coordinate becomes positive. All solutions decay
to zero as t grows. You can see that as t increases the trajectory of
$u_2(t)$ becomes tangent to the eigenline. It hooks around and comes
in towards zero from the other direction. This lets you fill out this
phase portrait, which is called a "defective nodal sink."

This isn't the only thing that can happen if you have a repeated eigenvalue,
though. What if $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, for example? This is upper triangular,
so the eigenvalues are the diagonal entries, 2 and 2 : repeated. To find
an eigenvector we subtract 2 from the diagonal entries - you get the zero
matrix. So every vector is an eigenvector! This is the "complete case."
Every ray from the origin is the trajectory of a solution. The phase portrait
is called a "star source."

To summarize: if there is a repeated eigenvalue (in the 2 x 2 case)
then either:
the matrix is diagonal, when the matrix is "complete" an the phase portrait
is a star, or
the matrix is not diagonal, when the matrix is "defective," you have to
use the algorithm described above to find a second independent solution,
and the phase portrait is a defective node.

Another marginal case is:

det = 0 : at least one of the eigenvalues is zero.
If v is an eigenvector corresponding to this eigenvalue, then
the constant vector valued function $u(t) = c \alpha$ is a solution for
any constant c: there is a line (at least) of constant solutions.
Several patterns are possible, and they are illustrated in the
Supplementary Notes.

The phase portrait in each one of these borderline cases shows some
features which are not determined purely by the eigenvalues.
Another feature for which you need to go back to the matrix is deciding
on whether a spiral is turning clockwise or counterclockwise.
(This is independent of the question of whether it is a sink or a source.)

We can tell which you get by thinking about what u' is at some point.
A convenient one to pick is $[1;0]$: then $u' = Au =$ (first column of A).
So if the bottom left entry is positive, the spiral is moving
counterclockwise; if negative, it is moving clockwise.

[4] (I did not talk about:)

Stability: All linear systems fall into one of the following categories:

Asymptotically stable: all solutions ---> 0 as t ---> infinity
These systems occupy the upper left quadrant, tr < 0 and det > 0,
so the eigenvalues have negative real part.

Unstable: most solutions ---> infinity as t ---> infinity
Saddles and unstable nodes and spirals are examples.

Neutrally stable: all solutions stay bounded as t ---> infinity
but most do not ---> 0 .
These systems occur along the boundary between the other two types:
-- Along the ray tr = 0 , det > 0, so the eigenvalues are nonzero and
purely imaginary. The phase portrait is a "center" and the trajectories
are ellipses.
-- Along the ray det = 0 , tr < 0 , so one eigenvalue is zero and the
other is negative. The phase portrait is a comb.

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18.03 Class 36, May 5, 2010

The matrix exponential: initial value problems.

1. Definition of e^{At}
2. Computation of e^{At}
3. Uncoupled example
4. Defective example
5. Exponential law

[1] Recall from day one:

(a) x' = rx with initial condition x(0) = 1 has solution x = e^{rt} .

x' = rx with any initial condition has solution x = e^{rt} x(0) .

Later, we decided to *define* e^{it} as the solution to

(b) x' = ix with initial condition x(0) = 1 .

Following Euler, a solution is given by cos t + i sin t , so we found that

$$e^{it} = \cos(t) + i \sin(t)$$

(c) Now we are studying u' = A u . Let's try to *define*

The solution to u' = Au with initial condition u(0) is u = e^{At}u(0). (**)

Note that the initial value u(0) is a vector, and u(t) is a vector valued function. So the expression e^{At} must denote a matrix, or rather a matrix valued function.

What could e^{At} be? For a start, what is its first column? Recall that the first column of any matrix B is the product B[1;0] , and [b_1 b_2] [1;0] = b_1 , so combining this with (**) we see:

The first column of e^{At} is the solution to u' = Au with u(0) = [1;0].

Similarly,

The second column of e^{At} is the solution to u' = Au with u(0)= [0;1].

This is the DEFINITION of e^{At} . It makes (**) true for all u(0) , because e^{At} u(0) is a solution (being a linear combination of the columns of e^{At} , which are solutions), and when t = 0 we get

$$e^{A0} u(0) = I u(0) = u(0) .$$

[2] Computation of e^{At}

We need a method for computing it, though. To explore this we'll use the

Example: A = [1 1 ; 0 2] .

This is upper triangular, so its eigenvalues are given by the diagonal entries: lambda = 1, lambda = 2. The (tr,det) pair lies in the upper right quadrant, below the critical parabola; the phase portrait is an unstable node.

Find eigenvectors:

  lambda_1 = 1 :   A - I : [0 1 ; 0 1] [?;?] = [0;0] : v_1 = [1;0]

  lambda_2 = 2 : A - 2I : [-1 1 ; 0 0] [?;?] = [0;0] : v_2 = [1;1]

Two independent solutions are given by

      u_1 = [e^t ; 0] , u_2 = [e^{2t} ; e^{2t}]

and the general solution is

      u = c_1 [e^t ; 0] + c_2 [e^{2t} ; e^{2t}]

We could go ahead and solve for c_1 and c_2 to get solutions with the desired initial conditions. What follows is a clever way to do that.

There is a compact way to write this linear combination: it is

      u = [e^t , e^{2t} ; 0 , e^{2t}] [c_1 ; c_2].   (***)

This matrix is a "fundamental matrix" for the system: its columns are independent solutions. Such a matrix will be denoted by Phi(t); so here

      Phi(t) = [e^t , e^{2t} ; 0 , e^{2t}]

Phi(t) behaves very much like we want e^{At} to behave; its columns are solutions, even independent ones, and the general solution is given by

      Phi(t) [c_1 ; c_2]

The matrix exponential e^{At} is a fundamental matrix: it is the fundamental matrix Phi(t) such that Phi(0) = I .

Our Phi(t) does not evaluate this way. To fix this, I claim we should form

      Phi(t) Phi(t)^{-1}

Explanation: If B is a square matrix, you can ask whether it has an *inverse* matrix, a matrix B^{-1} such that

       B B{-1} = I   and   B^{-1} B = I

(either implies both). The answer, as for numbers, is not always. It turns out that there is an inverse exactly when det(B) is not zero.

In the 2x2 case, B = [a b ; c d] and

      [a b ; c d]^{-1} = (1/det(A)) [d -b ; -c a].

We can check this:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = (\det B)\, I$$

Now let's see: each column $\Phi(t)\, \Phi(0)^{-1}$ is a linear combination of the columns of $\Phi(t)$, so it's a solution. What remains is to check the normalization; but $\Phi(0)\, \Phi(0)^{-1} = I$.

Conclusion:

$$e^{At} = \Phi(t)\, \Phi(0)^{-1}$$

where $\Phi(t)$ is ANY fundamental matrix for A.2A

In our example, $\Phi(0) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $\Phi(0)^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$, and so

$$e^{At} = \begin{bmatrix} e^t & e^{2t} \\ 0 & e^{2t} \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} e^t & e^{2t} - e^t \\ 0 & e^{2t} \end{bmatrix}.$$

[3] Uncoupled example: Suppose $A = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$.
The eigenvalues are $\lambda_1 = a$ and $\lambda_2 = d$

I can see the eigenvectors: $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Basic solutions are $e^{\lambda_1 t} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
$e^{\lambda_2 t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

so $\Phi(t) = \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix}$

and this is already normalized: so it is the matrix exponential.

[4] Defective example.

Sometimes the matrix exponential can be a bit unexpected. For example:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Then $\operatorname{tr} A = 0$ and $\det A = 0$, so the only eigenvalue is 0, with multiplicity 2. This is not a diagonal matrix, so it is defective, and we could find solutions by the standard method. However, it is also a companion matrix, for the second order equation $x'' = 0$.
Solutions of this are easy! $x_1 = 1$, $x_2 = t$. So basic solutions to

$$u' = A u$$

are $u_1 = \begin{bmatrix} x_1 \\ x_1' \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
$u_2 = \begin{bmatrix} x_2 \\ x_2' \end{bmatrix} = \begin{bmatrix} t \\ 1 \end{bmatrix}$

$$\Phi(t) = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

This satisfies $\Phi(0) = I$, so

e^{At} = [1 t ; 0 1]

[5] We can take t to be a specific value, of course: eg t = 1 :

e^[0 1 ; 0 0] = [1 1 ; 0 1]

and this lets us define e^A for any square matrix A .

Then e^0 = I , as you might expect, but watch out:

e^A e^B = e^{A+B} *provided that* AB = BA

So for example (e^A)^n = e^{nA}

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18.03 Differential Equations

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18.03 Class 37, May 7, 2010

Vector fields and nonlinear systems

1. Autonomous systems
2. Vector fields
3. Equilibria
4. Linearization

[1] Introduction to general nonlinear autonomous systems.

Recall that an ODE is autonomous if x' depends only on x and not on t:

$$x' = f(x)$$

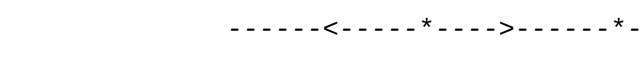
For example, I know an island in the St Lawrence River in upstate New York where there are a lot of deer [Slide]. When there aren't many of them, they multiply with growth rate k; $x' = kx$. Soon, though, they push up against the limitations of the island; the growth rate is a function of the population, and we might take it to be $k(1-(x/a))$ where a is the maximal sustainable population of deer on the island. So the equation is

$$x' = k(1-(x/a))x \quad , \quad \text{the "logistic equation."}$$

On this particular island, $k = 3$ and $a = 3$, so $x' = (3-x)x$. (Maybe the units are hundreds of deer, and years).

There are "critical points" at $x = 0$ and $x = 3$.
When $0 < x < 3$, $x' > 0$.
When $x > 3$, $x' < 0$, and, unrealistically, when $x < 0$, $x' < 0$ too.

I drew some solutions, and then recalled the phase line:



One day, a wolf [Slide] swims across from the neighboring island, pulls himself up the steep rocky coast, shakes the water off his fur, and sniffs the air. Two wolves, actually.

Wolves eat deer, and this has a depressing effect on the growth rate of deer. Let's model it by

$$x' = (3-x-y)x$$

where y measures the population of wolves.

Now, wolves in isolation follow a logistic equation too, say

$$y' = (1-y)y \quad \text{(no deer)}$$

But the presence of deer increases their growth rate, say

$$y' = (1-y+x)y$$

We have a *nonlinear autonomous system*

$$x' = (3-x-y)x$$
$$y' = (1-y+x)y$$

[2] Vector fields. The general case would be

$$x' = f(x,y)$$
$$y' = g(x,y)$$

We can visualize this system in the following way. The pair of populations determines a point on the plane, [x ; y]. The population evolves through time, tracing a curve on the plane: u(t) = [x(t) ; y(t)] . What we know about this curve is that its velocity vector,

$$u'(t) = [\ x'(t)\ ;\ y'(t)\ ]$$

is a function the current position on the plane:

$$u'(t) = F(u(t))\ ,\quad F(x,y) = [\ f(x,y)\ ;\ g(x,y)\ ]$$

This data associates a vector to each point in the plane. It is a "vector field," as studied in 18.02.

An autonomous system is the same thing as a vector field. A solution is a vector valued function u(t) whose velocity vector at each point it goes through is given by the value of the vector field at that point.

We have not been drawing the vector fields of the linear systems we've been studying, but they are there. For example A = [0 -1 ; 1 0] determines the vector field with value at [x;y] given by

$$A[x;y] = [-y;x]$$

This is the vector with the same length as [x;y] but which points at right angles to the radius vector. Tr(A) = 0, Det(A) = 1 , so we have a center. In fact the trajectories are circles.

I displayed the vector field associated to the deer/wolf system, using the web version of pplane, http://math.rice.edu/~dfield/dfpp.html .

The trajectories are more complicated, and I clicked up a bunch of them. This is the "phase portrait" of the autonomous system.

Note that the phase line of the deer-only system occurs along the x-axis, and the wolf-only system along the y-axis. The "phase line" we studied back in section 1 is a 1-dimensional phase diagram.

[3] Now, again, we are interested in equilibria. There seem to be four of them in this example. The behavior near them looks pretty similar to the linear phase portraits we have just been studying: I see a node near (0,0), saddles on the x- and y-axes, and a spiral out in the middle somewhere.

This is a deep and important fact: usually, near an equilibrium an autonomous system is well approximated by a linear system, which we can

then analyze using eigenvectors and eigenvalues. You can see the eigenvectors in the nonlinear phase diagram!

Let's find out where the equilibria, or critical points, are located.

First think about where the horizontal component of the vector field vanishes:

$x' = 0$, that is $(3-x-y)x = 0$.

This happens when either $x = 0$ or $3-x-y = 0$; that is, the vector field is vertical along those two lines.

The vertical component vanishes where

$y' = 0$, that is $(1-y+x)y = 0$.

This happens when either $y = 0$ or $1-y+x = 0$; that is, the vector field is horizontal along those two lines.

The vector field is both vertical and horizontal exactly when it vanishes. There are four places where that happens:

$(0,0)$, $(3,0)$, $(0,1)$,

and the place where both $y = 3-x$ and $y = 1+x$, which is at $(1,2)$.

[4] Linearization. Let's see what happens near (0,0), for a start:

Expanding out,

$x' = f(x,y) = 3x - x^2 - xy$
$y' = g(x,y) = y + xy - y^2$

Near the origin the quadratic terms in the vector field are insignificant, so "to first order"

$x' = 3x$
$y' = y$

The deer and wolf populations both expand by natural growth.
They are "decoupled." This is a
homogeneous linear system with matrix $[3 0 ; 0 1]$. The eigenvalues are 1 and 3 , so we have a node. The eigenline for the smaller eigenvalue is the y axis, so this is the line to which all solutions (except for the other normal mode!) all solutions become tangent as the approach the origin.

At some other point $P = (a,b)$, we have the "tangent line (or plane) approximation:

$f(x,y) \sim f(a,b) + (\partial f/\partial x)|P\ (x-a) + (\partial f/\partial y)|P\ (y-b)$

$g(x,y) \sim g(a,b) + (\partial g/\partial x)|P\ (x-a) + (\partial g/\partial y)|P\ (y-b)$

Suppose (a,b) is a critical point for, so the $f(a,b) = g(a,b) = 0$.
Use a new coordinate system:

$u = [\ x-a\ ;\ y-b\ ]$

Then

$u' \sim J(a,b)\ u$

where J is the Jacobian matrix,

$J(x,y) =$

$$\begin{matrix} \partial f/\partial x & \partial f/\partial y \\ \partial g/\partial x & \partial g/\partial y \end{matrix}$$

For us: $J = [\ 3-2x-y\ ,\ -x\ ;\ y\ ,\ 1+x-2y\ ]$

So $A = J(1,2) = [\ 3-2-2\ ,\ -1\ ;\ 2\ ,\ 1+1-4\ ] = [\ -1\ ,\ -1\ ;\ 2\ ,\ -2\ ]$.

The point is that near to $(1,2)$ the system behaves very much like the linear system with this matrix of coefficients.

The characteristic polynomial is

$$p_A(\lambda) = \lambda^2 + 3\lambda + 4 = (\lambda + (3/2))^2 + (7/4)$$

so the eigenvalues are $-(3/2) \pm i\ \sqrt{7}/2$.

We have a stable spiral, rotating counterclockwise since the lower left entry is positive. I sketched this and observed that it does correspond to the nonlinear phase portrait near that critical point.

We have learned that the populations of deer and of wolves converge to these stable levels no matter what the initial populations are (as long as they are positive). Also, as the population levels approach this limiting value, they oscillate about them, decaying exponentially like $e^{-3t/2}$ and with period

$(2\pi) / (\sqrt{7}/2) \sim 4.75.$

The entire upper right quadrant is a "basin,"
with the equilibrium $(1,2)$ a "sink" or "attractor."

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18.03 Class 38, May 10, 2010

Linearization: The nonlinear pendulum and phugoid oscillation

[1] Nonlinear pendulum
[2] Phugoid oscillation
[3] Buckling bridge

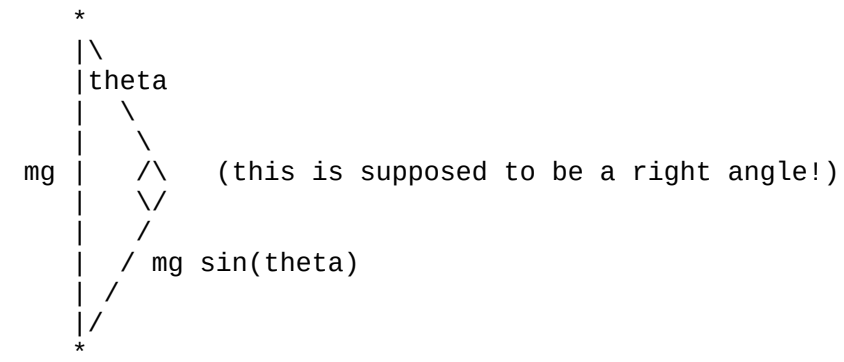
[1] The bob of a pendulum is attached to a rod, so it can swing clear around the pivot. This system is determined by three parameters:

L length of pendulum
m mass of bob
g acceleration of gravity

We will assume that the motion is restricted to a plane.

To describe it we need a dynamical variable. We could use a horizontal displacement, but it turns out to be easier to write down the equation controlling it if you use the angle of displacement from straight down. Write theta for that angle, measured to the right.

Here is a force diagram:



Write s for arc length along the circle, with s = 0 straight down. Of course,

$$s = L\,\theta$$

Newton's law says

$$ms'' = F$$

The force has the $-mg\sin(\theta)$ component of the force of gravity (and notice the sign!), and also a frictional force which depends upon

$$s' = L\,\theta'$$

Make the simplest model for friction, $-cs' = -cL\,\theta'$. So:

$$mL\,\theta'' = -mg\sin(\theta) - cL\,\theta'$$

Divide through by mL and we get

theta" + b theta' + k sin(theta) = 0

where k = g/L and b = c/m .

This is a nonlinear second order equation. It still has a "companion first order system," obtained by setting

x = theta   ,   y = x'

so y' = theta" = - k sin(theta) - b theta'   or

x' = y
y' = - k sin(x) - by

This is an autonmous system. Let's study its phase portrait.

Equilibria: y = 0 , sin(x) = 0 ; that is,

x = 0 , +-pi , +-2pi , ...

Let's compute the Jacobian:

J(x,y) = [0 , 1 ; - k cos(x) , -b]

When x = 0 , +-2pi , +-4pi , ..., cos(x) = 1 and

J(x,y) = [0 1 ; -k -b]

det = k , tr = -b . Suppose b is small, so we get spirals.

When x = +-pi , +-3pi , ...,   cos(x) = -1 and

J(x,y) = [0 1 ; k -b]

det = -k , tr = -b : saddles.

Then I revealed the entire phase portrait. It shows another feature common to all companion systems:

The trajectories above the x axis move right, those below move left.

Trajectories coming down from the left represent the pendulum swinging around in counterclockwise complete circles. Trajectories coming up from the right represent the pendulum swinging around in clockwise circles.

With a student I animated the pendulum swinging around. The successive dips represent passing through the vertical position. In very exceptional cases, the trajectory heads straight at the saddle equilibria; they converge to it exponentially, but most likely miss and move away from it exponentially. The saddles represent the unstable equilibria which are straight up. Eventually, the trajectory gets caught in a basin (actually it was always in that basin) and spiral in towards the attractor of that basin, which is straight down.

[2] Phugoid equation.

The forces on an airplane are like this: suppose the plane is moving
to the right. It is tipped up a bit, which is hard to draw using ascii.

Along the directon of motion we have the thrust, created by the engines
Directed downwards we have the force of gravity, AKA the weight of the plane.
There is a frictional force, which we decompose into two componets:
Drag is directed against the thrust,
Lift is directed perpendicularly to the direction of motion.

One can analyze the effect of these forces near equilibrium.
Write F for the thrust
g for the acceleration of gravity
m for the mass of the plane
v_0 for equilibrium speed.

At equilibrium the airplane is moving horizontally.
We study what happens when you are jarred off of equilibruium, by a downdraft
for example. Then the horizontal component of the velocity is v_0 + u
and the vertical component is w , where u and w are small relative
to v_0 .

$$\frac{d}{dt}\begin{pmatrix} u \\ w \end{pmatrix} = \begin{pmatrix} -2F/mv_0 & -g/v_0 \\ 2g/v_0 & 0 \end{pmatrix}\begin{pmatrix} u \\ w \end{pmatrix}$$

The characteristic polynomial is

$$p_A(\lambda) = \lambda^2 + (2F/mv_0)\lambda + (2g^2/v_0^2)$$

Let's use the language of natural frequency and damping ratio:

$$p_A(\lambda) = \lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2$$

so

$$\omega_n = (\sqrt{2})\, g/v_0$$

$$2F / mv_0 = 2\zeta\omega_n = 2\zeta(\sqrt{2})\, g / v_0$$

$$\zeta = (1/\sqrt{2})\, F/mg$$

Lessons to be learned from this linearization:

(1) The period of this "phugoid oscillation" depends *only on the equilibrium
speed* and not on the thrust or mass of the airplane. In units of meters and
seconds, it comes to about 0.45 v_0 .

Boeing 747 : 260 m/sec : 118 seconds
F15 : 838 m/sec : 380 seconds

(2) The damping ratio depends only on *thrust/weight*. Both zeta and
thrust/weight are dimensionless, the same in all units.

The system is underdamped as long as zeta < 1 , ie F/mg < sqrt(2)

Even an F15 doesn't come close to having F/mg > sqrt(2) ! - so the phugoid system is always underdamped:

747 :   F/mg ~ .27 , zeta ~ .19
F15:   F/mg ~ .67 , zeta ~ .47

[3] I want to show you this movie. In the summer of 1940, a bridge was built ove the Tacoma Narrows in Washington State. It was a mile long and just 39 feet across - an elegant suspension bridge. Very early it acquired the sobriquet "Galloping Gerdie," because it would oscillate vertcially in a very exciting way. The structural engineers studied this and concluded that it wasn't a problem. But in November, the first winter storm picked up. The bridge showed a different behavior now - it wasn't oscillating vertically now, it was twisting. Here's a video -

http://www.youtube.com/watch?v=P0Fi1VcbpAI

What happened here? Resonance? The wind was gusting back and forth at just the natural frequency of the bridge? Nah. What happens is this:

Suppose theta is the angle off of horizontal, at some point along the bridge.

$$\theta'' + b\,\theta' + \omega^2 \theta = F$$

F depends on the wind velocity, but look: When the edge of the bridge tips up, the torisonal force from the wind changes. So perhaps F looks like

$$F = a(v)\,\theta + c(v)\,\theta'$$

Actually, it turns out that the dominant contribution is from the theta' term, so let's simplify by just dropping the theta term.

So: $\theta'' + (b-c(v))\,\theta' + \omega_n^2 \theta = 0$

Now the characteristic polynomial is

$$s^2 + (b-c(v))\,s + \omega_n^2$$

$$= (s + (b-c(v))/2)^2 + \omega_d^2$$

with roots

$$r = (c(v)-b)/2 \pm \omega_d$$

(The experiment shows the roots are not real!)

It turns out that the shape the graph of c(v) is like a W . For v near zero, c(v) < 0 : the effect increases the damping and causes the bridge to be MORE stable. At some point c(v) = 0, which is still OK. But eventually it equals and then surpasses the damping constant b : then you get effective anti-damping, the real part of the root becomes positive, and we get solutions oscillating under increasing exponentials: disaster!

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18.03 Differential Equations

Spring 2010

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18.03 Class 39, May 12, 2010

Dangers of linearization; limit cycles; chaos; next steps in mathematics

1. Stability
2. Limits of linearization
3. Limit cycles
4. Strange attractors
5. Essential skills
6. Next steps

[1] Let's say that an equilibrium is
-- "stable" if all nearby trajectories stay near to it and converge to it as t --> infinity
-- "unstable" otherwise

We classified linear phase portraits using the (Tr,Det) plane.
In the (Tr,Det) plane, the stable region is the upper left quadrant.
It is bounded by the straight half-lines where either the eigenvalues are purely imaginary, or one is zero and the other is negative.

[2] The method we sketched on Friday and Monday works well "generically," i.e. almost all the time. It lets you predict what the phase portrait of an autonomous system looks like near equilibrium, most of the time, and whether the equilibrium is stable or not.

The facts: (1) If the linearization is not on the borderline of the stable quadrant, it correctly predicts the stability of the equilibrium.

(2) If the eigenvalues of the Jacobian have nonzero real parts - so (tr,det) is neither on the horizontal axis nor on the tr = 0, det > 0 line -- then the phase portrait is a small deformation of the linear phase portrait.

Example: x' = -y - y^2, y' = x + y^2 .
The linearization at (0,0) is x' = y , y' = -x : trace = 0 , det > 0, so you expect centers (which are not stable), but the fact is that you get stable spirals (seen on a pplane plot). Other examples might show stable spirals instead, or centers. You can't tell which you get from the linearization alone. [These spiral decay to zero more slowly than exponential, though.]

Example: x' = 4xy , y' = x^2 - y^2 .
The linearization at (0,0) is now x' = 0 , y' = 0 :

$$J(0,0) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Degenerate: det = 0. In fact *both* eigenvalues are zero (and it is complete). What is the phase portrait of this linear system? It just says u' = 0 so every point is a constant solution. The eigenvalues are both zero, every vector is an eigenvector, every solution is constant, it is a degenerate system.

On the other hand, a Matlab plot [Slide] of the phase portrait shows that it has six ray solutions, unlike any linear portrait. Weird. Second order effects dominate.

[3] In studying a nonlinear autonomous system, the most important thing, after equilibria, is the possibility of periodic solutions.

For example, the "van der Pol" equation is

$$x'' + a\,(x^2-1)\,x' + x = 0$$

for fixed a. When $a = 0$ this is just the harmonic oscillator, with $\omega_n = 1$; all the nonzero solutions are periodic of period 2π.

The companion system is

$$\begin{aligned} x' &= y \\ y' &= -x - a\,(x^2-1)\,y \end{aligned}$$

This system turns out to continue to have periodic solutions. When $a > 0$ the situation is in fact even better: there is ONLY ONE periodic trajectory, and all other nonzero solutions converge to it. This is a "limit cycle." The human heart is in fact controlled by an equation like this. This is why it dependably returns to a normal periodic pattern after being disturbed.

The limit cycle is an "attractor," in the sense that every nearby trajectory stays nearby and converges to it. I demonstrated the Mathlet "Vector Fields."

[4] Limit cycles are typical for 2D systems.
But when you move to 3D things are much more complicated. The first such system was discovered right here at MIT by Edward Lorenz [Slide], who was modeling "convection rolls" in the upper atmosphere. In 1963 he wrote down a fairly simple model, a nonlinear autonomous system in 3 dimensions:

$$\begin{aligned} x' &= -ax + by \\ y' &= -xz + rx - y \\ z' &= xy - bz \end{aligned}$$

for constants a , b , r. Here's what happens for certain values of these parameters - and I showed the IDE tool "Lorenz Equations, Phase Plane, $0 < r < 30$. <http://www.aw-bc.com/ide/idefiles/media/JavaTools/lrnzr320.html>.

The solutions don't ever settle down to a periodic orbit; but neither do they run off to infinity. They just wrap pretty crazily around the two nonzero unstable equilibria which exist provided that $r > a/b$.

Now the "attractor" is much stranger than just a loop, and, strangely, is called a "strange attractor." This is the fundamental characteristic of "chaos."

[5] In addition to these skills [Slide] there are a couple of big ideas: Linearity is one. The exponential function is another.

1. First order models. Euler's method. Autonomous equations.

2. First order linear equations: Integrating factors

3. Complex numbers and exponentials.

4. Second order LTI systems: Superposition; Exponential Response Formula; exponential or sinusoidal signal; amplitude gain and phase lag; Polynomial signal; Variation of parameter.

5. Delta functions, unit impulse response, convolution.

6. Fourier series, periodic solutions.

7. Laplace transform; transfer function and weight function.

8. Linear systems: eigenvalues, eigenvectors.

9. Linear phase portraits.

10. Nonlinear phase portraits; linearization at equilibria.

[6] Study resources:

Practice final.

OCW Final Exam and Practice Final.

Absent from this year's curriculum:

--variation of parameters as an alternative to integrating factors

--``Exponential Shift Law'' (We did these problems using a different ``variation of parameters: solve $p(D)x=q(t)e^{rt}$ by trying $x=u(t)e^{rt}$ and finding a differential equation for $u(t)$.)

--Laplace transform solution of second order equations with non-rest initial conditions

Other resources:

Hour exams and practice hour exams

Recitation problems and their solutions

Homework and solutions

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